HabCam SOP

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Chapter 1

Home

Download Full SOP as PDF

Welcome to the HabCam Operational Standard Operating Procedure (SOP) Manual.

[Insert homepage overview, welcome message, quick navigation links, and mission overview here]

Part I

Program Information

Chapter 2

Marine Development & Advanced Technology

NOAA Fisheries | Northeast Fisheries Science Center

Core Mission & Strategy

Develop surveys, advanced technologies, and analytical approaches to inform stock and ecosystem assessments.

2.0.1 Interactions with Offshore Wind

Offshore wind energy developments impact at least **14 long-term scientific surveys** in the U.S. Northeast region (e.g., bottom trawl, scallop, ecosystem monitoring) through preclusion, habitat change, and survey design disruptions.

2.0.2 6-Step Survey Mitigation Framework

1. **Evaluate Impacts:** Assess disruptions to spatial and statistical survey coverage.
2. **Design Methods:** Develop new survey methodologies for wind turbine environments.
3. **Calibrate Surveys:** Establish statistical cross-calibration between old and new tools.
4. **Bridge Solutions:** Maintain continuous data collection during construction phases.

5. **Conduct Surveys:** Deploy uncrewed surface/underwater systems and acoustic monitoring.
6. **Communicate Data:** Provide transparent data to fishery managers and stakeholders.

2.1 Key Technological Initiatives

2.1.1 1. Long-Range AUV (LRAUV) Development

- **Platform & Payload:** Developed in collaboration with MBARI and WHOI, incorporating dual stereo camera payloads to mimic the Habitat Camera (HabCam) setup.
- **Objectives:** Access wind farm lease areas and reduce total days at sea required for HabCam survey coverage.
- **2024 Deployments:** Sampled over 200 nautical miles in May and June 2024. Calibrations with dredge and HabCam V4 showed no significant bias in scallop shell height measurements.
- **Nearshore & Offshore Operations:** Successfully tested operating from small vessels nearshore (e.g., South Fork Wind Farm in November 2024).

2.1.2 2. Uncrewed Surface Vehicles (USV - DriX)

- **Operational Value:** Designed to sample close (20–30 meters) to wind turbine structures, operate 24/7 safely, and maneuver efficiently around complex installations.
- **Missions:** Conducted a 20-day survey covering three offshore wind areas (including Block Island, Cox Ledge, and Vineyard Wind) in late 2023. Continued sampling scheduled focusing on North Atlantic right whale forage habitat questions.

2.1.3 3. Habitat Camera (HabCam) Survey

- **System Overview:** The HABitat CAMera system (HABCAM) is a non-invasive, underwater imaging vehicle towed from the stern of research vessels approximately 2 meters above the seafloor, capturing high-resolution, overlapping stereo images.
- **Sensor Suite:** The camera array is additionally equipped with a compass, CTD, sonar, and sensors that survey the seafloor, which can be used for estimates of species abundance, habitat characterization, and biodiversity—giving a holistic view of benthic taxa.

- **Resource Monitoring:** This vehicle has been used extensively for surveying the sea scallop resource on Georges Bank and the Mid-Atlantic Bight, as well as collecting data on finfish and habitat.

2.1.4 4. Machine Learning & Optical Enhancements

- **Automated Processing:** Developing machine learning tools (e.g., VIAME software) under the NOAA Inflation Reduction Act (IRA) Optical Strategic Initiative to automate scallop identification and measurements, replacing manual annotation.
- **Ocean Acidification Monitoring:** Integrated a channelized optical system into HabCam in 2024 to measure bottom water dissolved inorganic carbon (DIC) and pCO , pairing scallop biomass with ocean chemistry data.

2.1.5 5. Active Acoustics & Cooperative Research

- **Long-Term Data:** Active acoustic data collected continuously since 1998 across multiple frequency bands (18, 38, 120 kHz).
- **Industry Collaboration:** Partnering with commercial fishing vessels to evaluate catchability, study fish life history, and develop acoustic survey methods for species like Atlantic mackerel.

2.2 Administrative & Future Directions

- **Proposals & Funding:** Submitted 7 proposals since May 2024 across VIAME machine learning, mackerel acoustics, oceanographic USV sampling, and AUV-ABIS development.
- **Staffing Updates:** Hiring underway for a HabCam Scientist, Biological Science Technician, and Electronics Engineers.
- **Future Priorities:**
  - Passive acoustics for groundfish (e.g., Atlantic cod, haddock).
  - Expanding AUV optical applications to complementary technologies like eDNA.
  - Model-based integration of traditional and advanced technology survey data.
  - Directed catchability and mitigation studies for marine energy development.

Chapter 3

Team Directory

M. Conor McManus, Ph.D. Supervisory Research Fish Biologist Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Marine Development & Advanced Technology Program	Col 2

Conor earned a BA in Marine Science from Boston University, and a MS and PhD in Oceanography from the University of Rhode Island's Graduate School of Oceanography. Previously, Conor worked for Applied Science Associates, and then the Rhode Island Department of Environmental Management's Division of Marine Fisheries. Conor joined the NEFSC in 2024 and leads the Marine Development and Advanced Technology Program.

Contact: - **Office:** (401) 859-3404 - **Email:** michael.conor.mcmanus@noaa.gov - **Links:** Google Scholar

Cameron Fairclough Advanced Technology Specialist Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Marine Development & Advanced Technology Program	Col 2

Cameron earned a BA in Marine Biology from Roger Williams University in 2019. Previously, Cameron worked for Woods Hole Oceanographic Institution in various roles in the Biology and Applied Ocean Physics and Engineering departments. Cameron joined the NEFSC in 2022 as an Electronics Engineer contractor before transitioning to an Advanced Technology Specialist in 2024.

Contact: - **Office:** (774) 704-2274 - **Email:** cameron.fairclough@noaa.gov

Nicholas M. Calabrese, Ph.D. Fish Biologist Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Marine Development & Advanced Technology Program	Col 2

Nicholas Calabrese is a fisheries scientist specializing in the development and application of advanced technologies for fisheries-independent surveys. His research focuses on integrating optical imaging systems, artificial intelligence, and innovative sampling approaches to improve marine resource assessments.

Prior to joining NOAA, Nicholas led the development of optical video trawl systems and other optical survey technologies at UMass Dartmouth's School for Marine Science and Technology (SMAST).

Education: - B.S. Marine Biology, Roger Williams University, RI - M.S. Living Marine Resource Science and Management, UMass Dartmouth SMAST, MA - Ph.D Living Marine Resource Science and Management, UMass Dartmouth SMAST, MA

Contact: - **Office:** (401) 307-1599 - **Email:** nicholas.calabrese@noaa.gov - **Links:** LinkedIn | ResearchGate | ORCID | Google Scholar

Shannen M. Allen Biological Technician III (Affiliate) Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Marine Development & Advanced Technology Program	Col 2

Shannen earned a BS in Marine Science, Safety, and Environmental Protection from Massachusetts Maritime Academy in 2023. Prior to joining NEFSC, Shannen worked as a Hydrographer and Marine Scientist supporting NOAA's Office of Coast Survey. She joined the NEFSC program in 2026 as a Biological Technician III Affiliate.

Contact: - **Office:** (508) 645-6731 - **Email:** shannen.allen@noaa.gov - **Links:** LinkedIn

Jeremy F. Jenrette, Ph.D. Data Scientist (Affiliate) Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Marine Development & Advanced Technology Program	Col 2

Jeremy earned a BS in Biochemistry in 2018 and a PhD in Fish and Wildlife Conservation from Virginia Tech in 2025. In 2026, Jeremy began working for NOAA's Advanced Technology Program developing image-recognition software for assessing sea scallop populations from Habitat Camera Trawl Surveys.

Contact: - **Office:** (540) 577-6500 - **Email:** jeremy.jenrette@noaa.gov - **Links:** LinkedIn | ResearchGate | ORCID | Google Scholar | Website

Dana Morton, Ph.D. Fishery Biologist Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Shellfish Survey Program	Col 2

Dana is a marine ecologist with research interests in shellfish biology, invertebrate zoology, food webs, parasite ecology, and spatial ecology. She earned a Ph.D. from UC Santa Barbara, an M.Sc. from San Diego State University, and a B.S. from UC Santa Cruz. She began working at NOAA in 2023.

Contact: - **Office:** (774) 238-7424 - **Email:** dana.morton@noaa.gov - **Links:** Google Scholar

---

Zachary Fyke Biological Science Technician Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Shellfish Survey Program	Col 2

Zachary graduated from Michigan State University with a B.S. in Fisheries and Wildlife in 2017. As part of ESB, Zachary primarily works on Shellfish Surveys including Atlantic Sea Scallop, Northern Shrimp, Ocean Quahog, and Atlantic Surfclam.

Contact: - **Office:** (508) 495-2002 - **Email:** zachary.fyke@noaa.gov

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Jonathan Duquette Biological Science Technician Population & Ecosystems Monitoring & Analysis Division / Ecosystems Surveys Branch / Shellfish Survey Program	Col 2

Jonathan Duquette is a Biological Science Technician with the Ecosystems Surveys Branch at the Northeast Fisheries Science Center. Specializing in fisheries science and marine data collection, Jonathan plays an integral role in critical research initiatives including HabCam and sea scallop dredge surveys.

Contact: - **Office:** (508) 495-2034 - **Email:** jonathan.duquette@noaa.gov

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Dvora Hart, Ph.D. Operations Research Analyst Resource Evaluation & Assessment Division / Populations Dynamics Branch / Stock Assessment Methods Task Program	Col 2

Dvora is the lead assessment scientist for Atlantic sea scallops, and secondary for spiny dogfish. She developed length-based and spatial stock assessment methods (such as the SAMS model) and tools to analyze HabCam towed camera survey data.

Contact: - **Office:** (508) 495-2369 - **Email:** deborah.hart@noaa.gov - **Links:** ResearchGate | Scopus | ORCID

Jui-Han Chang, Ph.D. Research Biologist (Affiliate) Resource Evaluation & Assessment Division / Populations Dynamics Branch / Stock Assessment Methods Task Program	Col 2

Education: - Ph.D. Interdisciplinary (Marine Sciences, Statistics, and Resource Economics), University of Maine - M.S. Marine Resource Assessment and Management, National Taiwan Ocean University - B.B.A. International Trade, Aletheia University

Contact: - **Office:** (508) 495-2052 - **Email:** jui-han.chang@noaa.gov

Alexander Hansell, Ph.D. Fish Biologist Resource Evaluation & Assessment Division / Populations Dynamics Branch / Stock Assessment Methods Task Program	Col 2

Alex earned a BS in Biology from Northeastern University and a PhD in Fisheries Oceanography from the University of Massachusetts. Alex is the lead assessment scientist for Georges Bank yellowtail flounder, Georges Bank winter flounder, and Southern New England Cod.

Contact: - **Office:** (508) 203-6894 - **Email:** alex.hansell@noaa.gov - **Links:** Google Scholar

Chapter 4

Interns

[Insert internship opportunities, student projects, application guidelines, and alumni records here]

Chapter 5

Publications

5.1 1. Peer-Reviewed Publications

- **Chang et al. (2016)**
  Chang, J.-H., Hart, D. R., Shank, B. V., Gallager, S. M., Honig, P., & York, A. D. (2016). Combining imperfect automated annotations of underwater images with human annotations to obtain precise and unbiased population estimates. *Methods in Oceanography*, 17, 169–186.
  https://repository.library.noaa.gov/view/noaa/52801

- **Chang et al. (2017)**
  Chang, J.-H., Shank, B. V., & Hart, D. R. (2017). A comparison of methods to estimate abundance and biomass from belt transect surveys. *Limnology and Oceanography: Methods*, 15(5), 480–494.
  https://repository.library.noaa.gov/view/noaa/52800

- **Kaplan et al. (2017)**
  Kaplan, K. A., Hart, D. R., Hopkins, K., Gallager, S., York, A., Taylor, R., & Sullivan, P. J. (2017). Evaluating the interaction of the invasive tunicate *Didemnum vexillum* with the Atlantic sea scallop *Placopecten magellanicus* on open and closed fishing grounds of Georges Bank. *ICES Journal of Marine Science*, 74(9), 2470–2479.
  https://repository.library.noaa.gov/view/noaa/53606

- **Miller et al. (2019)**
  Miller, T. J., Hart, D. R., Hopkins, K., Vine, N. H., Taylor, R., York, A. D., & Gallager, S. M. (2019). Estimation of the capture efficiency and abundance of Atlantic sea scallops (*Placopecten magellanicus*) from paired

photographic–dredge tows using hierarchical models. *Canadian Journal of Fisheries and Aquatic Sciences*, 76(6), 847–855.
https://repository.library.noaa.gov/view/noaa/67644

- **Hart and Chang (2022)**
  Hart, D. R., & Chang, J.-H. (2022). Estimating Natural Mortality for Atlantic Sea Scallops (*Placopecten magellanicus*) Using a Size-Based Stock Assessment Model. *NOAA Institutional Repository*.
  https://repository.library.noaa.gov/view/noaa/60704

- **Duskey et al. (2023)**
  Duskey, E., Hart, D. R., Chang, J.-H., & Sullivan, P. J. (2023). Partitioning spatial dynamics in abundance of marine fisheries stocks between fine- and broad-scale variation: a Bayesian approach. *NOAA Institutional Repository*.
  https://repository.library.noaa.gov/view/noaa/65742

- **Vanaki et al. (2025)**
  Vanaki, S., Hart, D. R., & Chang, J.-H. (2025). How many dollars are in the sea? Estimating sand dollar (*Echinarachnius parma*) abundance using an iteratively trained convolutional neural network and generalized additive models. *Ecological Informatics*, 90, 103311.
  https://repository.library.noaa.gov/view/noaa/72198

5.2 2. Resource Survey Reports

- **2012 Sea Scallop Resource Survey Report**
  NOAA Fisheries Service, Northeast Fisheries Science Center. (2012). *Resource Survey Report: Sea Scallop Survey, Cape Hatteras - Georges Bank, 1 June – 7 July 2012*. Woods Hole, MA: NOAA Fisheries Service.
  https://repository.library.noaa.gov/view/noaa/22541

- **2013 Sea Scallop Resource Survey Report**
  NOAA Fisheries Service, Northeast Fisheries Science Center. (2013). *Resource Survey Report: Sea Scallop Survey, Cape Hatteras - Georges Bank, 13 June – 20 July 2013*. Woods Hole, MA: NOAA Fisheries Service.
  https://repository.library.noaa.gov/view/noaa/22540

- **2014 Sea Scallop Resource Survey Report**
  NOAA Fisheries Service, Northeast Fisheries Science Center. (2014). *Resource Survey Report: Sea Scallop Survey, Mid-Atlantic Bight - Georges Bank, 30 June – 25 July 2014*. Woods Hole, MA: NOAA Fisheries Service.
  https://repository.library.noaa.gov/view/noaa/22539

- **2015 Sea Scallop Resource Survey Report**
  NOAA Fisheries Service, Northeast Fisheries Science Center. (2015). *Resource Survey Report: Sea Scallop Survey, Mid-Atlantic Bight - Georges Bank, 16 May – 21 June 2015*. Woods Hole, MA: NOAA Fisheries Service.
  https://repository.library.noaa.gov/view/noaa/22538

- **2016 Sea Scallop Resource Survey Report**
  NOAA Fisheries Service, Northeast Fisheries Science Center. (2016). *Resource Survey Report: Sea Scallop Survey, Mid-Atlantic Bight - Georges Bank, 19 May – 24 June 2016*. Woods Hole, MA: NOAA Fisheries Service.
  https://repository.library.noaa.gov/view/noaa/22537

- **2017 Sea Scallop Resource Survey Report**
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5.3 3. Cruise Results

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Chapter 6

Media

[Insert photo galleries, video clips, outreach materials, and press release links here]

Part II

Overview & Species

Chapter 7

Survey Overview

HabCam Survey Crew on Deck

7.1 Summary of Purpose

The Atlantic sea scallop is one of the most valuable fisheries in the United States (U.S.). The National Marine Fisheries Service (NMFS) is required to assess the status of sea scallop populations and set catch limits for sea scallops in Atlantic U.S. waters, typically implementing annual or biennial specifications through the New England Fishery Management Council (NEFMC). Monitoring the population status of sea scallops is a top priority for the Population & Ecosystems Monitoring & Analysis Division (PEMAD) at the Northeast Fisheries Science Center (NEFSC).

The HABitat CAMera system (HABCAM) is a non-invasive, underwater imaging vehicle towed from the stern of research vessels approximately 2 meters above the seafloor, capturing high-resolution, overlapping stereo images. The camera array is additionally equipped with a compass, CTD, sonar and sensors that survey the seafloor, which can be used for estimates of species abundance, habitat characterization, and biodiversity—giving a holistic view of benthic taxa. This vehicle has been used extensively for surveying the sea scallop resource on Georges Bank and the Mid-Atlantic Bight, as well as collecting data on finfish and habitat.

High-resolution stereo imagery capturing sea scallops on the seafloor.

Chapter 8

Why We Survey

Atlantic sea scallops are one of the largest commercial fisheries in the United States. In 2023, the value of the fishery was $359.5 million.

The Magnuson–Stevens Act fosters the long-term biological and economic sustainability of marine fisheries, including the scallop fishery. Under this act, we help: * Prevent overfishing * Rebuild overfished stocks * Increase long-term economic and social benefits derived from fishing * Ensure a safe and sustainable supply of seafood * Protect habitat that fishery species need to spawn, breed, feed, and grow to maturity

Fisheries-independent data from the HabCam Survey is used in the sea scallop stock assessment to infer how the stock size and composition has changed over time. Since sea scallops live on the sea floor, the HabCam's towed camera system is an ideal method for understanding the abundance and distribution of this species.

The survey began as a collaboration between the Northeast Fisheries Science Center and the Woods Hole Oceanographic Institution in 2004 through a Cooperative Institute of the North Atlantic Region agreement. The survey was transitioned to the Northeast Fisheries Science Center in 2012 as part of the Ecosystems Surveys enterprise. The survey typically runs between April and June sampling the Mid-Atlantic Bight and Georges Bank over two legs, each spanning 9 to 10 days.

For many years, the HabCam Survey and the Scallop Dredge Survey were integrated over three survey legs aboard the R/V *Hugh Sharp* homeported in Lewes, Delaware. With the transition of the Scallop Dredge Survey to an industry vessel in 2023, the HabCam Survey is now completed on the *NOAA Ship Henry B. Bigelow*.

The HabCam vehicle prepared on deck before deployment.

The primary equipment of the HabCam Survey is the HabCam vehicle. The vehicle houses a suite of sensors and electronics mounted on a frame towed behind the ship via a fiber optic cable. HabCam sensors include stereo cameras and strobes for seabed imaging, and sensors for measuring: * Conductivity * Temperature * Pressure * Oxygen concentration * Turbidity * Chlorophyll * Fluorescent dissolved organic matter (fDOM)

HabCam also uses a side-scan sonar to visualize the seabed and a forward-looking sonar for object avoidance. We tow the vehicle between 5 to 7 knots and it flies approximately 2 meters above the seabed. A pilot manually pays cable in and out to ensure clear visibility of the seafloor and achieve the desired height above the seabed. Scientists annotate images and, using the estimated field of view, derive sea scallop standardized abundances and size compositions.

Chapter 9

Data Use

The data is used to develop indices of abundance and size composition of Atlantic sea scallop for the stock assessments. Additional fisheries and ecosystem data collected from the images is also processed for use in future stock and ecosystem assessment needs.

A number of other surveys collect data on scallop distribution, abundance, and size supported by the Center's Scallop Dredge survey and the Research Set Aside Program (including dredge, video drop camera, and an additional HabCam survey). The intent of these multiple surveys is to increase the accuracy and precision of the sea scallop assessment through thorough fisheries-independent surveys.

9.1 Access Our Data & Scientific Documents

Our HabCam survey data are available by request. To request data, please contact **Cam Fairclough** (Cameron.Fairclough@noaa.gov) or **Nicholas Calabrese** (Nicholas.Calabrese@noaa.gov).

- Benthic habitat influences sea scallop distributions and swimming behavior based on underwater imagery and machine learning
- Combining imperfect automated annotations of underwater images with human annotations to obtain precise and unbiased population estimates
- Modeling the spatiotemporal effects of ocean acidification and warming on Atlantic sea scallop growth to guide adaptive fisheries management

Chapter 10

The Atlantic Sea Scallop

Atlantic Sea Scallop (Placopecten magellanicus)

10.1 Overview & Population Status

The primary Atlantic sea scallop fishery operates along the Atlantic coast from the Mid-Atlantic to the US/Canada border using predominantly paired or single scallop dredges and trawl gear[cite: 9]. Most vessels land scallops as shucked meats (the adductor muscle), though some land whole scallops[cite: 9]. Implementing regulations are found at **50 CFR part 648 subpart D**[cite: 9].

According to the 2020 stock assessment, Atlantic sea scallop is **not overfished** and **not subject to overfishing**[cite: 9].

10.2 Appearance & Biology

- **Appearance:** Scallops are bivalves held together by an adductor muscle[cite: 9]. Shells are saucer-shaped with smooth, scalloped edges[cite: 9]. The upper shell is reddish-pink/brown and the lower shell is white or cream (5–10% are albinos)[cite: 9].
- **Lifespan:** Can live up to 20 years, growing rapidly during their first few years up to ~6 inches[cite: 9].
- **Reproduction:** Reproduce by age 2, producing hundreds of millions of eggs per year during late summer/early fall or spring spawning[cite: 9]. Larvae remain in the water column for 4–6 weeks before settling[cite: 9].
- **Locomotion:** Propel themselves through the water by snapping top and bottom shells open and shut to escape predators like sea stars[cite: 9].

Large Atlantic sea scallop specimen.

10.3 Habitat, Range & Management

Atlantic sea scallops are found in the Northwest Atlantic Ocean from Newfoundland to Cape Hatteras, NC[cite: 9]. Adult scallops live in dense "beds" on sandy or gravelly ocean floors at depths of 100 to 300 feet on Georges Bank and the Mid-Atlantic[cite: 9].

Map of US and Canadian scallop resource aggregations and commercial fishing activity.

NOAA Fisheries and the New England Fishery Management Council manage the fishery under the **Atlantic Sea Scallop Fishery Management Plan** via Total Allowable Catch (TAC), crew limits, rotational closures, Vessel Monitoring Systems (VMS), and Individual Fishing Quotas (IFQs)[cite: 9].

10.4 Harvest & Gear Impacts

In 2023, commercial landings totaled **27.4 million pounds** of meats valued at **$360 million**[cite: 9]. Fishermen primarily use turtle deflector-style dredges equipped with 4-inch rings to increase efficiency and allow undersized scallops to escape[cite: 9].

Harvested sea scallops being sorted on deck.

10.5 Scientific Classification

Taxonomy Level	Classification
Kingdom	Animalia
Phylum	Mollusca
Class	Bivalvia
Order	Pectinida
Family	Pectinidae
Genus	*Placopecten*
Species	*magellanicus*

Anatomy of the Atlantic Sea Scallop.

Part III

Hardware & Schematics

Chapter 11

Component Reference Table

Attachment	Label	Description
Frame	A	Durable, protective and stable platform designed to carry imaging and sampling equipment close to the seafloor.
Tail	B	Supports the rudder, the elevator (vehicle lift) and the side scan sonar.
Blue View	C	An advanced multibeam imaging or scanning sonar system. It functions like an underwater acoustic camera, using high-frequency sound waves rather than light to generate real-time video-like imagery and 3D point clouds in zero-visibility conditions.

Attachment	Label	Description
CTD	D	It is the primary instrument package used to measure the essential physical properties of seawater such as Conductivity, Temperature, and depth.
EcoTriplet	E	A multi-parameter optical sensor used to measure light scattering and fluorescence in natural waters.
Main Electronics Bottle	F	A heavy-duty, waterproof cylinder that protects an underwater vehicle's delicate computers, navigation systems, and sensors from the crushing pressure of the ocean.
Expansion Bottle	G	A reservoir that maintains the internal pressure of a fluid-filled system slightly above the surrounding sea pressure. It allows internal oils to expand and contract as temperatures change, while preventing seawater from leaking into sensitive electronics or motors.

Attachment	Label	Description
SideScan Sonar	H	An acoustic imaging system mounted on an underwater vehicle. It emits fan-shaped sound waves to the left and right, mapping the seafloor to create highly detailed, photo-like 2D images of the seabed, shipwrecks, and debris.
Cable Tubes	I	Hold the cables going to the sensors, reducing cable drag/chaffing.
Cameras	J	Downward facing stereo cameras which capture images of the seafloor, synchronized with the strobes.
4 - Strobes	K	High-intensity, flashing lights used to aid in vehicle recovery, mark worksites, and provide synchronized illumination for high-resolution imaging.
Altimeter	L	An acoustic instrument that measures the exact vertical distance between the vehicle and the seabed. It sends down sound waves and times how long the pulse takes to bounce back from the ocean floor.

Attachment	Label	Description
Attitude Sensor	M	Measures its 3D orientation in the water, specifically tracking its pitch (nose up or down), roll (side-to-side tilt), and yaw/heading (compass direction).
Pinger	N	A device that emits repeatable acoustic sound pulses. They are used for tracking a vehicle's position, guiding it to specific subsea targets, or enabling emergency recovery if the vehicle is lost.
Doors	O	To keep the vehicle straight while towing, two doors are attached to HABCAM (act as a rudder).
Lead Weights	P	They allow the craft to achieve neutral buoyancy, fine-tune its center of gravity, and offset buoyant forces.
Weak Link	Q	An engineered mechanical failure point. It is designed to break intentionally when subjected to a predetermined threshold of force (e.g., getting snagged on rocks or debris) to prevent catastrophic damage to the rest of the vehicle or its tether.

Chapter 12

Frame Schematics & Physical Envelope

12.0.1 Structural Core & Design Principles

- **Primary Frame Architecture:** Built from 304 Stainless Steel welded tubular/pipe roll-cage construction for high tensile strength, impact toughness, and chassis rigidity.
- **Corrosion Mitigation:** Exposed stainless steel structures utilize sacrificial zinc/aluminum anodes and marine-grade anti-corrosion coatings to combat pitting and galvanic action.
- **Fasteners & Hardware:** Passivated 316/304 Stainless Steel hardware with anti-seize provisions prevents thread galling during deck maintenance.
- **Hydrodynamic Tow Envelope:** Engineered to minimize roll and drag at 5–7 knot tow speeds approximately 2 meters above the seabed.

12.0.2 Physical Specifications

PARAMETER	SPECIFICATION	NOTES/DETAILS
Frame Material	304 Stainless Steel	Welded tubular roll-cage assembly
Dry Weight (In Air)	3200 lbs	Fully loaded with primary payload
Wet Weight (In Water)	3200 lbs	Positive/negative buoyancy metrics

PARAMETER	SPECIFICATION	NOTES/DETAILS
Overall Dimensions (Without Tailside)	108.0in. x 62.0in. x 41.0in.	Max clearance envelope
Overall Dimensions (With Tailside)	155.52in. x 62.0in. x	
Max Depth Rating	110 to 113 meters (360 to 370 feet)	Frame & housing operational limit
Operational Tow speed	5 - 7 knots	Optimal hydrodynamic stability

12.1 HabCam Frame Schematics

12.1.1 System Overview

This slide contains the structural schematics and technical drawings for the frame of the Habitat Mapping Camera System (HabCam). The HabCam is an advanced, continuously towed underwater imaging vehicle utilized for high-resolution benthic habitat and marine population surveys. The frame serves as the foundational chassis, providing structural integrity and spatial orientation for a multi-million-dollar suite of oceanographic sensors.

12.1.2 Core Design

Because the HabCam operates just meters above the seafloor in challenging marine environments, its frame is engineered to meet strict operational tolerances. The core design prioritizes three major factors: * **Hydrodynamic Stability:** The geometry of the frame is designed to minimize drag and roll, ensuring a smooth, predictable flight path while being towed via the ship's winch. * **Durability and Impact Resistance:** Constructed from marine grade, corrosion resistant alloys, the frame features a robust roll-cage architecture. This protects sensitive internal housings from catastrophic damage in the event of an accidental seafloor strike or boulder impact. * **Modularity and Accessibility:** The frame must allow for rapid on deck maintenance. The open architecture design ensures that technical teams (such as the HabCam Lead) can easily access fiber optic terminations, adjust wiring if necessary, and calibrate sensors between deployments.

12.1.3 Payload Integration

These schematics detail the mounting points, brackets, and load-bearing structures required to securely integrate the vehicle's primary payload. When re-

viewing the drawings, note the specific integration zones for: * Primary and secondary high-resolution camera housings. * Illumination strobes and associated power manifolds. * Forward-looking obstacle avoidance sonar and altimeters. * The primary tow-point assembly and fiber optic telemetry housing.

12.1.4 Materials

To maintain structural integrity during tow operations and cushion mechanical impact, the HabCam chassis and mounting components use the following specifications: * **Primary Structural Frame:** 304 Stainless Steel tubular/pipe construction, selected for its high tensile strength, impact toughness, and weldability in structural roll-cage configurations. * **Corrosion Mitigation:** Exposed 304 stainless steel components are equipped with sacrificial zinc/aluminum anodes or marine-grade anti-corrosion coatings to protect against pitting and galvanic action in prolonged saltwater immersion. * **Fasteners and Brackets:** Passivated 316 / 304 Stainless Steel hardware with anti-seize provisions to prevent galling during on-deck servicing, or storage. * **Pressure Housings:** Hard-anodized aluminum or specialized pressure-rated housings for sensitive cameras, strobes, and oceanographic sensors.

12.1.5 Frame Body Schematics

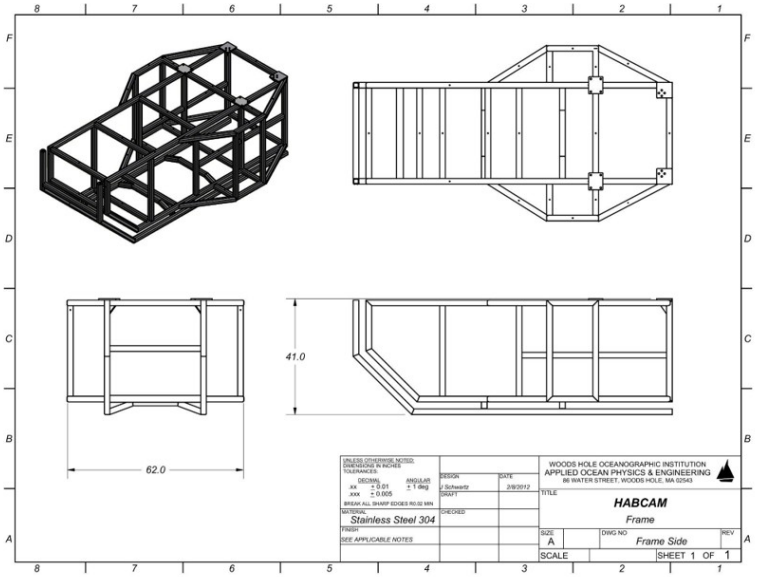


Figure 12.1: Frame Body 1

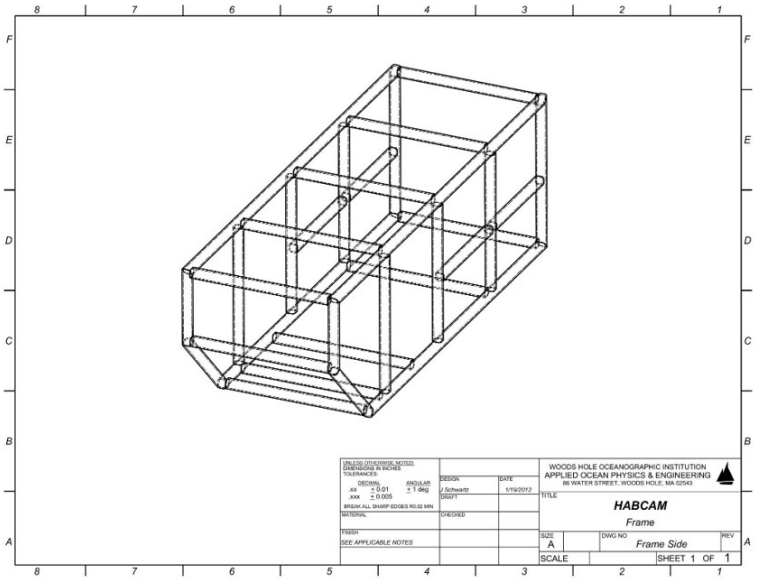


Figure 12.2: Frame Body 2

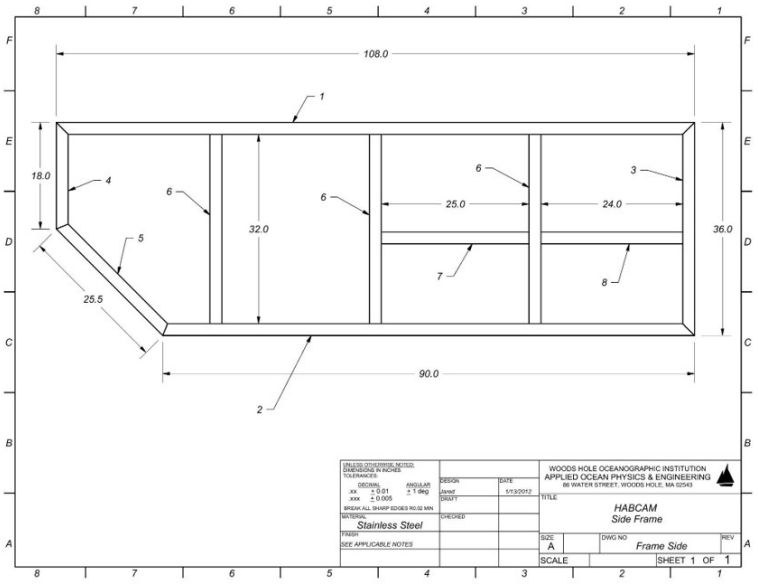


Figure 12.3: Frame Body 3

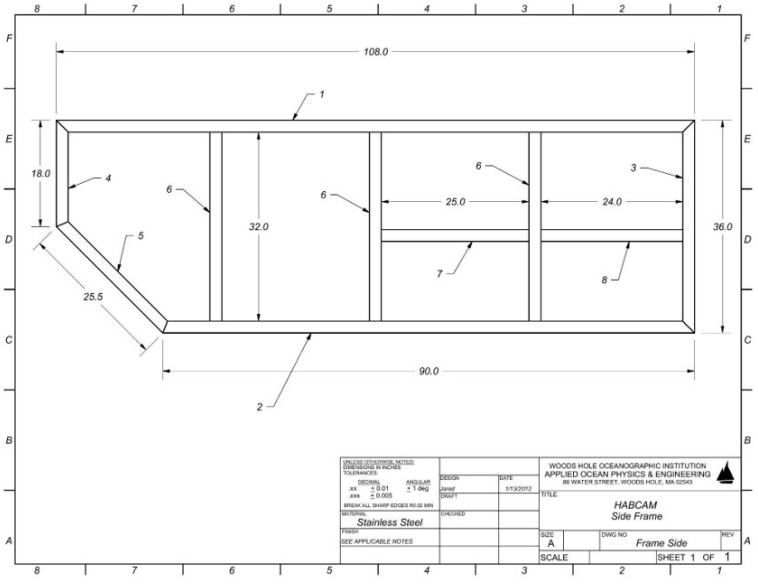


Figure 12.4: Frame Body 4

12.1.6 Tail Schematics

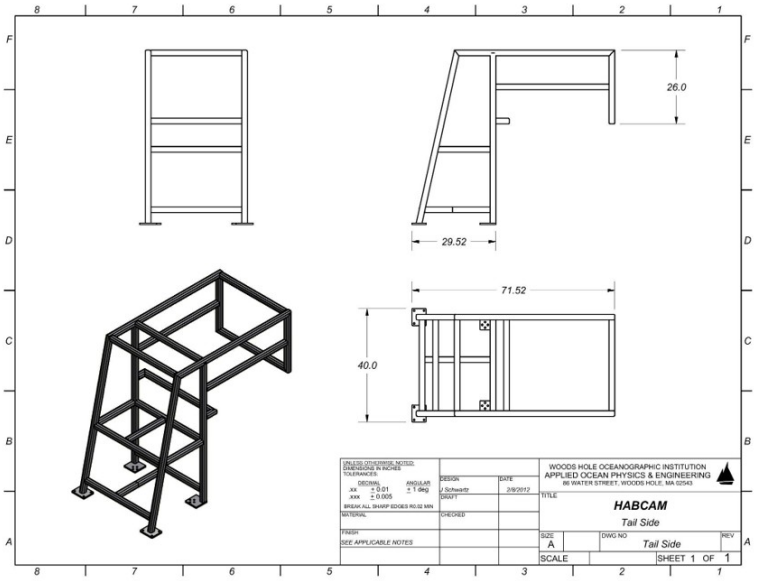


Figure 12.5: Tail 1

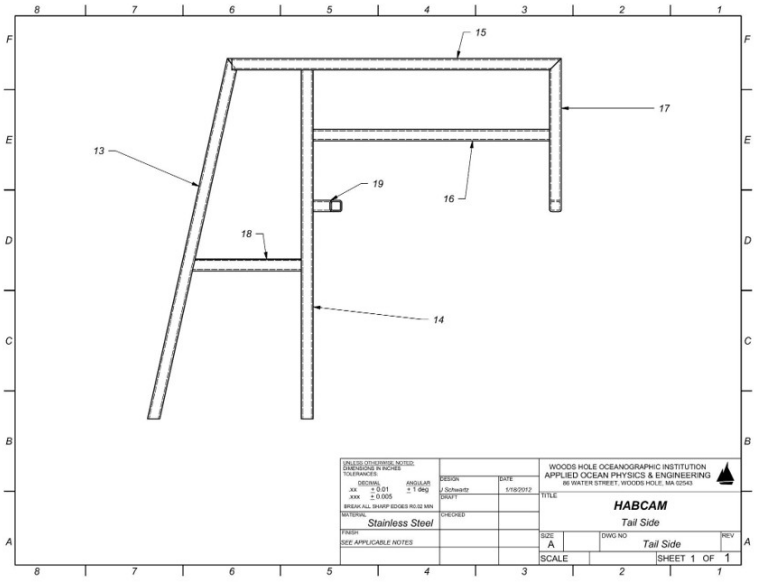


Figure 12.6: Tail 2

Chapter 13

Cable Harness Specifications

13.1 System Overview

This slide set contains the electrical schematics, pinout diagrams, cable harness routing plans, and interconnect specifications for the Habitat Mapping Camera System (HabCam).

As a continuously towed underwater imaging vehicle operating near the seabed, the HabCam relies on oceanographic sensors, imaging systems, and acoustic tools. The cable harnesses described herein provide the link between the centralized vehicle telemetry housing, power distribution manifolds, and individual payload sensors.

13.2 Cable Harness Design

All HabCam cable assemblies are engineered to withstand prolonged immersion in seawater under hydrostatic pressure, high tension, and continuous vibration. Design criteria include:

- **Wet-Pluggable Connectors:** Utilize marine-grade wet-mate circular connectors (e.g., SubConn, Impulse, Glenair) rated for deep-water hydrostatic pressures.
- **Mechanical Protection:** Cables feature heavy-duty polyurethane or neoprene outer jackets with embedded water-blocking compounds and Kevlar strain-relief strength members.

- **Noise Mitigation:** High-speed data pairs and sensitive optical feedback lines utilize individual foil wrapping, tinned-copper braided shielding, and isolated analog/digital ground returns to eliminate EMI and crosstalk from power lines.
- **Color-Coded Conductor Wiring:** Standardized color coding (following marine instrumentation standards) is maintained across all sub-assemblies to facilitate rapid deck repairs.

13.3 Maintenance

To ensure system reliability during sea trials and survey operations, technicians must adhere to the following servicing guidelines:

Connector Care & Lubrication: - Inspect rubber faces and O-rings for grit, salt crystals, or hair prior to deployment. - Apply a light coating of Dow Corning 111 silicone grease to neoprene connector faces. - Never use petroleum-based lubricants, as they degrade neoprene seals.

Bend Radius & Mechanical Anchoring: - Maintain a minimum static bend radius of 6x the cable diameter, and a dynamic bend radius of 10x the cable diameter. - Secure all cables along the 304 Stainless Steel frame using rubber-lined P-clamps and heavy-duty UV-resistant cable ties.

Dummy Plug Usage: - Always cap un-mated bulkhead connectors with protective dummy plugs immediately upon disconnect on deck to protect pin contacts from salt spray.

13.4 Cable Harness Master

Schematic Reference	Connector Model (Vehicle → Device Side)	Pin Count	Primary Protocol / Signal	Operating Voltage
HabCamV4 - Chanos Cable	13-Pin Circular → 13-Pin Circular	13 → 13	Biogeochemical Data + Dual Power Rails	+12 VDC

Schematic Reference	Connector Model (Vehicle → Device Side)	Pin Count	Primary Protocol / Signal	Operating Voltage
HabCamV4 Wiring - Camera Power Ethernet Cable	13-Pin Circular → 13-Pin Rectangular	13 → 13	Ethernet Data + Camera Power + Strobe Sync	System Power / Strobe Voltage
Cable Wiring - A-Sphere	SubConn MCIL4F → SubConn MCIL6F	4 → 6	RS232 Serial Data (Moxa Interface) + Power	+24 VDC
Microstrain 3DM-GX3-25 AHRS	SubConn MCIL6F → SubConn MCIL4F	6 → 4	RS232 Serial Data (Moxa Interface) + Power	+12 VDC
HabCamV4 - BlueView Cable	SubConn DIL13M → Teledyne Impulse Titan	13 → 10	Sonar TX/RX Differential + Power	12-48 VDC
Cable Wiring - Benthos PSA916 Sonar Altimeter	SubConn MCIL6F → Impulse RMG-6FS	6 → 6	RS-232 Serial Data (Moxa Interface) + Power	+12 VDC (+6–24V Range)
Cable Wiring - HabCam V4 Eco-Triplet	SubConn MCIL6F → SubConn MCIL6F	6 → 6	RS232 Serial Data (Moxa Interface) + Power	+12 VDC

13.5 Individual Cable Specs

13.5.1 Chanos Cable

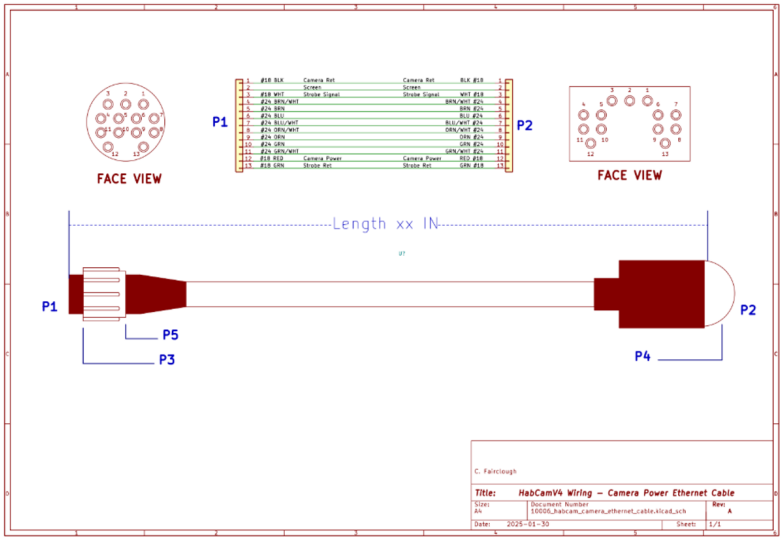


Figure 13.1: Chanos Cable Schematic

Schematic Reference: HabCamV4 - Chanos Cable
Primary Function: Connects the Chanos in-situ biogeochemical analyzer to the HabCamV4 system for ocean chemistry data collection and sensor power.

Key Specifications: - **Connector Configuration:** 13-pin circular connector on the vehicle hub end (P1) to a 13-pin circular connector on the Chanos sensor end (P2). - **Power Conductors:** Dual +12 VDC power rails using 18 AWG conductors (#18 WHT and #18 RED) paired with dual 18 AWG ground returns (#18 BLK and #18 GRN). - **Data & Signal Lines:** 24 AWG color-coded twisted pairs for differential communications and control lines. - **Cable Shielding:** Pin 2 on both ends is dedicated to the cable screen/shield grounding.

Pin Allocation Callouts: - **Power Supply (+12V DC):** P1 Pins 3 & 12 (#18 WHT & #18 RED) → P2 Pins 3 & 12 - **Power Ground (GND):** P1 Pins 1 & 13 (#18 BLK & #18 GRN) → P2 Pins 1 & 13 - **Screen / Cable Shield:** P1 Pin 2 → P2 Pin 2 - **Signal Lines (24 AWG):** - P1 Pin 4 (#24 BRN) → P2 Pin 4 - P1 Pin 5 (#24 BRN/WHT) → P2 Pin 5 - P1 Pin 6 (#24 BLU) → P2 Pin 7 - P1 Pin 7 (#24 BLU/WHT) → P2 Pin 6 - P1 Pin 8 (#24

ORN) → P2 Pin 8 - P1 Pin 9 (#24 ORN/WHT) → P2 Pin 9 - P1 Pin 10 (#24 GRN) → P2 Pin 10 - P1 Pin 11 (#24 GRN/WHT) → P2 Pin 11

13.5.2 Camera Power Ethernet Cable

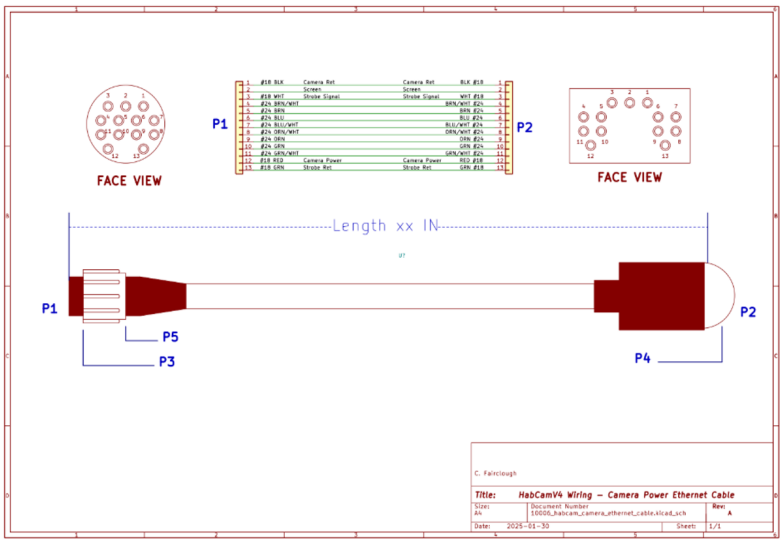


Figure 13.2: Camera Power Ethernet Cable Schematic

Schematic Reference: HabCamV4 Wiring - Camera Power Ethernet Cable **Primary Function:** Delivers primary system power, strobe synchronization signals, and high-speed Ethernet data between the central telemetry hub and the main camera system.

Key Specifications: - **Connector Configuration:** 13-pin circular connector on the hub end (P1) to a 13-pin rectangular connector on the camera housing end (P2). - **Power & Strobe Conductors:** Heavy-gauge 18 AWG conductors dedicated to camera power delivery and strobe pulse timing: - Camera Power: #18 RED (Camera Power) paired with #18 BLK (Camera Ret) - Strobe Control: #18 WHT (Strobe Signal) paired with #18 GRN (Strobe Ret) - **Ethernet Data Lines:** 8 color-coded 24 AWG conductors forming four twisted pairs for high-speed Ethernet telemetry. - **Cable Shielding:** Pin 2 on both connectors connects directly to the cable screen/shield for noise suppression.

Pin Allocation Callouts: - **Camera Return (GND):** P1 Pin 1 (#18 BLK) → P2 Pin 1 - **Screen / Cable Shield:** P1 Pin 2 → P2 Pin 2 - **Strobe Signal:**

P1 Pin 3 (#18 WHT) → P2 Pin 3 - **Ethernet Data Lines (24 AWG):** - P1 Pin 4 (#24 BRN/WHT) → P2 Pin 4 - P1 Pin 5 (#24 BRN) → P2 Pin 5 - P1 Pin 6 (#24 BLU) → P2 Pin 6 - P1 Pin 7 (#24 BLU/WHT) → P2 Pin 7 - P1 Pin 8 (#24 ORN/WHT) → P2 Pin 8 - P1 Pin 9 (#24 ORN) → P2 Pin 9 - P1 Pin 10 (#24 GRN) → P2 Pin 10 - P1 Pin 11 (#24 GRN/WHT) → P2 Pin 11 - **Camera Power:** P1 Pin 12 (#18 RED) → P2 Pin 12 - **Strobe Return:** P1 Pin 13 (#18 GRN) → P2 Pin 13

13.5.3 A-Sphere Cable Wiring

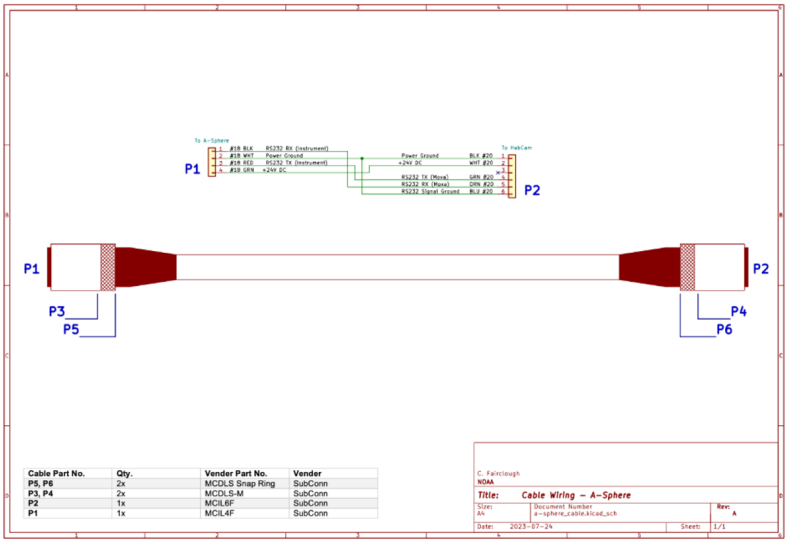


Figure 13.3: A-Sphere Cable Schematic

Schematic Reference: Cable Wiring - A-Sphere
Primary Function: Connects the A-Sphere instrument to the HabCam system to supply system power (+24V DC) and handle RS232 serial data communication.

Key Specifications: - **Connector Configuration:** SubConn 4-pin inline female connector (MCIL4F) with lock collar and snap ring (MCDLS-M / MCDLS Snap Ring) on the HabCam side (P1) to a SubConn 6-pin inline female connector (MCIL6F) on the A-Sphere instrument side (P2). - **Power Lines (P1 - HabCam Side):** Dual 18 AWG conductors for +24V DC power (#18 RED & #18 GRN) and dual 18 AWG conductors for Power Ground (#18 BLK &

#18 WHT). - **Serial Communication (P2 - Instrument Side):** 20 AWG color-coded conductors providing RS232 transmit, receive, and signal ground connections between the instrument and the internal Moxa serial server.

Pin Allocation Callouts: - **P1 (HabCam Side - Power):** - Pins 1 & 2 (#18 BLK & #18 WHT): Power Ground - Pins 3 & 4 (#18 RED & #18 GRN): +24V DC - **P2 (A-Sphere Side - RS232 Serial Data):** - Pin 1 (BLK #20): RS232 RX (Instrument) - Pin 2 (WHT #20): RS232 TX (Instrument) - Pin 3 (GRN #20): RS232 TX (Moxa) - Pin 4 (ORN #20): RS232 RX (Moxa) - Pin 5 (BLU #20): RS232 Signal Ground

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13.5.4 Microstrain Cable

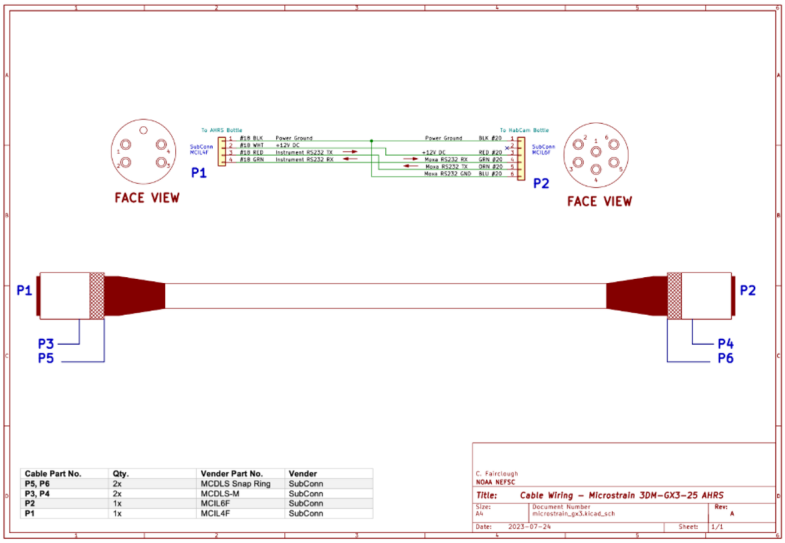


Figure 13.4: Microstrain Cable Schematic

Schematic Reference: Cable Wiring - Microstrain 3DM-GX3-25 AHRS
Primary Function: Connects the Microstrain 3DM-GX3-25 AHRS (Attitude and Heading Reference System) bottle to the HabCam telemetry bottle for real-time roll, pitch, and orientation tracking via an internal Moxa serial server.

Key Specifications: - **Connector Configuration:** SubConn 4-pin inline female connector (MCIL4F) on the AHRS Bottle side (P1) to a SubConn 6-pin inline female connector (MCIL6F) on the HabCam Bottle side (P2). Both connectors utilize locking collars and snap rings (MCDLS-M / MCDLS Snap

Ring). - **Power Supply:** +12 VDC power rail and dedicated power ground bridging the two bottles. - **Serial Communication:** RS232 serial telemetry links the AHRS instrument transmit/receive lines directly to the Moxa serial port interface.

Pin Allocation Callouts: - **P1 (AHRS Bottle Side):** - Pin 1 (#18 BLK): Power Ground - Pin 2 (#18 WHT): +12V DC - Pin 3 (#18 RED): Instrument RS232 TX (signals to Moxa RX) - Pin 4 (#18 GRN): Instrument RS232 RX (signals from Moxa TX) - **P2 (Electronics Bottle Side):** - Pin 1 (BLK #20): Power Ground - Pin 2: Unused / No Connection (NC) - Pin 3 (RED #20): +12V DC - Pin 4 (GRN #20): Moxa RS232 RX - Pin 5 (ORN #20): Moxa RS232 TX - Pin 6 (BLU #20): Moxa RS232 GND

13.5.5 BlueView Cable

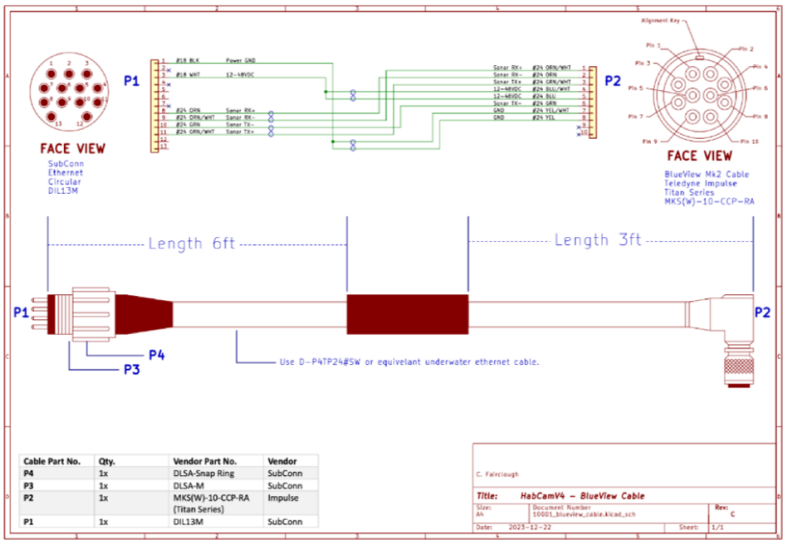


Figure 13.5: BlueView Cable Schematic

Schematic Reference: HabCamV4 - BlueView Cable
Primary Function: Connects the BlueView Mk2 multibeam acoustic imaging sonar to the HabCam central telemetry hub for real-time forward obstacle avoidance and seabed monitoring.

Key Specifications: - **Connector Configuration:** SubConn 13-pin circular male connector (DIL13M) with a locking sleeve collar (DLSA-M / DLSA-Snap

Ring) on the vehicle hub end (P1) to a Teledyne Impulse Titan Series 10-pin right-angle connector (MKS(W)-10-CCP-RA) on the sonar end (P2). - **Cable Stock:** Underwater ethernet-rated cable (D-P4TP24#SW or equivalent). - **Signal Protocol:** Differential Sonar TX/RX data communications using 24 AWG color-coded conductors. - **Operating Voltage:** Wide-range input (12–48 VDC) utilizing 18 AWG primary power conductors feeding split 24 AWG pin terminations at the sonar connector.

Pin Allocation Callouts: - **Power Ground:** P1 Pin 1 (#18 BLK) → P2 Pins 7 & 8 (#24 YEL/WHT & YEL) - **Power Supply (12–48 VDC):** P1 Pins 2 & 3 (#18 WHT) → P2 Pins 4 & 5 (#24 BLU/WHT & BLU) - **Sonar RX+ / RX-:** P1 Pin 8 (#24 ORN) & Pin 9 (#24 ORN/WHT) → P2 Pin 1 & Pin 2 - **Sonar TX+ / TX-:** P1 Pin 11 (#24 GRN/WHT) & Pin 10 (#24 GRN) → P2 Pin 3 & Pin 6 - **Unused Terminations:** P1 Pins 4, 5, 6, 7, 12, 13; P2 Pins 9, 10

13.5.6 Sonar Altimeter Cable

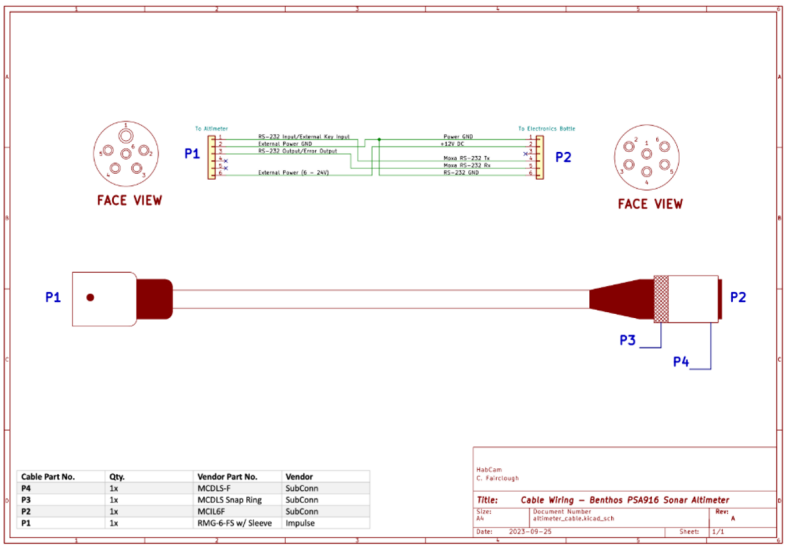


Figure 13.6: Sonar Altimeter Cable Schematic

Schematic Reference: Cable Wiring - Benthos PSA916 Sonar Altimeter
Primary Function: Connects the Teledyne Benthos PSA-916 Sonar Altimeter

directly to the electronics bottle to deliver height-above-seabed measurements to an internal Moxa serial interface.

Key Specifications: - **Connector Configuration:** Impulse 6-pin female connector with locking sleeve (RMG-6FS -w/ Sleeve) on the altimeter end (P1) to a SubConn 6-pin inline female connector (MCIL6F) with snap ring and locking collar (MCDLS Snap Ring / MCDLS-F) on the electronics bottle end (P2). - **Power Supply:** Accepts wide input range (+6 to 24V) at the altimeter, powered via a +12V DC rail and external power ground line from the electronics bottle. - **Serial Communication:** RS-232 input/output lines route directly to the Moxa serial port interface (Rx, Tx, and RS-232 GND).

Pin Allocation Callouts: - **P1 (Altimeter Side - Impulse RMG-6FS):** - Pin 1: RS-232 Input / External Key Input - Pin 2: External Power GND - Pin 4: RS-232 Output / Error Output - Pin 5: External Power (+6–24V) - Pins 3 & 6: Unused / No Connection (NC) - **P2 (Electronics Bottle Side - SubConn MCIL6F):** - Pin 1: Power GND - Pin 2: Unused / No Connection (NC) - Pin 3: +12V DC - Pin 4: Moxa RS-232 Tx - Pin 5: Moxa RS-232 Rx - Pin 6: RS-232 GND

13.5.7 Eco-Triplet Cable

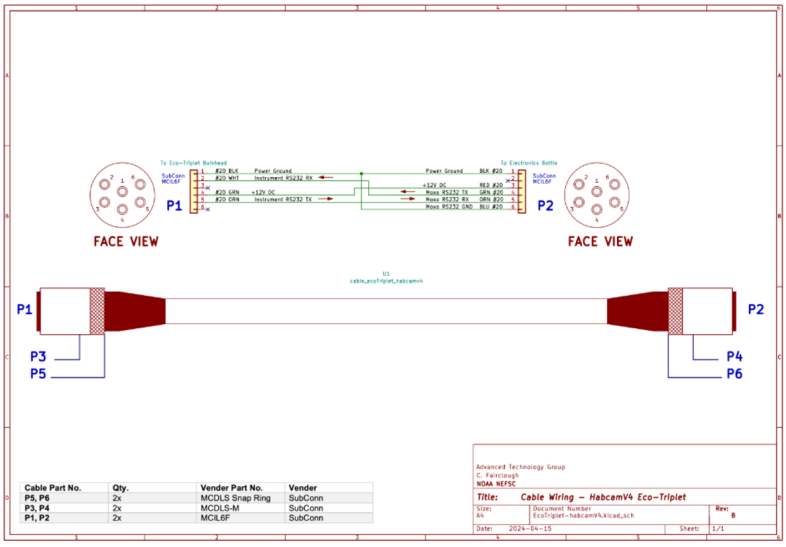


Figure 13.7: Eco-Triplet Cable Schematic

Schematic Reference: Cable Wiring - HabCamV4 Eco-Triplet
Primary Function: Connects the Sea-Bird / WET Labs ECO Triplet fluorometer bulkhead directly to the HabCam electronics bottle for environmental optical sampling and data transmission.

Key Specifications: - **Connector Configuration:** SubConn 6-pin inline female connector (MCIL6F) on the ECO Triplet bulkhead side (P1) to a matching SubConn 6-pin inline female connector (MCIL6F) on the electronics bottle side (P2). Both ends are secured with locking collars and snap rings (MCDLS-M / MCDLS Snap Ring). - **Power Supply:** Delivered over a +12 VDC power line with a shared power ground return using 20 AWG conductors. - **Serial Communication:** RS232 serial interface routing instrument transmit and receive lines to the internal Moxa serial server.

Pin Allocation Callouts: - **P1 (ECO Triplet Bulkhead Side):** - Pin 1 (#20 BLK): Power Ground - Pin 2 (#20 WHT): Instrument RS232 RX (connects to Moxa TX) - Pin 3: Unused / No Connection (NC) - Pin 4 (#20 GRN): +12V DC - Pin 5 (#20 ORN): Instrument RS232 TX (connects to Moxa RX) - Pin 6: Unused / No Connection (NC) - **P2 (Electronics Bottle Side):** - Pin 1 (BLK #20): Power Ground - Pin 2: Unused / No Connection (NC) - Pin 3 (RED #20): +12V DC - Pin 4 (GRN #20): Moxa RS232 TX - Pin 5 (ORN #20): Moxa RS232 RX - Pin 6 (BLU #20): Moxa RS232 GND

Chapter 14

HabCam Electrical Schematics & Network Architecture

14.1 Master Table

Section	Description	Primary Components / Schematics
Network Connectivity & CWDM	Copper-to-fiber multiplexing and subsea telemetry pipeline.	Mako Cameras, BlueView Sonar, SFP Media Converters, OptiLink
Ethernet Electrical Routing	Dual switch architecture inside the Electronics Bottle.	Ubiquiti Switches, SubConn DBH13M Data Breakouts, Test Ports
Bulkhead Specifications	Pinouts, O-rings, and thread specs for Electronics Bottle bulkheads.	SubConn MCBH3M, MCBH4M, MCBH6M, DBH13M, OptiLink A-04-BCR
Internal Fiber Cabling	Media conversion, SFP transceivers, and CWDM channels.	IMC IE-ModeConverter, AFL CWDM Mux, L-Com ST Barrels

Section	Description	Primary Components / Schematics
Bottle Endcap Layouts	External port assignments for Electronics & Expansion bottles.	Front/Back Endcaps (J1–J22, J50–J52), Expansion Bottle (J1–J11)
Serial Port Arrays (INST 1–12)	Moxa N-Port servers and regulated DC power buses.	Moxa 5650-8-DT, Terminal Blocks T5–T16, SubConn MCBH6M
AC Power & Strobe Control	AC distribution, DC rail rectifiers, and strobe timing bus.	Astrodyne PS1–PS4, Sealevel CC320, MVS 5000 Strobes
Sensor Bottles & Housings	Wiring schematics for individual pressure housings.	Strobe Bottle, Attitude Bottle, Prosilica & Mako Camera Housings
CTD & Environmental Cables	Dedicated cable assemblies for oceanographic sampling.	Sea-Bird SBE 37/49 CTD, ECO-Triplet Fluorometer Cable

14.2 HabCam Network

The HabCam network architecture details the end-to-end data conversion and multiplexing pipeline. High-resolution subsea imaging, acoustic sonar, and serial instrument telemetry convert from copper electrical signals into Coarse Wavelength Division Multiplexing (CWDM) optical signals for long-distance transmission across the primary tow cable umbilical.

Figure 1.1: HabCam Subsea Network Connectivity & CWDM Conversion Schematic.

14.2.1 CWDM Wavelength Allocation & Subsystem Routing

Subsystem / Device	Data Output Type	Wavelength Allocation (Tx / Rx)	Intermediate Conversion	Primary Optical Link
Allied Vision Mako G-234C (Cam 1)	Copper Gigabit Ethernet	Tx 1470 nm / Rx 1490 nm	SFP Media Converter	CWDM Multiplexer → OptiLink J2
Allied Vision Mako G-234C (Cam 2)	Copper Gigabit Ethernet	Tx 1510 nm / Rx 1530 nm	SFP Media Converter	CWDM Multiplexer → OptiLink J2
Sonar / Data Switch Bus	RS-232 / Copper Ethernet	Tx 1590 nm / Rx 1610 nm	Moxa Server → SFP Converter	CWDM Multiplexer → OptiLink J2

14.3 Ethernet

Routing across dual Ethernet switches inside the Electronics Bottle handles internal data switching, diagnostic test ports, and subsea 13-pin connector breakouts (Data1Net and Data2Net).

Figure 2.1: Dual Switch Ethernet Architecture & Subsea Pinout Breakout.

14.3.1 Switch Port Assignment Matrix

Switch Unit	Port	Target Network Line	Destination / Assigned Function
Ethernet Switch #1 / (Ubiquiti 60-105-ProSafe)	Eth1 / Eth2 / Eth3 / Eth4 / Eth5	Moxa1 Net / Moxa2 Net / Timing Net / Unused / Inter Net	Moxa Serial Server Interface #1 / Moxa Serial Server Interface #2 / Precision Timing / Sync Network / Terminated / Spare / Inter-switch trunk line to Switch #2 (Eth1)
Ethernet Switch #2 / (Ubiquiti 60-105-ProSafe)	Eth1 / Eth2 / Eth3 / Eth4 / Eth5	Inter Net / Data1 Net / Data2 Net / Test Net / Data Net	Trunk link from Switch #1 (Eth5) / SubConn 13-pin Connector (Data1Net) / SubConn 13-pin Connector (Data2Net) / Internal L-com RJ45 Test Jack (ECF504-8SA) / Central Data Bus

14.4 Bulkhead Connectors

Master technical specifications for all SubConn bulkhead connectors installed on the Electronics Bottle endcaps.

Figure 3.1: SubConn Bulkhead Face Views and Hardware Specifications.

14.4.1 Bulkhead Hardware & Pinout Specifications

- **3-Pin AC Input Bulkhead (J1):**
  - **Hardware Spec:** SubConn MCBH3M-SS | **Thread:** 7/16-20 UNF-2A | **O-Ring:** 2-014

  - **Pinout:** Pin 1: Ground Return | Pin 2: AC (Hot) | Pin 3: AC (Neutral)

- **4-Pin Strobe Bulkheads (J5–J8):**
  - **Hardware Spec:** SubConn MCBH4M-SS | **Thread:** 7/16-20 UNF-2A | **O-Ring:** 2-014
  - **Pinout:** Pin 1: Strobe Signal Return | Pin 2: Strobe Signal | Pin 3: AC (Hot) | Pin 4: AC (Neutral)
  - **Power Rating:** MVS-5002 Strobe (12V @ 4.5A) & fan (12V @ 0.5A) via Astrodyne LPR75-12 (75W total).

- **6-Pin Serial Bulkheads (J11–J22):**
  - **Hardware Spec:** SubConn MCBH6M-SS | **Thread:** 7/16-20 UNF-2A | **O-Ring:** 2-014
  - **Pinout:** Pin 1: Power Return | Pin 2: Power Output #1 | Pin 3: Power Output #2 | Pin 4: Instrument Rx (Tx by Moxa) | Pin 5: Instrument Tx (Rx by Moxa) | Pin 6: Comms Return
  - **Power Limits (Per Port):** 100W @ 48V (2A), 75W @ 24V (3A), 35W @ 12V (3A), 15W @ 5V (3A).
  - **Total Serial Bus Power Available:** 100W (48V), 90W (24V), 100W (12V), 70W (5V).

- **13-Pin Ethernet / Camera Bulkheads (J3, J4, J9, J10):**
  - **Hardware Spec:** SubConn DBH13M-SS | **Thread:** 1/2-20 UNF-2A | **O-Ring:** 2-015
  - **Camera Pinout (J3, J4):** Pin 1: Camera Return | Pin 3: Strobe Signal | Pins 4–11: Ethernet Color Pairs | Pin 12: Camera Power | Pin 13: Strobe Return
  - **Ethernet Pinout (J9, J10):** Pins 1–3: Power Conductors (#18 BLK, #18 ORN, #18 WHT) | Pins 4–11: Cat6 Ethernet Pairs | Pins 12–13: Main Power Rails (#18 RED, #18 GRN)

- **SubConn OptiLink Fiber Bulkhead (J2):**
  - **Hardware Spec:** SubConn OptiLink A-04-BCR | **Thread:** 7/8-14 | **O-Rings:** 2-019 & 2-022
  - **Capacity:** 4 Single-Mode (SM) Optical Fiber channels.

14.5 Internal Fiber Cabling

Details internal optical media conversion, CWDM wavelength multiplexing, and fiber pass-through routing inside the Electronics Bottle to bridge copper Ethernet devices to the SubConn OptiLink subsea bulkhead.

Figure 4.1: Internal Optical Media Conversion & CWDM Routing.

14.5.1 Fiber Channel Routing Map

Channel Source	Input Protocol	Optical Transceiver	CWDM LC Port Wave-lengths	Barrel Coupling	SubConn OptiLink Pin
Camera 1 Ethernet	RJ45 Copper	Aaxeon SFP-C 1490	Tx 1470 nm / Rx 1490 nm	ST-to-ST Barrel 1 (FOA-011A)	Fiber 1
Camera 2 Ethernet	RJ45 Copper	Aaxeon SFP-C 1490	Tx 1510 nm / Rx 1530 nm	ST-to-ST Barrel 2 (FOA-011A)	Fiber 2
Sonar Ethernet	RJ45 Copper	Aaxeon SFP-C 1490	Tx 1570 nm / Rx 1550 nm	ST-to-ST Barrel 3 (FOA-011A)	Fiber 3
Data Net Ethernet	RJ45 Copper	Aaxeon SFP-C 1490	Tx 1610 nm / Rx 1590 nm	ST-to-ST Barrel 4 (FOA-011A)	Fiber 4

14.6 Bottle Endcap

External bulkhead connector placements, designations, part numbers, and gender specifications for both the Electronics Bottle and Expansion Bottle.

Figure 5.1: Electronics Bottle Front and Back Endcap Bulkhead Layouts.

14.6.1 Electronics Bottle Bulkhead Assignments

Port Designation	Assigned Function	Endcap Location	SubConn Bulkhead Part	Mating Cable Plug	Pin / Fiber Count
J1	AC Power Input	Front Endcap	MCBH3M-SS	MCIL3F	3 Pins
J2	Main Fiber Input	Front Endcap	OptiLink A-04-BCR	OptiLink Plug	4 SM Fibers

Port Designation	Assigned Function	Endcap Location	SubConn Bulkhead Part	Mating Cable Plug	Pin / Fiber Count
J3, J4	Camera #1, Camera #2	Front Endcap	DBH13M-SS	DIL13F	13 Pins
J5–J8	Strobe #1 through #4	Front Endcap	MCBH4M-SS	MCIL4F	4 Pins
J9, J10	Ethernet #1, Ethernet #2	Back Endcap	DBH13M-SS	DIL13F	13 Pins
J11–J22	Serial Ports #1 through #12	Back Endcap	MCBH6M-SS	MCIL6F	6 Pins
J50–J52	Direct Optical ST Ports	Front Endcap	Lancer 500-BST-S1-B02-000	ST-Dry Penetrator	Single ST Fiber

Figure 5.2: Expansion Bottle Endcap Bulkhead Layout.

14.6.2 Expansion Bottle Bulkhead Assignments

Port Designation	Assigned Subsystem	SubConn Bulkhead Part	Mating Cable Plug	Pin Count & Gender
J1	AC Power In	MCBH3F-SS	MCIL3M	3 Pins (Female Bulkhead)
J2	Data Net In	DBBH13F-SS	DIL13M	13 Pins (Female Bulkhead)
J3	Ethernet #2	DBBH13M-SS	DIL13F	13 Pins (Male Bulkhead)

Port Designation	Assigned Subsystem	SubConn Bulkhead Part	Mating Cable Plug	Pin Count & Gender
J4	BlueView Sonar	DBBH13M-SS	DIL13F	13 Pins (Male Bulkhead)
J5	Ethernet #3	DBBH13M-SS	DIL13F	13 Pins (Male Bulkhead)
J6	Unused / Spare Port	DBBH13M-SS	DIL13F	13 Pins (Male Bulkhead)
J7–J11	Serial #1 through #5	MCBH8F-SS	MCIL8M	8 Pins (Female Bulkhead)

14.7 Serial Port Arrays

Internal power distribution buses, Moxa N-Port serial server interfaces, and pin routings for auxiliary instrumentation ports INST 1 through INST 12 (J11–J22) inside the Electronics Bottle.

Figure 6.1: Electrical Schematic for Serial Ports 1 through 6 (J11–J16).

Figure 6.2: Electrical Schematic for Serial Ports 7 through 12 (J17–J22).

14.7.1 Serial Port Wiring & Terminal Block Matrix

Instrument Port	Back Endcap Port	Moxa Server & DB9 Connector	Power Bus Terminal Blocks	SubConn MCBH6M Pin Allocation
INST 1	J11	Moxa Server #1 (Port 1 / J38)	Terminal Block T5	Pin 1: Power Return (BLK #22) / Pin 2: Power Output #1 (WHT #22) / Pin 3: Power Output #2 (RED #22) / Pin 4: Instrument Rx / Moxa Tx (GRN #22) / Pin 5: Instrument Tx / Moxa Rx (ORG #22) / Pin 6: Comms Return (BLU #22)
INST 2–6	J12–J16	Moxa Server #1 (Ports 2–6 / J39–J43)	Terminal Blocks T6–T10	Identical 6-pin wiring structure to INST 1
INST 7	J17	Moxa Server #2 (Port 1 / J44)	Terminal Block T11	Identical 6-pin wiring structure to INST 1
INST 8–12	J18–J22	Moxa Server #2 (Ports 2–6 / J45–J49)	Terminal Blocks T12–T16	Identical 6-pin wiring structure to INST 1

14.8 AC Power & Strobe

Governs high-voltage AC input distribution, internal DC power generation (5V, 12V, 24V, 48V), system cooling fans, camera power control, and flash pulse triggers for four MVS 5000 strobe units.

Figure 7.1: AC Power Distribution, Rectifiers, and Strobe Control Schematic.

14.8.1 Internal Rectifiers & Interface Allocation

- **AC Main Input (J1):** High-voltage AC enters via SubConn MCBH3M-SS with a 4A slow-blow fuse (F1: Littelfuse 03540821ZXBL) guarding the main AC rails.
- **Internal DC Regulators:**
  - **PS1 (Astrodyne RS150-48):** Generates +48V DC (Fused via F2 @ 1A Fast) → Terminal Strip T1.
  - **PS2 (Astrodyne RS150-24):** Generates +24V DC (Fused via F3 @ 1A Fast) → Terminal Strip T2.
  - **PS3 (Astrodyne RS150-12):** Generates +12V DC (Fused via F2-3 @ 8A Fast) → Terminal Strip T3.
  - **PS4 (Astrodyne RS100-5):** Generates +5V DC (Fused via F4-5 @ 8A Fast) → Terminal Strip T4.
- **System Cooling:** Dual Mechatronics GDA6025-12BB 12V cooling fans equipped with tachometer feedback lines.
- **Timing Unit (U1):** Sealevel CC320 timing interface on the Timing Net with optical trigger outputs (Out1–Out8) pulled up via 1.2kΩ resistors.

14.8.2 Strobe & Camera Bulkhead Connections

Port Name	Endcap Port	SubConn Model	Assigned Function	Pin Allocation Callouts
STR-1	J5	MCBH4M-SS	Strobe Unit #1 Power & Trigger	Pin 1: Strobe Return (#20 BLK) / Pin 2: Strobe Signal (#20 WHT) / Pin 3: AC Hot (#20 RED) / Pin 4: AC Neutral (#20 GRN)
STR-2, 3, 4	J6, J7, J8	MCBH4M-SS	Strobe Units #2, #3, #4 Power & Trigger	Identical 4-pin mapping to STR-1
CAM-1	J3	DBH13M-SS	Main Camera #1 Interface	Pin 1: Camera Return (#18 BLK) / Pin 3: Strobe Signal (#18 WHT) / Pins 5–11: Cam1 Ethernet (#24 Pairs) / Pin 12: Camera Power (#18 RED) / Pin 13: Strobe Return (#18 GRN)
CAM-2	J4	DBH13M-SS	Main Camera #2 Interface	Identical 13-pin mapping to CAM-1

14.9 Sensor Bottles

14.9.1 Strobe Bottle Internal Schematic

WHOI Strobe Bottle Internal Electrical Schematic.

Details AC power conversion, thermal regulation, opto-isolated trigger pulse amplification, and driving circuit connections for the Perkin-Elmer MVS-5002 strobe unit:

- **Internal Power Converter (PS1):** Astrodyne LPR75-12 converting high-voltage AC into regulated +12V DC.
- **Trigger Conditioning:** Dual NPN transistors (Q1, Q2 2N3904) driving a Perkin-Elmer MVS-5002 optocoupler stage with a 150Ω current-limiting resistor. Nominal trigger: 5V @ 20mA (4.5V @ 18mA minimum).
- **Internal Cooling:** Sunon KDE1206PHV2 12V fan drawing air out through housing side vents.

14.9.2 Attitude Bottle Internal Schematic

Figure WHOI Attitude Bottle (Microstrain 3DM-GX3-25 IMU) Schematic.

Bulkhead Pin (SubConn MCBH4M-SS J1)	Assigned Signal	Wire Specification	Sensor Connector (P2) & Function (U1)
Pin 1	Power Return	20 AWG Violet	Pin 9 / Pin 6 (Power Ground)
Pin 2	+12V Power Input	20 AWG Green	Pin 8 (+5–16V Power Input)
Pin 3	RS232 Out From Housing	20 AWG Yellow	Pin 4 (RS232 Transmit - Tx)
Pin 4	RS232 In To Housing	20 AWG Orange	Pin 5 (RS232 Receive - Rx)

14.9.3 Camera Synchronization Bus Circuit Analysis

Optically Isolated Shared Camera / Strobe Synchronization Bus.

- **Bus Bias:** Single 2050Ω (0.5W) pull-up resistor to +24V DC in the Electronics Bottle.

- **Active Low State (Trigger Pulled Low):** Line near 0V; 11.7mA through pull-up resistor; 0mA through camera LEDs; 0.3mA through each strobe; total collector sinking current = 12.9mA.
- **Inactive High State (Floating High):** Line at 3.4V; 10.1mA through pull-up resistor; 5.3mA through each camera input LED (exceeds drive threshold).
- **Worst-Case Single Camera Load:** Single-camera / 4-strobe line rises to 5.0V, delivering 9.5mA to the camera LED.

14.9.4 Allied Vision Mako G-234C Camera Housing Wiring

Allied Vision Mako G-234C Camera Housing Electrical Wiring Diagram.

Vehicle Bulkhead Port (SubConn DBH13MSS J1)	Wire Color / Spec	Internal Target / Power PCB (1A1)	Camera Receptacle (Hirose HR25-7TR-8PA)
Pin 1	Black (#18)	Power PCB 1A1P1 Pin 1 (GND)	Hirose Pin 4 (SyncIn 1 Gnd) & Pin 5 (Sync Gnd)
Pin 12	Red (#18)	Power PCB 1A1P1 Pin 2 (+12V DC)	Hirose Pin 7 (+12V Power)
Pin 3	White (#18)	Power PCB 1A1P1 Pin 4 (SYNC IN)	Optical Trigger Input Line
Pin 13	Green (#18)	Power PCB 1A1P1 Pin 3 (SYNC GND)	Optical Trigger Return Line
Pins 4–11	Cat6 Ethernet Pairs	Aaxeon FCU3002A Media Converter	Gigabit Ethernet Port (PoE Capable)
Optical Output (J2)	ST-Dry Penetrator	Lancer 500-BST-S1-B02	Aaxeon SFP Module (Tx 1550 nm / Rx 1310 nm)

14.10 Environmental Sensors

14.10.1 Sea-Bird SBE 37 / SBE 49 CTD Cable Assembly

Sea-Bird SBE 37/49 CTD Sensor Cable Schematic.

SubConn MCIL4F Pin (SBE CTD Side)	Wire Color	Signal Function	SubConn MCIL6F Pin (HabCam Side)
Pin 1	Black	Power Ground	Pin 1 & Pin 6 (Ground)
Pin 2	White	RS-232 Rx (Connects to HabCam Tx)	Pin 5 (Orange Wire - RS-232 Tx)
Pin 3	Red	RS-232 Tx (Connects to HabCam Rx)	Pin 4 (Green Wire - RS-232 Rx)
Pin 4	Green	9–24V DC Operating Power	Pin 2 (White Wire - 9–24V DC)

14.10.2 ECO-Triplet Fluorometer Cable Assembly

Figure 9.2: Sea-Bird / WET Labs ECO-Triplet Cable Schematic.

SubConn MCIL6F Pin (ECO-Triplet Side)	Wire Color	Signal Function	SubConn MCIL6F Pin (Bottle / INST 4 Side)
Pin 1	Black	Power Ground	Pin 1 (Black Wire - Ground)
Pin 2	White	RS-232 Rx	Pin 4 (Green Wire - RS-232 XMT)
Pin 3	Red	Not Used (N/U)	Pin 2 (White Wire - Open / N/U)
Pin 4	Green	+12V DC Operating Power	Pin 3 (Red Wire - +12V DC)

SubConn MCIL6F Pin (ECO-Triplet Side)	Wire Color	Signal Function	SubConn MCIL6F Pin (Bottle / INST 4 Side)
Pin 5	Orange	RS-232 Tx	Pin 5 (Orange Wire - RS-232 Rx)
Pin 6	Blue	Analog #1 (N/U)	Pin 6 (Blue Wire - RS-232 Rx)

Chapter 15

Vehicle Sensor Preparation & Assembly Protocols

HabCam Vehicle Submerged in Sundance Tank for Camera Calibration

15.1 Mounting Summary

Subsystem	Mounting Location	Required Hardware & Fasteners	Required Tools
Electronics Bottle *(1x)*	Aft Vehicle Frame	• (4x) Black Aluminum Brackets • (4x) Barrier Rubber Pieces (Pliobond 35 glued) • (8x) 9/16"-12 (3 ½") Partial Thread Bolts • (8x) 9/16"-12 (2 ¼") Partial Thread Bolts • (4x) 32" 11mm Fully Threaded Rods • (6x) Plastic Washers, (10x) Steel Washers • (8x) Steel Lock Nuts, (8x) Hex Nuts	• 9/16" Socket & Wrench • 7/16" Wrench • 14mm Wrench
BlueView Sonar *(1x)*	Forward Vehicle Frame	• (4x) Steel Brackets • (4x) Barrier Rubber Pieces • (6x) 9/16"-12 (3 ½") Steel Partial Thread Bolts • (4x) 9/16"-12 (2 ¼") Steel Partial Thread Bolts • (20x) 3/16" Steel Washers • (10x) Steel Lock Nuts, (10x) Hex Nuts	• 9/16" Socket & Wrench • 14mm Wrench • Aqua Shield Grease

Subsystem	Mounting Location	Required Hardware & Fasteners	Required Tools
CTD 39 Oxygen *(1x)*	Starboard Forward Rail	• (2x) UHMW Brackets, (2x) Plastic Brackets • (4x) Barrier Rubber Pieces • (4x) 7/16”-14 (4 ¾”) Steel Partial Thread Bolts • (2x) 7/16”-14 (1”) Steel Full Thread Bolts • (3x) 3/16” (1”) Socket Head Screws • (10x) Steel Washers, (6x) Lock Nuts, (2x) Hex Nuts	• 7/16” Socket & Wrench • 11mm Wrench • Hex Key Set (1/4”–1/16”)
CTD 49 Sound Speed *(1x)*	Port Forward Rail	• (6x) Plastic Brackets (Sensor, Center, Frame) • (8x) Barrier Rubber Pieces • (8x) 9/16”-12 (2 ¾”) Steel Partial Thread Bolts • (16x) Steel Washers • (8x) Steel Lock Nuts, (8x) Hex Nuts	• 9/16” Socket & Wrench • 14mm Wrench

Subsystem	Mounting Location	Required Hardware & Fasteners	Required Tools
Eco-Triplet Puck *(1x)*	Center Forward Expansion Rail	• (2x) Steel Brackets (Top & Bottom) • (2x) Barrier Rubber Pieces • (2x) 9/16”-12 (2 ½”) Steel Partial Thread Bolts • (2x) 9/16”-12 (2”) Steel Partial Thread Bolts • (6x) Steel Washers, (4x) Lock Nuts, (2x) Hex Nuts	• 9/16” Socket & Wrench • 14mm Wrench
Habitat Cameras *(2x)*	Center Frame Plate	• (1x) Steel Mounting Plate (16 in x 16 in) • (6x) Metal Brackets • (4x) M7-1.0 x 45mm Threaded Bolts • (4x) 1/4”-20 (2”) Socket Head Screws • (8x) 1/4”-20 (1”) Socket Head Screws • (6x) Steel Washers, (6x) Steel Lock Nuts	• Hex Key Set (1/4”–1/16”) • M7 / 3/16” Allen Keys

Subsystem	Mounting Location	Required Hardware & Fasteners	Required Tools
Strobe Lights *(4x)*	Forward, Aft, Port, Starboard	• (8x) Steel Brackets • (8x) Barrier Rubber Pieces • (8x) 9/16”-12 (3 ½”) Steel Partial Thread Bolts • (8x) 7/16”-14 (1 ¾”) Steel Partial Thread Bolts • (32x) Plastic Washers, (32x) Steel Washers • (16x) Steel Lock Nuts, (16x) Hex Nuts	• 9/16” Socket & Wrench • 7/16” Socket & Wrench • 14mm / 11mm Wrenches
Attitude Sensor *(1x)*	Internal Housing Chassis	• Housing Mounting Brackets & Isolation Collars • Stainless Steel Attachment Fasteners	• Standard Wrench & Socket Set
Sonar Altimeter *(1x)*	Center Camera Plate	• (2x) Plastic Brackets • (4x) Barrier Rubber Pieces • (4x) 9/16”-12 (6 ½”) Steel Partial Thread Bolts • (8x) Steel Washers • (4x) Steel Lock Nuts, (4x) Hex Nuts	• 9/16” Socket & Wrench • 14mm Wrench

15.2 Preparation & Assembly Instructions

15.2.1 BlueView Forward Looking Sonar (1x)

Hardware Inventory:

- (4x) Steel Brackets
- (4x) Barrier Rubber Pieces
- (6x) 9/16"-12 (3 ½") Steel Partial Thread Hex Bolts
- (4x) 9/16"-12 (2 ¼") Steel Partial Thread Hex Bolts
- (20x) 3/16" Steel Washers
- (10x) Steel Lock Nuts
- (10x) 9/16"-12 Steel Hex Nuts

Tools Needed:

- 9/16" Socket Wrench
- 9/16" Combination Wrench
- 14mm Combination Wrench (Optional)
- Aqua Shield Grease

System Overview: A forward-facing sonar (forward-looking sonar or forward imaging sonar) is designed to provide high-resolution imaging of the underwater environment in front of a vessel or vehicle. Unlike traditional sonar systems that measure depth directly below or capture data in a single direction, forward-facing sonar scans the area ahead to detect obstacles, submerged structures, and features during poor visibility, night operations, or turbid water surveys. Widely used in navigation, underwater inspections, search-and-rescue, and operations in complex marine environments.

Sonar Preparation & Mounting Steps:

1. Prior to attaching BlueView to the front of the vehicle, ensure the bracket has rubber gaskets glued to them (brackets should already be attached to the HabCam vehicle).
2. Attach BlueView to the bracket, with the Impulse connector facing towards the back of the vehicle.
3. Thread four stainless steel screws through the bracket.
4. Prior to adding the stainless steel washer, lock washer, and nut, apply Aqua Shield over all of the threads (there should be 4 screws).
5. Add the stainless steel washer, lock washer, and nut to all four screws (washer on inside, lock washer in middle, nut on outside).
6. Use 9/16" ratcheting wrenches to tighten the nuts.

15.2.1.1 Teledyne Marine BlueView M450 Mk2 System Specifications

Parameter	Specification	Parameter	Specification
Operating Frequency	450 kHz	Supply Voltage	12–28 VDC
Field of View	130°	Max Power Consumption	24 W
Max Range	300 m (984 ft)	Connectivity	Ethernet
Optimum Range	2–150 m (6.5–197 ft)	Connector	Impulse MKS(W) Splash-mate (standard); Impulse MKS, Burton (optional)
Beam Width (Horizontal / Vertical)	1° / 10°	Range Resolution	50.8 mm (2.0 in)
Number of Beams (Maximum)	768 (0.18° beam spacing)	Update Rate	Up to 20 Hz

15.2.1.2 BlueView Mechanical Options

Version	Depth Rating	Dimensions (L × W)	Can Diameter	Weight (Air / Water)
BlueView M450 S Mk2	1,000 m	203.4 mm (8.0 in) × 195.6 mm (7.7 in)	Ø101.6 mm (4.0 in)	3.5 kg / 1.6 kg
BlueView M450 D6 Mk2	6,000 m	253.7 mm (10.0 in) × 195.6 mm (7.7 in)	Ø127.0 mm (5.0 in)	8.5 kg / 5.5 kg

15.2.2 CTD-39 Dissolved Oxygen Sensor (1x)

Hardware Inventory:

- (2x) UHMW Brackets
- (2x) Plastic Brackets
- (4x) Barrier Rubber Pieces
- (4x) 7/16"-14 (4 ¾") Steel Partial Thread Hex Bolts
- (2x) 7/16"-14 (1") Steel Full Thread Hex Bolts
- (3x) 3/16" (1") Steel Socket Head Screws Full Thread
- (10x) Steel Washers
- (6x) Steel Lock Nuts
- (2x) 7/16"-14 Steel Hex Nuts

Tools Needed:

- 7/16" Socket Wrench
- 7/16" Combination Wrench
- 11mm Combination Wrench (Optional)
- Hex Key Set (1/4"–1/16")

System Overview: The SBE 39plus is a high-accuracy, fast-sampling temperature (pressure optional) recorder with USB interface, internal batteries, and onboard memory. Designed for moorings or long-duration fixed-site deployments, as well as nets, towed vehicles, or ROVs. Memory capacity exceeds 9.5 million samples excluding pressure data, or 5.5 million samples including pressure. Sampling every 0.5 sec yields approximately 55 days of data without pressure or 32 days with pressure (battery endurance exceeds memory capacity).

CTD Preparation Steps:

1. Prior to attaching Seabird 39 CTD to the Starboard forward vertical frame rail, ensure bracket has rubber gaskets glued to them (brackets should already be attached to HabCam vehicle).
2. Attach Seabird 39 CTD to the bracket, with the bulkhead connector facing down by threading screw through bracket.
3. Prior to adding stainless steel washer, lock washer, and nut, apply Aqua Shield over all of the threads (there should be 4 screws).
4. Add the stainless steel washer, lock washer, and nut to all four screws (washer on inside, lock washer in middle, nut on outside).
5. Use 7/16" ratcheting wrenches to tighten the nuts.

15.2.2.1 Sea-Bird Scientific SBE 39plus System Specifications

Parameter	Specification	Parameter	Specification
Communication Options	RS-232	Sampling Rate	Up to 2 Hz (2 samples/sec)
Depth Rating	600 m	Memory Capacity	9.5 million samples T; 5.5 million samples TD
Power Source	Internal, External	Power Consumption	Lithium battery pack (4 AA Saft LS 14500): > 11,000,000 samples TD
Housing Material Options	Plastic, Titanium	Accuracy	Temperature: ±0.002 °C (-5 to +35 °C); ±0.01 °C (+35 to +45 °C) / Optional Pressure: ±0.1% of full scale range
Onboard Memory	Yes	Resolution	Temperature: 0.0001 °C / Optional Pressure: 0.002% of full scale range
Connector	MCBH	Software / Biofouling	Seasoft V2 | Biofouling Protection: No

Documentation Links:

- SBE 39plus DataSheet Zip
- SBE 39plus Manual (Rev B)
- SBE 39 Methods and Procedures Zip
- SBE 39plus Configurations Overview

15.2.3 CTD-49 Sound Speed Sensor (1x)

Hardware Inventory:

- (6x) Plastic Brackets
- (8x) Barrier Rubber Pieces

- (8x) 9/16”-12 (2 ¾”) Steel Partial Thread Hex Bolts
- (16x) Steel Washers
- (8x) Steel Lock Nuts
- (8x) 9/16”-12 Steel Hex Nuts

Tools Needed:

- 9/16” Socket Wrench
- 9/16” Combination Wrench
- 14mm Combination Wrench (Optional)

System Overview: The SBE 49 FastCAT is an integrated CTD sensor intended for use as a modular component in towed vehicles, ROVs, AUVs, or other autonomous platforms that supply DC power and acquire serial data. It is an easy-to-use, light, and compact instrument suited to small vehicle payloads. FastCAT must be externally powered, and its RS-232C data saved or telemetered by the vehicle. It does not support auxiliary sensors directly. FastCAT’s pump-controlled, TC-ducted flow feature minimizes salinity spiking, and its 16 Hz sampling provides very high spatial resolution of oceanographic structures and gradients. Measured data and derived variables of salinity and sound velocity are output in real-time in engineering units or raw HEX.

CTD Preparation Steps:

1. Prior to attaching Seabird 49 CTD to Port forward vertical frame rail, ensure bracket has rubber gaskets glued to them (brackets should already be attached to HabCam vehicle).
2. Attach Seabird 49 CTD to the bracket, with the pump facing down by threading screw through bracket.
3. Prior to adding stainless steel washer, lock washer, and nut, apply Aqua Shield over all of the threads (there should be 4 screws).
4. Add the stainless-steel washer, lock washer, and nut to all four screws (washer on inside, lock washer in middle, nut on outside).
5. Use 9/16” ratcheting wrenches to tighten the nuts.

15.2.3.1 Sea-Bird Scientific SBE 49 FastCAT CTD System Specifications

Parameter	Specification	Parameter	Specification
Communication Options	RS-232C	Sampling Rate	16 Hz (16 samples/sec)

Parameter	Specification	Parameter	Specification
Depth Rating	350 m	Power Consumption	Input: 0.75 A @ 9–24 VDC / Turn-on transient: 750 mA / Sampling/transmitting (with pump): 350 mA @ 9V, 285 mA @ 12V, 180 mA @ 19V
Power Source	External	Accuracy	Conductivity: ±0.0003 S/m / Temperature: ±0.002 °C / Pressure: ±0.1% of full scale range
Housing Material Options	Plastic, Titanium	Resolution	Conductivity: 0.00005 S/m (0.4 ppm salinity) / Temperature: 0.0001 °C / Pressure: 0.002% of full scale range
Connector	MCBH	Software / Biofouling	Seasoft V2 | Biofouling Protection: Yes

Documentation Links:

- SBE 49 DataSheet PDF
- SBE 49 Manuals Zip
- SBE 49 Methods and Procedures Zip

15.2.4 Eco-Triplet Puck - Fluorometer (1x)

Hardware Inventory:

- (2x) Steel Brackets
- (2x) Barrier Rubber Pieces
- (2x) 9/16”-12 (2 ½”) Steel Partial Thread Hex Bolts
- (2x) 9/16”-12 (2”) Steel Partial Thread Hex Bolts

- (6x) Steel Washers
- (4x) Steel Lock Nuts
- (2x) 9/16”-12 Steel Hex Nuts

Tools Needed:

- 9/16” Socket Wrench
- 9/16” Combination Wrench
- 14mm Combination Wrench (Optional)

System Overview: The next generation of optical monitoring with Sea-Bird Scientific’s cutting-edge ECO V2 Series. Offers dynamic range transitioning from deep blue ocean to coastal waters. Delivers 16-bit resolution and enhanced signal-to-noise ratios across up to four channels. Ideal for biological monitoring and dye trace studies. Robust potted optics block ensures long-term sensor stability. Optional anti-biofouling technology enables extended field measurements.

Fluorometer Preparation Steps:

1. Prior to attaching Eco-Triplet Puck to Center Forward Horizontal expansion bottle rails, ensure the bracket has rubber gaskets glued to them (brackets should already be attached to HabCam vehicle).
2. Attach Eco-Triplet to the bracket, with the bulkhead connector facing towards the center of HabCam and resting above the crossbar, by threading screw through bracket.
3. Prior to adding the stainless-steel washer, lock washer and nut, apply Aqua Shield over all of the threads (there should be 2 screws).
4. Add the stainless-steel washer, lock washer, and nut to both screws (washer on inside, lock washer in middle, nut on outside).
5. Use 9/16” ratcheting wrenches to tighten the nuts.

15.2.4.1 HabCam Sensor Configuration (Model BBFL2, Serial Number: 627)

Configured via joint agreement between Population Dynamics Branch and Oceanography Branch.

Channel	Measurement Parameter	Wavelength	Sensitivity	Typical Range
Channel 1	Scattering (NTU)	650 nm	~0.003 m -1	0–5 m -1

Channel	Measurement Parameter	Wavelength	Sensitivity	Typical Range
Channel 2	Chlorophyll (EX/EM)	695 nm	0.025 g/l	0–50 g/l
Channel 3	fDOM	460 nm	0.28 ppb/count	0–375 ppb

15.2.4.2 Sea-Bird Scientific ECO V2 Optical Sensor System Specifications

Parameter	Specification
Communication Options	RS-232, Analog
Depth Rating	600 m, 2,000 m, 6,000 m
Power Source	Internal, External
Housing Material Options	Aluminum, Titanium, ABS, Acetal, Copolymer
Connector	MCBH-6M
Sampling Rate	User-selectable to 8 Hz
Memory Capacity	1 Ch: 1,048,000 samples | 2 Ch: 762,000 | 3 Ch: 599,000 | 4 Ch: 493,000
Software / Biofouling	UCI | Biofouling Protection: Yes

Documentation Links:

- ECO V2 Single Channel DataSheet Zip
- ECO V2 Manuals Zip

15.2.5 Main Electronics Bottle - WHOI Engineered (1x)

[DROP ELECTRONICS BOTTLE MOUNTED / EXTRA PHOTO HERE]

Hardware Inventory:

- (4x) Black Aluminum Brackets
- (4x) Barrier Rubber Pieces
- (8x) 9/16”-12 (3 ½”) Steel Partial Thread Hex Bolts
- (8x) 9/16”-12 (2 ¼”) Steel Partial Thread Hex Bolts
- (4x) 32” 11mm Fully Threaded Rods

- (6x) Plastic Washers
- (10x) 3/16" Steel Washers
- (8x) Steel Lock Nuts
- (8x) 9/16"-12 Steel Hex Nuts

Tools Needed:

- 9/16" Socket Wrench
- 9/16" Combination Wrench
- 7/16" Socket Wrench
- 14mm Combination Wrench (Optional)
- 3/16" Hex Wrench
- Small Phillips Screwdriver
- Rubber Mallet
- Multimeter
- Molykote 55 Grease & Dielectric Grease
- Pliobond 35 Rubber Adhesive
- Aqua Shield Grease
- Vacuum Pump Fixture

Main Electronics Bottle Removal, Disassembly, Maintenance & Reassembly Procedure:

1. Before removing the main electronics bottle from HabCam, ensure it is powered off.
2. Disconnect all cables from bulkhead connectors on both sides of the main electronics bottle.
3. Remove main bottle top clamps with 9/16" ratcheting wrench.
4. Place the main bottle cradle on the table.
5. Two people are needed to gently remove the main bottle from HabCam. One person climbs on top of the vehicle and gently slides the main bottle out of the back of the vehicle. The second person holds the weight in the back. Then both carry the bottle to the cradle.

6. Check the condition of top and bottom clamps on HabCam. If replacement is needed due to wear/damage, remove brackets from HabCam using 9/16" ratcheting wrenches.
7. If replacing the main electronics bottle clamp, ensure rubber is applied to the inside of the clamp before reattaching to HabCam. Rubber can be attached to clamps using Pliobond 35.

8. To open the main electronics bottle, remove threaded rods along the outside of the main bottle using 7/16" ratcheting wrenches (there should be 4 of them).

9. Use a 3/16" hex wrench to pop-out the vacuum purge plug on the back of the main bottle. You should hear air escape upon loosening if it has not been opened since coming back from the field.
10. Remove the end cap on the forward end cap (opposite side of the purge plug). There should be a label that says "open this end first". This end is opened first as it is NOT connected to the electronics chassis.

11. Use a rubber mallet to lightly bang on the back end of the cap's lip to loosen it.
12. To remove the forward end cap, unplug ethernet cables, two mini amp-style and the 3 large amp-style connectors. Now take the cap completely out; when setting it down avoid bending the bulkheads as much as feasibly possible.
13. Use a mallet on the opposite cap's rim to loosen it from housing. Once loose, set on a soft pad on the floor upright with that cap facing upward, remove it with the electronic housing, and place it on the cradle.

14. On the underside of the main bottle, there is a removable prototype board, attached with screws. Using a small Phillips screwdriver, remove the screws from the standoffs. Take the plate off and replace the desiccant packs with fresh ones (4 large packs).

15. Check the suite of fuses to ensure they are properly connected with a multimeter. Set multimeter to continuity mode; you should hear a beep if connected properly.
16. Turn on the vehicle by plugging in 3-pin power cable and fiber optic bulkhead (currently using Port 3). On the step-up transformer, turn on input, then output (in that order).
17. Check that you can connect to both Moxas, both Guardasofts, and whatever sensors are plugged into the bottle (i.e. attitude sensor).
18. If working, turn off main bottle by turning off output, then input on step-up transformer. Unplug 3-pin power cable and fiber optic bulkhead.
19. Add a cap to the fiberoptic bulkhead on the main bottle to protect it from scratches.
20. Check if desiccant needs replaced. Attach 4 packets to underside of prototype board using painter's tape. Recommended replacement is once a year.
21. Reattach prototype board to standoffs.
22. Prior to reattaching housing, slip off O-rings on back end cap, wipe off excess grease, apply isopropyl alcohol to paper towel and quickly wipe down O-rings. Do not over apply alcohol as it eats through rubber.
23. Gently check O-rings for nicks or cuts, and apply an even coat of Molykote 55.
24. Apply an even coat of Molykote 55 to inside edge of housing that meets back end cap O-rings when assembled.

25. To reassemble main electronics bottle, gently invert inside of main bottle and hover over housing, ensuring nothing is pinched as you close. Requires substantial pressure to snap closed.
26. On front end cap, remove O-rings, wipe off excess grease, apply isopropyl alcohol to paper towel and quickly wipe down O-rings.
27. Flip electronics bottle and apply even coat of Molykote 55 to inside edge of housing meeting front end cap O-rings.
28. Before attaching front end cap, plug in ethernet cables, two mini amp-style and 3 large amp-style into respective ports.
29. Hover end cap over housing, ensuring nothing is pinched as you close. Align handles and rod holes as close as possible; press firmly until snapped closed.
30. Turn on vehicle by plugging in 3-pin power cable and fiber optic bulkhead (Port 3). On step-up transformer, turn on input, then output. Listen for beep (if purge plug is out). Verify link light and attitude output on Engineering GUI.
31. If reporting well, power down by turning off output, then input on step-up transformer. Unplug 3-pin power cable and fiber optic bulkhead.
32. Before pulling a vacuum, ensure rod holes are aligned.
33. To pull a vacuum, attach vacuum plug to purge plug port. Attach hose to pump and turn pump on. Shut off valve at 16 inHg of mercury, then shut off pump.
34. Wait 24 hours, and confirm that pressure is still held at 16 inHg.
35. Check handle integrity, replace if necessary.
36. To secure main electronics bottle clamps, thread a 2.5" partially threaded screw through plastic insert. Apply Aqua Shield over all threads prior to adding stainless washer, lock washer, and nut.
37. Apply dielectric grease to female end of connector faces of camera cable and strobe. Listen for cracking sound as you align male and female connectors. Hand-tighten locking collars across all front/back connectors.
38. Ensure all desired sensors work by turning on vehicle (3-pin power and fiber optic bulkhead Port 3; step-up transformer input then output).

15.2.6 Allied Vision Mako G-234 - Cameras (2x)

/ Camera Housings & Frame Mount	/ Down-Looking Stereo Optical Ports

Hardware Inventory:

- (1x) Steel Mounting Plate (16 in × 16 in)
- (6x) Steel Brackets

- (4x) 1/4”-20 (2”) Steel Socket Head Screws Partial Thread
- (8x) 1/4”-20 (1”) Steel Socket Head Screws Full Thread
- (6x) Steel Washers
- (6x) Steel Lock Nuts

Tools Needed:

- Hex Key Set (1/4”–1/16”)

System Overview: Mako G-234 is a 2.35 megapixel GigE machine vision camera incorporating a high quality Type 1/1.2 (13.4 mm diagonal) Sony IMX249 CMOS sensor. At full resolution, this camera runs 41.5 frames per second (10-bit sensor readout). Higher frame rates are possible with a smaller region of interest (ROI).

15.2.6.1 Allied Vision Mako G-234 System Specifications

Parameter	Specification	Parameter	Specification
Resolution	1,936 × 1,216 (2.40 MP)	Sensor Size	11.34 mm × 7.13 mm (13.4 mm, Type 1/1.2)
Framerate	41 fps	Pixel Size	5.86 m × 5.86 m
Digital Interface	IEEE 802.3 1000BASE-T, IEEE 802.3af (PoE)	Exposure Time	16 s to 85 s
Sensor Type / Arch	Area Scan | CMOS	Image Buffer (RAM)	64 MByte
Sensor Model / Shutter	Sony IMX249 | Global Shutter (GS)	Power Supply	10.8 to 26.4 VDC AUX or 802.3at Type 1 PoE

Documentation Link: Allied Vision Mako G-234 DataSheet PDF

15.2.7 Arctic Rays (INKFISH) Dragonfish - Mini LED Strobe Lights (4x)

/ Rear Bracket Mount & Cable Harness	/ Front LED Array & Housing Assembly

Hardware Inventory:

- (8x) Steel Brackets
- (8x) Barrier Rubber Pieces
- (8x) 9/16”-12 (3 ½”) Steel Partial Thread Hex Bolts
- (8x) 7/16”-14 (1 ¾”) Steel Partial Thread Hex Bolts
- (32x) Plastic Washers
- (32x) Steel Washers
- (16x) Steel Lock Nuts
- (16x) 9/16”-12 Steel Hex Nuts

Tools Needed:

- 9/16” Socket Wrench
- 9/16” Combination Wrench
- 7/16” Socket Wrench
- 7/16” Combination Wrench
- 14mm Combination Wrench (Optional)
- 11mm Combination Wrench (Optional)

System Overview: High-output, 30,000-lumen compact LED strobe light rated down to 6,000 meters for HOV/ROV/AUV digital still photography or wherever high-output compact strobes are required.

Patented Driver Technology Features:

- Extremely compact package with integral driver electronics
- Wide-flood or focused-beam options
- Simple trigger-controlled pulse length
- Fast repetition rate
- Optional independent secondary quench line
- Customizable color temperature, pulse intensity, rep rate, and max pulse duration

15.2.7.1 Dragonfish Mini Technical Specifications

Category	Parameter	Specification
Optical	Operating Mode / Color Channels / Luminous Output / Color Temperature / CRI / Beam Angle (FWHM)	Pulsed Strobe / One / 30,000 Lm @ 5,700K / 2700K, 3000K, 3500K, 4000K, 5700K standard, 6500K / 70 min. Standard (optional 80 min, 90 min) / 100° standard (65° optional)
Electrical	Voltage / Power / Max Input Current / Pulse Duration / Pulse Rep Rate	9–35 VDC standard (other ranges optional) / 16 W pulsed max / 650 mA @ 24V during recharge / 0.1–5 ms standard / 2 Hz standard (10 Hz optional)
Interface	Connector / Control Signal	SubConn MCBH4M / Logic Trigger, 5V TTL, active-low
Mechanical & Env.	Weight / Materials / Depth Rating / Temperatures	In Air: 350g | In Water: 175g / 6061-T6 Aluminum Housing, Acrylic Window / 300 m (984 ft) / -10 °C to 40 °C operating | -25 °C to 85 °C storage

Documentation Link: Dragonfish-Mini Strobe Spec Sheet PDF

15.2.8 Microstrain AHRS Sensor (3DM-GX5-25-HRS) - Attitude Sensor (1x)

[DROP ATTITUDE SENSOR MOUNTED / EXTRA PHOTO HERE]

Hardware Inventory:

- Housing Mounting Brackets & Isolation Collars
- Stainless Steel Attachment Fasteners

Tools Needed:

- Standard Wrench & Socket Set

System Overview: Industrial-grade Attitude and Heading Reference System (AHRS). Relies on internal barometric pressure sensing alongside triaxial accelerometers, gyroscopes, and magnetometers processed through an onboard Auto-Adaptive Extended Kalman Filter (EKF) to achieve high-accuracy orientation estimates (pitch, roll, yaw).

15.2.8.1 Microstrain 3DM-GX5-25-HRS Specifications

Parameter	Specification	Parameter	Specification
Mounting Style	Chassis Mount	Supply Voltage Min / Max	4.25 V / 36 V
Sensor Type	9-axis IMU / AHRS	Sensing Axis	X, Y, Z
Interface Type	RS-232, USB	Part Aliases	6253-4220
Output Type	Digital	Product Type	IMUs - Inertial Measurement Units
Resolution	0.02 mg	Trade Name	MicroStrain
Operating Temp Range	-40 °C to +85 °C	Manufacturer	HBK / Hottinger Brüel & Kjær

Documentation Links:

- Microstrain 3DM-GX5-25 DataSheet PDF
- UK Declaration of Conformity Certificate
- 3DM-GX5-25 User Manual PDF

15.2.9 Teledyne Benthos PSA 916 - Sonar Altimeter (1x)

Hardware Inventory:

- (2x) Plastic Brackets
- (4x) Barrier Rubber Pieces
- (4x) 9/16”-12 (6 ½”) Steel Partial Thread Hex Bolts
- (8x) Steel Washers
- (4x) Steel Lock Nuts
- (4x) 9/16”-12 Steel Hex Nuts

Tools Needed:

- 9/16” Socket Wrench
- 9/16” Combination Wrench
- 14mm Combination Wrench (Optional)

System Overview: The Teledyne PSA-916 Programmable Sonar Altimeter is a lightweight altimeter designed for marine applications. Transmits a narrow-beam acoustic signal then measures round trip travel time for sound pulses reflecting off the seabed. Knowing the nominal speed of sound in water (1,500 m/s), travel time calculates precise distance to the ocean floor. Constructed from hard anodized, marine-grade aluminum housing and electronically isolated to eliminate signal interference with host sensors. The 200 kHz operating frequency achieves accurate measurements down to 6,000 meter depths. Applications include wave height measurement, obstacle avoidance, shallow water surveys, ice thickness, sediment transport, and scour studies.

15.2.9.1 Teledyne Benthos PSA 916 System Specifications

Parameter	Specification	Parameter	Specification
Operating Frequency	200 kHz	Range Outputs	Analog: 0–5 VDC | Digital: RS-232
Beam Width	14° conical typical	Operating Depth	6,000 m
Pulse Length	250 s standard	Power Required	7 to 24 VDC
Repetition Rates	5 pps or external	Power Consumption	60 mA @ 15 VDC
Range	100 m full scale	Size	2.25 in. diameter × 9.75 in. long
Resolution	RS-232: 1 cm | Analog: 2.5 cm	Weight	In Air: 0.63 kg (1.4 lbs) | In Water: 0.36 kg (0.8 lbs)
Required Topside Unit	UDB-9000 Universal Deck Box	Housing Material	Hard Anodized Marine-Grade Aluminum

Documentation Links:

- Teledyne Benthos PSA-916 Manual PDF
- Teledyne PSA 916 DataSheet PDF

Chapter 16

Mechanical Weak Link Assembly

The mechanical weak link protects the primary tow cable and ship winch from catastrophic over-tension during seafloor snags[cite: 8]. It is installed between the towing bridle and the mechanical cable clevis[cite: 6, 8].

[DROP WEAK LINK EXPLODED VIEW PHOTO HERE] / Weak Link Housing & Shear Pin Alignment	[DROP WEAK LINK ASSEMBLED PHOTO HERE] / Assembled Weak Link Attached to Tow Bridle

16.0.1 Assembly Procedure

1. Align the large holes on the outer housing with the large holes on the padeye inner housing (bridle), positioning set screws facing upwards[cite: 8]. Each side of the bridle and outer shell features one large hole and one small hole[cite: 8].
2. Slightly lift the bridle to slip the outer shell over the inner housing[cite: 8].
3. Rotate components until all matching holes align perfectly[cite: 8].
4. Insert the shear pin, wiggling the bridle and outer shell until the pin slides fully into position[cite: 8].
5. Ensure the shear pin is centered inside the body and sits completely flush on both sides[cite: 8].
6. Using a 3mm hex bit, tighten the set screws to **7.8 Nm** to lock the pin flush[cite: 8]. (Note: Set torque on torque wrench prior to tightening)[cite: 8].

7. Attach the mechanical clevis to the weak link using the primary termination bolts[cite: 8]. Ensure the set screws remain secured and protected by a hose clamp[cite: 8].

Chapter 17

Fiber Optic Termination

17.1 Purpose

Underwater fiber optic cables are used in environments where standard communication cables cannot provide the required durability, data capacity, or long-term reliability. The underwater sensors, subsea equipment, and HabCam need a cable that can handle water exposure, pulling force, bending, abrasion, pressure, and repeated deployment.

On the Henry B. Bigelow specifically, the fiber optic cable (run through the winch cable) is cut yearly, so a new fiber optic termination is required every time before deployment for a new survey.

17.2 Hardware

17.2.1 Part 1: Inserting the Fiber Optic into the Clevis

1. Mark and cut about 10m (33ft) off of the ship's fiber optic cable with a grinder and cutting wheel; this process disposes of any wire that was exposed to additional stress by going up and around the hanging block. Prior to cutting, put a hose clamp just after the 33ft mark (down the wire) to ensure that the cable does not start to unwrap.
2. After cutting the wire, leave the hose clamp on the open end and pay out enough wire length to reach into the starboard side of the ship's CTD lab and onto a table, where the fiber optic connectors will be applied.
3. Measure and mark 24ft from the open end. Install another hose clamp on this end.
4. Down at the 24 ft mark, take a dremel and score the outermost shield wire. *(Note: There are 3 layers of shielding).*

5. Starting from the open end, start peeling back the outer shield wire until you have reached the scored mark. Once you reach that mark, the shield wire should break off.
6. Repeat step 5 for the remaining shield wires until the black plastic shield is exposed.
7. Starting at the 24 ft mark, measure 12 inches up the wire and make a mark with a sharpie. *(Note: This is 25 ft from open end of the cable).*
8. Remove hose clamps and insert wire through the narrow end of sand-blasted clevis and pull through until the 12 inch mark is visible on the wider end of the clevis.
9. Refasten hose clamp only at the 12 inch mark.

17.2.2 Part 2: Creating the Broom

Steel Armor Wire Broom Formed and Seated inside Clevis Well

1. Starting at the 24 ft mark, start peeling the strands of shield wire (all three layers) until you reach the hose clamp (12 inches up the wire).
2. Bend each wire at a 90 degree angle against the hose clamp. Try to evenly space the wires around the cable. (All the wires will form a "broom").
3. The black rubber part of the cable should now be visible up to this mark.
4. Remove the hose clamp at the 25ft mark and move it right after the smaller opening of the clevis. (This prevents the shield wires from unwinding/bird-caging up the cable).
5. Walk the clevis out to the back deck.
6. Install two screw eyes in the deck, 6 feet from each other.
7. Take a 2 ft strap. Then feed a 1in diameter pin through the two holes at the opening of the clevis. Make sure one end of the strap is fed through the pin.
8. Attach the other side of the strap to the screw eye on deck with a 5/8th shackle.
9. Attach a Yale Grip rope line down the cable just after the clevis using several half hitches.
10. Put a 5/8th shackle on the rope end that is going away from the open end of the cable. This shackle will attach to the ship's manual come-along winch and pull the broom into the clevis' well.
11. Secure the come-along winch and place the shackle that is on the half-hitched rope (holding the cable) into the come-along winch hook. Make sure the come-along, rope, and cables are arranged in a straight line.
12. Attach the shackle to the other screw eye.
13. Use the come-along to pull down the broom into the clevis well. Try to get about 4-6 inches of broom inside the well; it is important, as we do not want the termination slipping during deployment, even with epoxy.
14. Once the broom is fully seated in the clevis, move and tighten the hose clamp to the cable at the base of the clevis (end with small opening).

17.2.3 Part 3: Epoxy

1. Hang the termination by the clevis bolt holes so that the broom is facing up.
2. Take 15ft of Tygon tubing and slide it over the black wire. Attach the tubing to the black rubber wire, and move it so that one end of the tubing is inserted down into the epoxy well. This tubing protects the black rubber wire from trimmed metal ends, which will occur later.
3. Try to bend the broom ends away from the Tygon tubing as much as possible, so that they will not poke into it. *(Note: Bend the black cable at a 45 degree angle so you do not block the pathway of the load pin).*
4. Using splicing tape, followed by electrical tape, create a reservoir around 2-3 in high around the openings of the clevis, where the broom becomes exposed.
5. Place newspaper or cardboard underneath where the clevis is hanging to prevent any epoxy mixture from spilling onto the ship's floors.
6. You will need gloves, Socketfast resin and its catalyst, a zip tie, and a long wooden mixing stick. Put on your gloves, take your catalyst and pour it into the Socketfast resin. Immediately take a wooden mixing stick and mix epoxy thoroughly for at least 2 minutes; the resin container should feel warm.
7. Carefully pour epoxy into the clevis well until the mixture comes up to the top of the duct tape reservoir. Use a funnel if available. A zip tie can be used to guide the flow and pop bubbles as the epoxy is poured. *(Tip: Lightly tap the clevis with a hammer as the resin is being poured to allow bubbles to rise).*
8. While the epoxy is wet, bend back any remaining broom strand ends away from the Tygon tubing.
9. Let the epoxy stand for a minimum of 1-2 hours before trimming back the broom strands to be as flush as possible with the hardened epoxy.
   **Important: If left to harden at room temperature, the epoxy should set for at least 24 hours before deploying into water. If time is of the essence, you may uniformly heat the clevis at 180-200° F with two heat guns for about 30 minutes to fully cure it.*
10. Before attaching the clevis to the HabCam towing bridle, perform a pull test with the ship's winch and the towing block to ensure that the epoxy will hold under tension. Aim for a tension of 8,000 lbs.

17.2.4 Part 4: The Boot

1. The boot uses 6 rolls of 2in self-vulcanizing tape and 2-3 layers (not rolls!) of 2in electrical tape.
2. Be sure when applying the tape that you pull tightly to ensure that the tape stretches and adheres to itself properly.
3. Taper self-vulcanizing tape so that the area closest to the clevis is thickest.

4. Entire boot length will be about 2.5ft or just under, measured from where the grooves in the clevis are.

17.2.5 Part 5: Fairing on Cable

Applying Zip Tie Fairing along Tow Cable on Deck

1. Use the wide, 18" zip ties.
2. Starting from where the boot ends, apply zip ties along the wire for about 15-20 feet until 8-10 bags of 100 zip ties are consumed.
3. Fairing wrap could be used as a substitute if handy; secure the wrap with electrical tape.

17.3 Fiber Optics Power Terminals

17.3.1 Part 1: Preparing Power and Fiber Optic Wires

1. From the open end of the fiber optic wire, measure 2 feet down and make a mark. VERY CAREFULLY take a razor knife and score the black rubber down the length of the wire to that mark; you do not need to cut all the way through the rubber to remove it. Use pliers to peel back the stiff rubber before cutting it off at the 2-foot mark.
   (Tip: Squeezing the end of the wire with pliers can help split the rubber).
2. After the 2 feet of black rubber has been removed, carefully separate the exposed wires. You should have 3 power wires (2 brown, 1 green), 3 fiber optic wires (1 red, 1 gray, 1 black), flexible rubber spacing material, and a thin, white-gray spacer strand.
3. Remove the flexible rubber spacing material and cut the thin white-gray spacer strand.
4. Snip about an inch off the fiber and power wires to eliminate any broken regions resulting from initial cable cuts.

17.3.2 Part 2: Inserting the Fiber Optic Wire into the J-Box

1. Unscrew the threaded cap of the empty connector on the side of the J-Box, removing the metal ring collar and rubber gasket.

Figure 2.1: Open J-Box Interior showing Empty Connector Port and Terminal Block

2. Guide the power and fiber optic wires through the threaded cap (threads facing the open end), followed by the metal ring collar and the rubber gasket (flattest part facing the open wire end). ***DO NOT MISS THIS STEP!**
3. Guide the wires through the open connector until about half an inch of black rubber is visible inside the J-Box. Secure a hose clamp around the black rubber flush against the inner wall to prevent slipping.

/ Threading Cap, Collar & Rubber Gasket onto Wire	/ Internal Hose Clamp Secured Flush to Inner Wall

4. Secure the gasket, metal ring collar, and threaded cap to the J-Box. Tighten the cap firmly with a pipe wrench without twisting the connector body.

/ Seating Rubber Gasket & Metal Collar into Connector	/ Tightening Threaded Cap with Pipe Wrench

17.3.3 Part 3: Connecting Ring Terminals to the Power Wires

1. Locate the 2 thick brown wires (ground/neutral) and 1 thick green wire (power driver).
2. Strip less than ½" of insulation from the green wire using a 12-10 AWG wire stripper.

12-10 AWG Ring Terminal and Gauged Wire Stripper/Crimping Tool

3. Place a yellow 12-10 AWG ring terminal over the exposed copper wire.
4. Crimp the upper half of the yellow jacket securely using the crimping tool. Tug on the terminal to verify connection strength.

Crimped 12-10 AWG Yellow Ring Terminal

5. Repeat steps 2–4 for both brown wires.

17.3.4 Part 4: Connecting the Power Wires to the J-Box

1. Connect the green power wire across from the center red terminal screw inside the J-Box. Gently loop the wire around the interior avoiding sharp 90° bends.

Green Power Wire Connected to Center Terminal Screw

2. Attach the brown wires to either side of the green wire across from their corresponding red terminal counterparts.

Green and Brown Conductors Connected to J-Box Terminal Strip

17.4 Fiber Optic ST Connectors

17.4.1 Part 1: Safety, Cleanliness, and Pre-Heating Oven

1. Wear safety glasses **AT ALL TIMES** when handling fiber optic wire ends.
2. Clean all tools and work surfaces with isopropyl alcohol and lint-free wipes prior to stripping or polishing.
3. Plug in and turn on the 3M Hot Melt oven kit; allow at least 6 minutes for uniform heating.

17.4.2 Part 2: Exposing the Protective Jacketing

4. Unpack a 3M Hot Melt Singlemode ST Yellow Boot connector packet. Discard the thin plastic tube.

3M ST Yellow Boot Packet & Disassembled Component Parts

5. Slide the yellow boot narrow-end first onto the gray fiber optic wire.

Yellow Boot Slid onto Gray Fiber Wire

6. Measure cut lengths on the 3M oven hotplate scale guide and mark the wire.

Measuring Cut Lengths on 3M Hotplate Scale Guide

17.4.3 Part 3: Trimming Metal Strands

7. Fan out the metal jacketing strands evenly without bending the central buffered glass strand.
8. Fold the metal strands back along the gray wire jacket as flat as possible.

/ Fanning Out Metal Jacketing Strands	/ Bending Metal Strands Backward Flat

9. Trim the folded metal strands to approximately 1/4 inch in length.

17.4.4 Part 4: Removing the 1st Buffer Layer

10. Reference the hotplate guide to determine buffer removal length.
11. Clean a 3-hole stripper tool with alcohol. Using the middle hole, gently strip away the primary buffer layer in short, straight pulls. Clean the tool after every strip.

17.4.5 Part 5: Removing the Final Buffer Layer

12. Using the smallest hole on the stripper tool, gently remove the inner buffer coating until flush with the thick buffer.

/ 3-Hole Fiber Optic Stripper Tool	/ Bare Glass Fiber Strand After Stripping

13. Gently wipe the bare glass strand with an alcohol-moistened low-lint wipe to remove microscopic buffer debris.

17.4.6 Part 6: Preparing to Heat the ST Connector

14. Remove cap from ST connector and place connector ferrule-side down into an oven ST holder.
15. Confirm the oven is fully heated (~6 minutes).

17.4.7 Part 7: Inserting Fiber into ST Connector

16. Place an ST holder into the hot oven hole while monitoring with a stopwatch.

/ Placing ST Connector Ferrule into ST Holder	/ 3M Hot Melt Oven & Hotplate Setup

17. Observe internal blue epoxy liquefy (~60–90 seconds). When epoxy reaches near the top of the ferrule hole, transfer holder to hotplate and immediately insert fiber strand until metal strands click flush into the connector.
18. Allow assembly to cool for 3 minutes undisturbed.

17.4.8 Part 8: Removing ST Connector from Holder

19. Slide yellow boot over connector rear housing until it snaps in place.
20. Pull boot straight up to remove connector from holder.
21. Verify optical continuity using a visual fault locator (red laser).

17.4.9 Part 9: Scoring and Removing Fiber Nipple

22. Position ruby scribe perpendicular to fiber strand 1/16th inch above ferrule tip.

Scribing Fiber Perpendicularly One Fiber Diameter from Ferrule Tip

23. Gently score the glass strand and pull straight off. Discard glass tip in sharps container.

17.4.10 Part 10: Air Polishing

24. Perform 15–20 figure-8 passes on a 9-micron gray polishing film until glass stub is flush with blue epoxy bead.

17.4.11 Part 11: Pad Polishing with Green Film

25. Clean black rubber polishing pad with alcohol. Place wet 1-micron green lapping film on pad.
26. Mount ferrule into polishing puck and perform figure-8 rotations until blue epoxy turns clear and white ceramic ferrule is exposed.

/ Failed Scope Views: Chipped Core, Epoxy Rings & Scratches	/ Dirty Ferrule View (Clean with Alcohol Wipe)

27. Inspect ferrule endface under 200x fiber scope to verify clean core surface.

17.4.12 Part 12: Final Polish with White Film

28. Perform 2–3 light figure-8 passes on 0.05-micron white film for final surface finish.

Ideal/Clean Fiber Ferrule Core View under Viewing Scope

17.4.13 Part 13: Terminating Backup Fibers

29. Repeat Parts 2–12 for the black and red backup optical fiber lines.

17.4.14 Part 14: Connecting Main Fiber to J-Box Cable

30. Connect primary gray fiber ST connector to the metal optical coupler inside the J-Box.

J-Box Layout and Cable Entry Positioning on Worktable

Coupling Primary Gray ST Fiber to Metal Adapter Assembly (Left: Aligning Connector; Right: Mated in Position)

31. Coil red and black backup fibers neatly inside J-Box housing and secure loosely with zip ties.

Red and Black Backup Fibers Coiled and Zip-Tied inside J-Box

17.5 Buttoning Up the J-Box

17.5.1 Part 1: Securing the Urethane Bladder

1. Align yellow urethane bladder indentation face-down against J-Box rim.
2. Align metal retaining frame over bladder.
3. Thread 5mm Allen screws with lock and flat washers through frame and bladder.

/ 1. Aligning Yellow Bladder Indentation Face Down	/ 2. Metal Retaining Plate Alignment
/ 3. 5mm Allen Screw with Lock & Flat Washers	/ 4. Threading 5mm Allen Screws into Perimeter

4. Evenly torque perimeter screws in a cross-pattern using 5mm Allen wrench until snug without over-compressing bladder material.

17.5.2 Part 2: Filling J-Box with Compensating Oil

1. Connect oil fill tube to lower J-Box quick-disconnect valve.
2. Connect purge tube from upper valve to overflow collection bucket.
3. Pump dielectric compensating oil into J-Box until all air bubbles are purged.

Compensating Oil Vacuum Pump & Purge Reservoir Setup

17.5.3 Part 3: Securing J-Box to HabCam Vehicle

1. Route oil-filled J-Box and main tow cable along the inside/rear of towing bridle structure.
2. Mount J-Box inverted on port-aft frame rail.
3. Connect optical plug to telemetry bottle optical bulkhead port; connect power plug to telemetry bottle power port.

Chapter 18

Fiber Optics Pull Test Protocol

18.1 Purpose

A pull test (or proof load test) using a water bag on a hydraulic winch line is performed to safely verify that the winch and its mounting structure can handle their maximum rated weight capacity. This test uses weights or simulated loads to prove that the mechanical, hydraulic, and brake systems work perfectly in real working conditions and that your winch is safe to use for anchoring, towing, and mooring operations.

1. Load Testing:
Purpose: To ensure that the winch is capable of pulling or holding loads according to its nominal capacity without structural failure. The winch is tested using actual loads of up to 110–125% of its design capacity.

2. Brake Test:
Purpose: To ensure that the main and emergency brake systems can safely hold loads, even under extreme conditions. The brake test is performed by suddenly stopping the winch while pulling a specific load. Technicians will evaluate whether the brakes can hold the load without slipping or dropping.

3. Functional Test:
Purpose: To evaluate the overall operational performance of the winch in real conditions. The functional test is performed after the load and brake tests to ensure that all components work together and that your winch is ready for safe and efficient use at sea.

18.2 Preparation

After the termination is complete (and tested) and the epoxy inside the clevis has cured you can proceed to the pull test. Attach the clevis to the water bag in the order shown in the image below. Note the hose clamp at the top, up against the clevis (Don't forget to install this).

You will need to coordinate with the Bigelow ahead of time to borrow the water bag from the U.S. Coast Guard and help from the Chief Bosun to perform the test.

Order of Operation: / / Clevis → / Swivel Shackle → / Shackle → / Load Sensor → / Water Bag	/ Rigging Linkage: Clevis, Swivel Shackle, Shackle, Load Sensor, & Water Bag

18.3 Test Execution & Verification

HabCam J-Box and cables positioned safely out of the way for testing.

Water Bag Suspended over Deck during Pump Filling

With the water bag in position, the Chief Bosun will coordinate with the engineers to turn on the pumps and begin filling the bag.

Monitor the tension on the load sensor and let the bosun know when you get to 9,000 lbs so they can slowly close the pumps with the intention to fully close the pumps at 10,000 lbs tension.

Next hold the tension at 10,000 lbs for approximately 1 minute.

/ Load Sensor Readout Displaying 10,000 lbs Test Tension	Once complete, the water bag can be emptied and brought back on the deck. Inspect the hose ring that was cinched above the clevis and verify that the clevis has not slipped under tension. If the location has changed and the clevis slipped then adjustments must be made, and the test must be run again. If its location hasn't changed, attach the J-Box cables and retest the fiber optics with the HabCam vehicle. If everything is working you are good to go.

Part IV

Software & Data

Chapter 19

Software & Data Overview

19.1 Stereo Acquisition Suite

19.1.1 Topside Status

Application Control	Control / Field	Example Value	Definition
Co-Pilot	Survey/Cruise (HabCam) ID	`2026/HabCam/Bigelow`	Designates the survey type and vessel (typically `Year/ID`).
Co-Pilot	Survey/Cruise (AUV) ID	`2026/AUV/Smallboat`	Designates the survey type and vessel for AUV operations.
Co-Pilot	OP Area	`Newport, RI to Newport, RI`	Operation Area where data is currently being collected.

19.1.2 Sensor Status

Application Control	Sensor	Status Light	Definition
Co-Pilot	**AHRS**	Green = ONRed = OFF	Attitude and Heading Reference System; computes 3D orientation (roll, pitch, yaw) and direction.
Co-Pilot	**CTD**	Green = ONRed = OFF	Conductivity, Temperature, and Depth sensor; determines salinity, density, and sound velocity.
Co-Pilot	**ECO Triplet**	Green = ONRed = OFF	Environmental Characterization Optics; highly customizable 3-channel optical sensor.
Co-Pilot	**SeaFET**	Green = ONRed = OFF	Ocean pH sensor designed for long-term, high-precision marine monitoring.
Co-Pilot	**GPS**	Green = ONRed = OFF	Global Positioning System; tracks physical location (latitude, longitude) and altitude.
Co-Pilot	**Fathometer**	Green = ONRed = OFF	Active sonar device measuring water depth beneath the vessel via acoustic echo travel time.

Application Control	Sensor	Status Light	Definition
Co-Pilot	**Altimeter 1**	Green = ONRed = OFF	Sensor measuring vehicle altitude above a fixed reference point (seafloor).

19.1.3 Vehicle Metrics

Application Control	Parameter	Example Value	Definition
Co-Pilot	Heading	`226.75°`	Compass direction in which the front of the vehicle is pointing.
Co-Pilot	Pitch	`-5.0°`	Vehicle attitude along the lateral axis in decimal degrees.
Co-Pilot	Roll	`46.3°`	Vehicle attitude along the longitudinal axis in decimal degrees.
Co-Pilot	Depth	`60.72m`	Depth of the vehicle in the water column.
Co-Pilot	Altitude	`2.32m`	Vertical distance between the vehicle and the ocean floor.

19.1.4 CTD Module

Application Control	Parameter	Example Value	Definition
Co-Pilot	Conductivity	`3.208770 S/m`	Measures the water's ability to conduct an electrical current.
Co-Pilot	Salinity	`32.836000 psu`	Concentration of dissolved salts in the water column.
Co-Pilot	Sound Speed	`1471.20 m/s`	Rate at which acoustic waves travel through the water column.
Co-Pilot	Temperature	`5.55°C`	Water temperature measured directly using thermistors or platinum thermometers.
Co-Pilot	Pressure	`61.147000 dbar`	Barometric pressure used to calculate precise vehicle depth.

19.1.5 Vessel Data

Application Control	Parameter	Example Value	Definition
Co-Pilot	Latitude	`40° 0.8764`	Geographic north-south position coordinate.
Co-Pilot	Longitude	`-72° 35.5802`	Geographic east-west position coordinate.

Application Control	Parameter	Example Value	Definition
Co-Pilot	Depth Below Transducer	`65.76m`	Vertical distance from the sonar sensor to the seafloor.
Co-Pilot	Heading	`0.00°`	Compass direction in which the vessel's bow is pointing.
Co-Pilot	COG	`59.6°`	Course Over Ground; actual direction moving across Earth's surface (0° to 360°).
Co-Pilot	SOG	`5.559 knots`	Speed Over Ground; actual speed relative to Earth's surface.

19.1.6 Oxygen Sensor

Application Control	Parameter	Example Value	Definition
Co-Pilot	Conductivity	`3.208770 S/m`	Water's ability to transmit electrical current; indicates total dissolved ions/salts.
Co-Pilot	Salinity	`32.836000 psu`	Total concentration of dissolved salts in water.
Co-Pilot	Dissolved Oxygen	`6.418000 ml/L`	Measurement of free, non-compound O molecules mixed into liquid.

Application Control	Parameter	Example Value	Definition
Co-Pilot	Temperature	`5.56°C`	Temperature of the water environment.
Co-Pilot	Pressure	`61.603000 dbar`	Ambient pressure measurement.

19.1.7 SeaFET (pH Sensor)

Application Control	Parameter	Example Value	Definition
Co-Pilot	Int pH	`N/A`	Internal pH calculated via closed KCl reference electrode.
Co-Pilot	t	`N/A`	Water temperature.
Co-Pilot	Int V	`N/A`	Raw voltage from internal reference electrode.
Co-Pilot	Therm V	`N/A`	Thermistor voltage reading for internal sensor temperature.
Co-Pilot	Supply V	`N/A`	Main supply voltage.
Co-Pilot	5V	`N/A`	Regulated DC power for integrated system components.
Co-Pilot	Ext pH	`N/A`	External pH calculated using seawater-exposed reference electrode.

Application Control	Parameter	Example Value	Definition
Co-Pilot	hum	`N/A`	Internal moisture/humidity level inside housing.
Co-Pilot	Ext V	`N/A`	Raw voltage from external solid-state reference electrode.
Co-Pilot	stat	`N/A`	Instrument operational status and configuration state.
Co-Pilot	Supply 1	`N/A`	Sensor power usage / current draw.
Co-Pilot	Iso V	`N/A`	Isolation voltage limit for safety components.

19.1.8 Attitude Plots

Application Control	Parameter	Range Limits	Definition
Co-Pilot	Heading	No Min/Max	Compass direction of the vehicle's bow.
Co-Pilot	Pitch	Max: 0 / Min: -12	Vehicle lateral axis tilt limits in decimal degrees.
Co-Pilot	Roll	Max: 50 / Min: 45	Vehicle longitudinal axis tilt limits in decimal degrees.

19.1.9 Dsvimview (Cameras & Winchfly)

Application Control	Control / Setting	Example Setting	Definition
Co-Pilot	Show Histogram	Toggled off	Visual chart grouping pixel response values across brightness ranges.
Co-Pilot	Histogram Equalize	Toggled off	Contrast adjustment technique spreading pixel intensities across full dynamic range.
Co-Pilot	Histogram Stretch	Toggled off	Contrast adjustment technique expanding cramped brightness ranges to highlight shadows/details.
Co-Pilot	Gain	`6.00 hz`	Signal amplification control to distinguish seabed substrate types.
Co-Pilot	Exposure	Max: 10.00 msMin: 0.00 ms	Camera exposure control balancing brightness while preventing glare or shadow detail loss.
Co-Pilot	RCVD	Gain: 18.0Exposure: 1.5 ms	Received signal strength control filtering out water column acoustic noise.

Application Control	Control / Setting	Example Setting	Definition
Co-Pilot	Show Winchfly	Toggled on Green	Enables automated vehicle steering to maintain set altitude above seafloor.
Pilot	Bottom	`-50 m`	Strongest acoustic return indicating the physical seafloor.
Pilot	Fish Depth	`59.9 m`	Calculated vertical distance from water surface to vehicle.
Pilot	Computed Bottom	`-75 m`	Algorithmically calculated seabed boundary correcting for false echoes.
Pilot	Alt	`3.5 m`	Vertical distance between vehicle and seafloor.
Pilot	Fathometer	`65.5 m`	Echo sounder water depth measurement.
Pilot	Fathometer Filter	`0.8`	Acoustic noise filter smoothing depth readings.
Pilot	Depth Filter	`0.8`	Restricts sonar/mapping display to specific vertical bounds.
Pilot	Calc Depth	`63.3 m`	Depth compensation setting adjusting image brightness/color for light attenuation.

19.2 Winch Joystick Control

19.2.1 Manual Winch Panel

Application Control	Field	Example Value	Definition
Pilot	Cast Duration	`26:26`	Elapsed time since the current cast was initiated.
Pilot	Set Length	`500 meter`	Maximum allowable winch cable payload payout limit.
Pilot	Tension	`5.77 tons`	Mechanical load threshold protecting cable from over-strain.
Pilot	Linespeed	`0 meter/min`	Rate at which cable spools on/off the winch drum.
Pilot	Length	`91.2 meter`	Total cable paid out in meters.

Examples referenced from HabCam Survey 2026 operations.

Chapter 20

Data Acquisition

20.1 Overview of Software Components

- **Part 1: Engineering GUI**
  - **Sensor Status:** Displays real-time operational status for all acquiring sensors (indicated in green). At sea, all sensors except the deprecated SeaFET will show green.
  - **Monitored Sensors:**
    - **AHRS:** Attitude sensor
    - **CTD:** Sea-Bird 49
    - **Oxygen:** Sea-Bird 39
    - **GPS:** Ship's GPS feed
    - **Fathometer:** Ship's fathometer feed
    - **Altimeter:** Benthos PSA 916
- **Part 2: Image Viewer (Both Cameras Online)**
  - Displays the live feed of the stereo cameras in real time as frames are acquired.
  - Controls the winchfly plot display as well as the trigger rate for both cameras and strobes.
- **Part 3: Stereo Acquisition**
  - A terminal script that loads configuration files onto each camera and initiates the acquisition feed.
- **Part 4: Event Logger**
  - Utilized by the co-pilot to record timestamped operational notes, trackline changes, image highlights, shift changes, and system events.

20.2 HabCam Vehicle Network Addresses

Component / Device	IP Address
Port Camera	`192.168.125.120`
Starboard Camera	`192.168.125.121`
Moxa Nport #1	`192.168.135.200`
Moxa Nport #2	`192.168.135.202`
Gardasoft #1	`192.168.135.91`
Gardasoft #2	`192.168.135.92`
BlueView Sonar	`192.168.1.45`

20.3 At-Sea Data Management Workflow

20.3.1 System Initialization & Time Synchronization

- All HabCam computers and servers are time-synchronized to the NOAA time server using Network Time Protocol (NTP).
- The main topside computer (`Dixon`) connects directly to the HabCam vehicle, onboard servers, ship GPS, and ship fathometer.
- Raw unprocessed environmental data is recorded directly into text files with associated timestamps for every entry.
- A processed CSV file is continuously generated to store vehicle sensor measurements, GPS coordinates, and corresponding stereo-paired image filenames formatted as `cruiseID.Timestamp.image#.tif`.

20.3.2 Starting Acquisition

1. Power on the HabCam vehicle and launch the `habcamDataAq` software suite.



Figure 20.1: Topside Workstation showing Engineering GUI and Image Viewer

3. Launch the following desktop shortcuts: **Engineering GUI**, **Image Viewer**, and **StereoAq**.
4. Verify in the **Engineering GUI** that all active sensor indicators display a green light. > **Note:** A red light indicates an offline or inactive sensor. The only expected red indicator during sea operations is the 5th sensor down on the left sidebar (the deprecated SeaFET sensor).
5. Open the **Image Viewer GUI** to confirm live video streams from both Port and Starboard cameras. Ensure the 3 status indicators relating to **recording**, **writing**, and **acquisition** are green.

20.4 Metadata Directories & Verification

20.4.1 Data File Formats

- **`.hab` Files:** Hourly raw data logs containing raw output streams from every sensor connected to the vehicle according to the HabCam v4 architecture.

- **`.img` Files:** Processed metadata CSV files containing timestamped image names, geospatial location coordinates, and aligned sensor metrics.

20.4.2 Directory Structures

Verify that hourly raw data (`.hab`) and processed metadata (`.img`) files are actively being written to the following topside paths:

- **Hourly Raw Sensor Data (`.hab`):** “‘bash /mnt/dixon-data/YYYY/hab/

Chapter 21

Server Framework

[Insert server rack specifications, network architecture, Apache web server configs, and storage mounts here]

Chapter 22

Annotation Training Guide

This section provides background on the annotator interface design, key workflow considerations, and standard operating procedures for how optical data is reviewed and annotated within the HabCam program.

Review the interactive slide deck below for step-by-step annotation guidelines and species identification standards.

(Interactive slide deck available on the online SOP website.)

22.1 Annotation Quiz

Put your training into practice before starting live annotation. Work through the self-assessment quiz below to verify your understanding of species identification criteria, bounding box conventions, and quality control flags.

(Interactive quiz available on the online SOP website.)

22.2 Species Guide: Seastars

PLACEHOLDER Explain when and why we annotate these…

(Species identification carousel available on the online SOP website.)

22.3 Species Guide: Fishes

PLACEHOLDER Explain when and why we annotate these…

(Species identification carousel available on the online SOP website.)

22.4 Species Guide: Invertebrates

PLACEHOLDER Explain when and why we annotate these…

(Species identification carousel available on the online SOP website.)

Chapter 23

Database Architecture & Operations

23.1 1. Archiving and Data Processing Workflow

23.1.1 Database Archiving

- **Realtime Image Check:** During realtime processing, the HabCam software periodically checks for newly created `.img` files.
- **Write Verification Buffer:** To ensure a file is fully written before processing, the software never archives the newest `.img` file immediately; it waits until the subsequent `.img` file is created (e.g., `20250110_2010.img` is not archived until `20250110_2020.img` exists).
- **Database Ingestion:** Archiving ingests the `.img` file into the active survey's PostgreSQL database under the `img` table, which serves as a virtual archive for all metadata and parameters pertaining to each image.
- **Access Tools:** Software applications used to access and query this metadata include **pgAdmin** (PostgreSQL client), **R**, and **Matlab**.

23.1.2 Realtime Processing & Annotation Pipeline

23.1.2.1 Processing GUI Functions

The HabCam realtime processing GUI is used for image processing, creating annotation assignments, generating lightmaps, and serving images to annotators.

The PostgreSQL database provides the GUI with metadata required for: 1. **Lightmap Construction:** Building lightmaps using survey images collected at vehicle altitudes between 1 and 4 meters above the seafloor. 2. **Stereo Image Processing:** Processing images using vehicle altimeter depth (height above seafloor), combining intrinsic/extrinsic parameters calculated from pre-survey stereo camera calibrations with the survey's generated lightmap.

23.1.2.2 Annotation Workflow

- Processed images are organized into **primary**, **secondary**, and **special assignments**.
- Assignments are served via the HabCam annotation webserver, allowing scientists to annotate and analyze images in near-realtime while at sea.
- All annotation records and metadata are immediately written back to the PostgreSQL database.

23.1.3 Data Review and Quality Assurance (QA)

- **Watch Chief Responsibilities:** At sea, the watch chief is responsible for training scientific party members and conducting periodic QA reviews on completed annotations.
- **Validation Criteria:** QA personnel review annotated images for completeness, taxonomic accuracy, and bounding box consistency.
- **Error Tracking:** Any discrepancies or errors identified during review are logged and reviewed directly with the annotator to ensure proper corrections are executed.

23.1.4 Survey Archival Workflows

Phase	Operational Steps
At-Sea Archival	Data managers continuously back up all metadata and raw optical data onto external hard drives. Processed images are redundantly backed up across multiple onboard HabCam servers.

Phase	Operational Steps
Post-Survey / Inter-Leg	External hard drives and a complete PostgreSQL database export are handed over to NOAA ITD for backup on landservers. The database is restored on land to allow processing and annotation to continue seamlessly between survey legs.

23.1.5 Data Archival Infrastructure

23.1.5.1 Physical & Physical Server Methods

- **Physical Media:** Raw optical data and metadata are continuously written to formatted external hard drives at sea. Raw metadata is also continuously `rsync`-ed across onboard servers.

23.1.5.2 Landserver Infrastructure (Habcam-DS Series)

Server	Primary Purpose & Workflow
`Habcam-DS1`	**Backup Server:** Automatically and recursively archives all relevant data directories from `Habcam-DS2` every 24 hours, maintaining daily execution log files.
`Habcam-DS2`	**Processing & Annotation Duplicate:** Mirrors the seagoing realtime processing and annotation server. Used on land to continue processing optical data, serving assignments, and providing assessment scientists with relational database access.

Server	Primary Purpose & Workflow
`Habcam-DS3`	**Storage & Media Swap Server:** Functions as a software backup for `DS2` and serves as the primary node for swapping hard disks containing images and metadata. Processed images from `DS2` are transferred here before being backed up offline.

23.1.5.3 Relational Database Architecture

- **PostgreSQL (pgAdmin):** Serves as the primary operational database engine for active HabCam survey data and image metadata.
- **Oracle Cloud Infrastructure (`ORAPROD`):** Databases are migrated into an Oracle cloud database (`HABCAM` schema) to provide an additional layer of security, disaster recovery, and long-term multi-cruise archival safety.

23.2 2. Two-Stage Database Lifecycle

```
[ At-Sea Acquisition & Annotations ]


  Stage 1: Active PostgreSQL DB (pgAdmin)
     Real-time Image Telemetry (img / img_lrauv)
     Assignment Manifests (assignments)
     Raw Annotations (annotations)


       [ QA/QC & Validation Phase ]
     Watch Chief Annotation Review
     Bounding Box & Taxonomic Verification
     Data Quality Flagging (DATA_IDENTIFIER)


  Stage 2: Enterprise Oracle Cloud (ORAPROD)
     Historical Multi-Cruise Warehouse (2009-Present)
     Stock Assessment Ingestion (CASA / SAMS)
     Centralized GIS & Public Data Products
```

23.2.1 Stage 1: Active Operations (PostgreSQL)

During sea operations and active survey legs, all real-time sensor streams, image manifests, and manual/AI annotations are written directly to a year-specific PostgreSQL database (`pgAdmin`). * **Real-time Ingestion:** Image telemetry (`img` / `img_lrauv`) is staged via `temp_img` and written in 10-minute intervals. * **Web Serving:** The active PostgreSQL instance connects directly to the HabCam web annotator to serve image assignments (`assignments`, `imagelist`) to scientists at sea or on land.

23.2.2 Stage 2: QA/QC & Validation

Before permanent archival, watch chiefs and QA teams review all completed annotations: * **Validation Criteria:** Taxonomic accuracy, bounding box precision, percent cover estimates, and completeness. * **Flagging:** Deprecated or incorrect records are marked (`deprecated = true`), and the assignment is assigned a specific `data_identifier` code.

23.2.3 Stage 3: Enterprise Archival (Oracle Cloud)

Once QA/QC is complete, the finalized current-year dataset is migrated into the NOAA Oracle Cloud database (`ORAPROD.HABCAM` schema). * **Multi-Cruise Linkage:** Every record is assigned composite keys (`HABCAM_PK`, `CRUISE_ID`, `DATA_IDENTIFIER`) to link it within the multi-decade historical repository (2009–present). * **Stock Assessment Ingestion:** Data from Oracle feeds directly into Catch-At-Size-Analysis (CASA) models and the Scallop Area Management Simulator (SAMS).

23.3 3. PostgreSQL vs. Oracle Schema Comparison

While the relational structures mirror each other logically, technical schema differences exist between the operational PostgreSQL database and the enterprise Oracle warehouse.

Metric / Feature	Active PostgreSQL Database (pgAdmin)	Enterprise Oracle Database (ORAPROD.HABCAM)	Architectural & Query Impact
Table Naming	Lowercase, unprefixed (e.g., annotations, assignments, img)	Uppercase, HABCAM_ prefixed (e.g., HABCAM_ANNOTATIONS, HABCAM_ASSIGNMENTS)	SQL queries targeting Oracle must use table prefixes (e.g., HABCAM.HABCAM_ANNOTATIONS).
Primary Keys & Linkage	Local text/integer IDs (e.g., annotation_id, assignment_id)	Composite surrogate keys (HABCAM_PK, CRUISE_ID, DATA_IDENTIFIER)	Oracle ties every record to a master cruise ID for multi-year historical queries.
URL Identifiers	Text strings in primary ID fields	Separate text and URL columns (ANNOTATION_URL_ID, IMAGE_URL_ID)	Oracle explicitly separates numeric database IDs from web-facing URL links.
Spatial / Geometry	Native PostGIS geometry types (USER-DEFINED / POLYGON)	CLOB text strings (GEOMETRY_TEXT, THE_GEOMETRY)	Oracle stores spatial bounding boxes as JSON/WKT CLOBs rather than PostGIS types.
Data Types	text, integer, numeric, boolean	VARCHAR2(255/4000), NUMBER, CLOB, TIMESTAMP WITH TIME ZONE	Oracle uses NUMBER for all integers/floats and size-constrained VARCHAR2 strings.
Deprecation Flags	Generic deprecated (boolean)	Table-prefixed ANNOTATION_DEPRECATED, CLASS_DEPRECATED (NUMBER 0/1)	Oracle uses numeric flags (0 = active, 1 = deprecated) with table-prefixed column names.

23.4 4. PostgreSQL Database Schema Architecture

The active PostgreSQL database contains 20 tables categorized into four functional modules.

23.4.1 Schema Overview

Category	Table Name	Columns	Purpose / Description
Annotation Core	`annotations`	17	Master bounding box annotations, measurements, and species classifications per image.
Annotation Core	`assignments`	17	Master records of annotation assignments (`YYYYNN`) and project goals.
Annotation Core	`imagelist`	5	References specific images assigned to an active annotation assignment.
Annotation Core	`imagelist_offline`	5	Image manifest lists assigned for offline / disconnected at-sea annotation.
Telemetry & Payload	`img`	24	Master telemetry and sensor metadata table for every acquired HabCam V4 image.

Category	Table Name	Columns	Purpose / Description
Telemetry & Payload	`img_lrauv`	25	Master telemetry for LRAUV AUV images and payload sensors (*Stella* / *Polaris*).
Training Workflow	`annotations_training`	17	Annotations produced during annotator training and calibration assignments.
Training Workflow	`assignments_training`	17	Training assignment records for onboarding annotators.
Training Workflow	`imagelist_training`	5	Image manifest lists assigned for training assignments.
Data Staging	`temp_img`	19	Staging table for temporary ingestion of HabCam V4 image telemetry.
Data Staging	`temp_lrauv_img`	18	Staging table for temporary ingestion of LRAUV AUV image telemetry.
Metadata & Lookup	`auth`	3	Annotator user credentials, access permissions, and authentication salts.
Metadata & Lookup	`classes`	5	Master taxonomic and target classification lookup table.

Category	Table Name	Columns	Purpose / Description
Metadata & Lookup	`facets`	4	Maps facet IDs to scope IDs to dictate how classes are evaluated in the GUI.
Metadata & Lookup	`idmodes`	4	Assignment-specific lists of annotation classes enabled in the GUI.
Metadata & Lookup	`scopes`	3	Defines scope categories (`1` = target, `2` = dominant substrate, `5` = % cover).
Metadata & Lookup	`substrate`	10	Automated % cover model predictions for seafloor substrate classes.
PostGIS System	`geometry_columns`	7	PostGIS spatial reference metadata view for geometry columns.
PostGIS System	`geography_columns`	7	PostGIS spatial reference metadata view for geography columns.
PostGIS System	`spatial_ref_sys`	5	PostGIS coordinate reference systems (CRS) definitions (e.g., EPSG:4326).

23.4.2 Key Table Definitions

23.4.2.1 `annotations` (Annotation Core)

Stores individual bounding box detections, percent cover estimates, and species labels.

Col	Field Name	Data Type	Description	Example Value
1	`annotation_id`	`text`	Unique HTTP ID	`http://habcam.noaa.gov/ann_9000207e13f`
2	`image_id`	`text`	Image URL	`http://habcam-data.whoi.edu/data/20250`
3	`scope_id`	`integer`	Scope code (1=target, 2=substrate, 5=% cover)	`1`
4	`category_id`	`text`	Category identifier URL	`http://habcam-data.whoi.edu/data/1102`
5	`geometry_text`	`text`	Bounding box coordinates (JSON format)	`"boundingBox":[[1054.67, 51.88], [1253.33, 346.54]]`
6	`thegeom`	`USER-DEFINED`	PostGIS polygon geometry	`SRID=4326;POLYGON((...))`
7	`annotator_id`	`text`	Annotator username	`ncalabrese`
8	`assignment_id`	`text`	Assignment HTTP URL ID	`http://habcam-data.whoi.edu/data/20250`
9	`timestamp`	`timestamp with tz`	Timestamp when annotation was created	`2025-05-22 14:12:57+00`
10	`class_id`	`integer`	Unique class ID mapping to `classes.class_id`	`185` (*Placopecten magellanicus*)
11	`deprecated`	`boolean`	Flag indicating if annotation is invalidated	`false`

Col	Field Name	Data Type	Description	Example Value
12	geometry_id	integer	Geometry identifier reference	12
13	imagename	text	Raw image filename reference	202501.20250528.100444831.772211.png
14	assignment_num	integer	Assignment ID in form YYYYNN	202501
15	percent_cover	integer	Recorded percent cover (if applicable)	25
16	comment	text	Annotator notes or QA/QC comments	High quality scallop identification
17	source	text	Source model or interface version	manual_annotator_v4

23.4.2.2 assignments (Annotation Core)

Defines structured annotation tasks assigned to scientists during a survey leg.

Col	Field Name	Data Type	Description	Example Value
1	assignment_id	integer	Unique assignment ID (YYYYNN)	202501
2	idmode	text	Enabled species class list name	NMFS 2025 ScallopsNFish
3	site_description	text	Assignment location or objective	At-Sea Primary Leg 1

Col	Field Name	Data Type	Description	Example Value
4	project_name	text	Survey project title	NEFSC Scallop Survey
5	priority	text	Assignment priority (HIGH, MEDIUM, LOW)	HIGH
6	initials	text	Creator initials or AUTO	BVS
7	idmode_id	integer	ID referencing idmodes.idmode_id	202501
8	added_timestamp	timestamp with tz	Creation timestamp	2025-05-22 10:00:00+00
9	num_images	integer	Total images in assignment	39000
10	comment	text	Assignment description notes	Primary Leg 1 assignment
11	status	text	Status (ready, in_review, done)	ready
12	tracknum	integer	Trackline sequence number	101
13	startimage	text	Starting image filename	202501.20250522.000001.png
14	stopimage	text	Ending image filename	202501.20250522.039000.png
15	subsample	text	Subsampling interval rule	1:10
16	date	text	Creation date (YYYY-MM-DD)	2025-05-22
17	time	text	Creation time (HH:MM:SS)	10:00:00

23.4.2.3 `img` (HabCam V4 Telemetry)

Stores physical sensor measurements, geospatial coordinates, and calculated optical field-of-view for every HabCam frame.

Col	Field Name	Data Type	Description	Example Value
1	`imagename`	`text`	Mapped PNG/TIF image filename	`202501.20250528.100444831.772211.png`
2	`lat`	`numeric`	Latitude in decimal degrees (NAD83)	`40.645128`
3	`lon`	`numeric`	Longitude in decimal degrees (NAD83)	`-72.065175`
4	`head`	`numeric`	Vehicle compass heading (degrees)	`322.77`
5	`pitch`	`numeric`	Vehicle pitch angle (degrees)	`4.83`
6	`roll`	`numeric`	Vehicle roll angle (degrees)	`-0.20`
7	`alt1`	`numeric`	Altitude off seabed in meters (Altimeter 1)	`2.22`
8	`alt2`	`numeric`	Altitude off seabed in meters (Altimeter 2)	`-99.99`
9	`vehicle_depth`	`numeric`	Water depth in meters (CTD pressure)	`47.44`

Col	Field Name	Data Type	Description	Example Value
10	s	numeric	Practical Salinity (PSS-78) from CTD	32.90
11	t	numeric	Water temperature (°C) from CTD	8.02
12	o2	numeric	Dissolved Oxygen (mg/L or raw counts)	5.92
13	cdom	numeric	CDOM fluorescence (RFU)	75.00
14	chlorophyll	numeric	Chlorophyll-a fluorescence (RFU)	64.00
15	backscatter	numeric	Optical backscatter reading	254.00
16	therm	integer	Raw thermistor voltage reading	549
17	internal_ph	numeric	SeaFET internal pH reading	8.05
18	external_ph	numeric	SeaFET external pH reading	8.07
19	timestamp	timestamp with tz	Capture timestamp (UTC)	2025-05-28 10:04:44.831+00
20	st_alt	numeric	Stereo-calculated altitude off bottom (m)	2.10

Col	Field Name	Data Type	Description	Example Value
21	`bottom_depth`	`numeric`	Ship fathometer seafloor depth (m)	`47.88`
22	`fov`	`numeric`	Calculated field-of-view width (m)	`1.02`
23	`fov_src`	`text`	Altitude source used for FOV (`alt1` or `st_alt`)	`st_alt`
24	`mm_px`	`numeric`	Image resolution (millimeters per pixel)	`0.55`

23.4.2.4 `img_lrauv` (LRAUV AUV Telemetry)

Dedicated telemetry schema for long-range autonomous underwater vehicle payloads (*Stella* and *Polaris*).

Col	Field Name	Data Type	Description	Example Value
1	`imagename`	`text`	LRAUV image filename	`LRAUV_Stella.20260601.120000.png`
2	`lat` / `lon`	`numeric`	Decimal degree coordinates (NAD83)	`39.81234`, `-73.12345`
3	`head` / `pitch` / `roll`	`numeric`	Vehicle 3D orientation (degrees)	`180.5`, `-1.2`, `-8.1`

Col	Field Name	Data Type	Description	Example Value
4	alt1 / pathfinder_alt	numeric	Altimeter / Pathfinder DVL altitude (m)	2.50, 2.51
5	vehicle_depth	numeric	LRAUV water depth (m)	52.3
6	o2 / cdom / chlorophyll	numeric	Calibrated environmental sensor metrics	5.24 mg/L, 4.79 RFU, 1.43 ug/L
7	dq_energy / dq_corr	numeric	Data quality energy & correlation metrics	0.98, 0.95
8	sea_water_salinity / temp	numeric	Calibrated salinity (PSS-78) & temp (°C)	32.85, 7.85
9	vehicle	text	Vehicle platform tag	LRAUV_Stella

23.5 5. Oracle Query Filtering (HABCAM_DATA_IDENTIFIER)

When querying the enterprise Oracle Cloud database (ORAPROD.HABCAM), queries **must filter by DATA_IDENTIFIER** to separate assessment-grade data from training, calibration, and historical RSA surveys.

DATA_IDENTIFIER	Operational Category & Assessment Status	Description & Usage Notes
1.0	**NEFSC V4 Scallop Assessment (QA/QCed)**	Official NOAA HabCam V4 survey data used in scallop stock assessments. Fully QA/QCed (2019–present).

DATA_IDENTIFIER	Operational Category & Assessment Status	Description & Usage Notes
1.1	**NEFSC V4 Invalid / Excluded**	Data NOT used in stock assessments (e.g., equipment malfunction or invalid trackline).
1.2	**NEFSC V4 Assessment Usable (QA/QCed)**	Supplemental survey data cleared for assessment use.
1.3	**NEFSC V4 Assessment Usable (Pending QA/QC)**	Valid survey data pending final QA/QC sign-off.
2.0	**Annotator Training**	Internal training assignments used for onboarding and calibrating annotators.
3.0	**Vehicle Calibration**	Optical, lightmap, or altimeter calibration runs.
4.0 – 4.2	**Special Projects**	Special research projects (e.g., 4.1 = 2022 Sand Dollar project; 4.2 = 2018 Testset selection).
6.0	**Historical RSA Data (No Metadata)**	Research Set-Aside (RSA) annotations from 2009, 2011, and 2012 without image metadata.
6.1	**RSA CFF HabCam V3 (2016)**	Coonamessett Farm Foundation V3 survey data used in scallop assessments.
6.2 – 6.3	**RSA Richard Taylor V2 (2012, 2014)**	Historical V2 HabCam survey data used in scallop assessments.
7.0	**NEFSC LRAUV Assessment Data (QA/QCed)**	Official LRAUV AUV survey data cleared and QA/QCed for stock assessment use.
7.1 – 7.3	**NEFSC LRAUV Secondary / Pending**	LRAUV AUV datasets categorized by assessment usability and QA/QC status.

Standard Stock Assessment Query Filter

For official stock assessment queries targeting HabCam V4 optical counts, filter for `DATA_IDENTIFIER = 1.0` (or `7.0` for LRAUV AUV data) in `HABCAM_ANNOTATIONS` and `HABCAM_IMAGE_METADATA`.

Chapter 24

NEFSC HabCam & AUV Data Query Pipeline

This pipeline extracts, cleans, and structures image annotations and associated telemetry metadata for NEFSC HabCam (V4, V3) and Tethys-Class LRAUV (*Stella* and *Polaris*) surveys from the NEFSC Oracle database (`ORAPROD`, `HABCAM` schema). The workflow filters out unusable images, applies environmental quality thresholds, isolates target taxa (such as live sea scallops or specific fish/invertebrate phyla), and generates standardized outputs for species analyses.

24.1 1. HABCAM Database Architecture & Reference Tables

24.1.1 1.1 Relational Table Architecture

The enterprise Oracle database (`ORAPROD.HABCAM`) consists of core relational tables used for tracking cruise assignments, image metadata, and species annotations.

Table Name	Years Available	Description & Usage Notes
`HABCAM_ANNOTATIONS`	2009 – Present	Master image annotation records. Includes manual, AI, and Research Set-Aside (RSA) records.
`HABCAM_ASSIGNMENTS`	2012 – Present	Web annotator assignment records and project goals (information incomplete prior to 2018).
`HABCAM_CLASSES`	2012 – Present	Species names and class ID lookup table (mapped via `SpeciesList_V6.csv`).
`HABCAM_DATA_IDENTIFIER`	2012 – Present	Defines gear, purpose, and QA/QC status codes for query filtering.
`HABCAM_ID_MODES`	2012 – Present	Target species lists enabled for specific annotation assignments.
`HABCAM_IMAGE_LIST`	2012 – Present	Image manifest lists assigned per annotation assignment.
`HABCAM_IMAGE_METADATA`	2012 – Present	Vehicle telemetry and sensor metadata for NEFSC HabCam V4.
`HABCAM_IMAGE_METADATA_LRAUV`	2021 – Present	Vehicle telemetry and sensor metadata for NEFSC Tethys LRAUVs (*Stella* / *Polaris*).
`HABCAM_IMAGE_METADATA_V3`	2017	Image metadata for CFF HabCam V3 surveys (contains no annotations).

24.1.2 1.2 Data Identifier Reference Table (DATA_IDENTIFIER)

The DATA_IDENTIFIER attribute is the primary quality-control filter used when querying HabCam and AUV records.

DATA_IDENTIFIER	Operational Category & Assessment Status	Description & Usage Notes
1.0	**NEFSC V4 Assessment (QA/QCed)**	Official HabCam V4 survey data used in scallop stock assessments; fully QA/QCed since 2019.
1.1	**NEFSC V4 Invalid / Excluded**	Data NOT used and should not be used in scallop assessments.
1.2	**NEFSC V4 Assessment Usable (QA/QCed)**	Supplemental survey data cleared for assessment use.
1.3	**NEFSC V4 Assessment Usable (Pending QA/QC)**	Valid survey data pending final QA/QC sign-off; use with caution.
2.0	**Annotator Training**	Internal training assignments used for onboarding annotators.
3.0	**Vehicle Calibration**	Vehicle, optical, or sensor calibration runs.
4.0 – 4.2	**Special Projects**	Special research projects (4.1 = 2022 Hollings Sand Dollar project; 4.2 = 2018 Testset selection).
5.0	**Unknown / Unverified**	Unverified annotation data (most likely wrong assignment numbers).
6.0	**Historical RSA (No Metadata)**	Historical RSA annotations (2009, 2011, 2012) without image metadata.
6.1	**RSA CFF V3 (2016)**	RSA CFF HabCam V3 2016 annotation data used in scallop assessments (no metadata).
6.2 – 6.4	**RSA Richard Taylor V2**	Historical V2 RSA annotation data (2012, 2014).

DATA_IDENTIFIER	Operational Category & Assessment Status	Description & Usage Notes
7.0	**NEFSC LRAUV Assessment Data (QA/QCed)**	Official LRAUV AUV survey data cleared and QA/QCed for stock assessment use.
7.1 – 7.3	**NEFSC LRAUV Secondary / Pending**	LRAUV AUV datasets categorized by assessment usability and QA/QC status (7.3 pending QA/QC).

24.2 2. Setup & Pipeline Prerequisites

24.2.1 2.1 Required R Packages

Ensure the following R libraries are installed and loaded prior to executing the pipeline:

```
install.packages(c("tidyverse", "dbplyr", "DBI", "ROracle", "stringr", "tictoc", "arrow"))
```

```
library(DBI)
library(ROracle)
library(dplyr)
library(dbplyr)
library(stringr)
```

24.2.2 2.2 Configure Query Parameters

Set the Oracle connection credentials, target survey year, taxonomic target, vehicle gear type, altitude cutoffs, and output preferences in the parameter block:

```
# 1. Oracle Credentials
ora_username <- "your_oracle_username"
ora_password <- "your_oracle_password"
```

```r
tns_alias    <- "NEFSC_pw_oraprod"

# 2. Target Year & Species Query Mode
year <- 2024  # Target survey year (some years include multiple cruises)
using_what_to_query <- c("class_id_list", "class_phylum_list")[2]

# Option A: Query by specific numeric class_ids
class_id_list <- c(258, 268)

# Option B: Query by broader taxonomic phylum/category
# Note: "Live Scallop" must be queried independently on its own
class_phylum_list <- c("Live Scallop", "Scallop", "Fish", "Roundfish")[3:4]

# Available phylum options in SpeciesList_V6.csv:
# "Live Scallop", "Scallop", "Fish", "Roundfish", "Flatfish", "Skate", "Algae",
# "Proteobacteria", "Chordata - Tunicate", "Invertebrate", "Invertebrate - Echinodermata",
# "Invertebrate - Crustacean", "Invertebrate - Cnidaria", "Invertebrate - Mollusca",
# "Invertebrate - Annelida", "Invertebrate - Porifera", "Invertebrate - Brachiopoda",
# "Invertebrate - Bryozoa", "Invertebrate - Ctenophora", "Invertebrate - Chelicerata",
# "Substrate", "Message", "Misc", "Man-made Object"

# 3. Vehicle Gear & Environmental Cutoffs
gear <- c("Habcam-V4", "AUV", "Habcam-V3")[1]

# Match DATA_IDENTIFIER list to selected gear:
# HabCam: c(1, 1.2, 1.3) | AUV: c(7, 7.2, 7.3)
data_identifier_list <- list(
  Habcam = c(1, 1.2, 1.3),
  AUV    = c(7, 7.2, 7.3)
)[[1]]

# Altitude Cutoff (m): <= 2.5m for HabCam V4; >= 3.0m for AUV (flies ~2.6m)
alt_cutoff <- 2.5

# 4. Input & Output Configuration
species_list <- read.csv("SpeciesList_V6.csv")
output_dir   <- "C:/Users/jui-han.chang/Desktop/"
target_label <- "live_scallop"
output_name  <- paste0(gear, "_", year, "_", target_label)
output_type  <- c("rds", "csv")[1]
```

24.3 3. Oracle Database Queries

24.3.1 3.1 Open Oracle Connection

Connects to the NEFSC Oracle database instance using `ROracle` and `DBI`:

```
connec <- tryCatch({
  conn <- dbConnect(
    DBI::dbDriver("Oracle"),
    username = ora_username,
    password = ora_password,
    dbname   = tns_alias
  )
  print("Database connected successfully!")
  conn
}, error = function(e) {
  message("Unable to connect to database.")
  message(paste("Error details:", e$message))
  return(NULL)
})
```

24.3.2 3.2 Execute Database Queries

Extracts cruise metadata, data identifiers, image telemetry, and annotations. Applies environmental and operational filters directly within the lazy query before downloading data to local memory (`collect()`):

```
if (is.null(connec)) {
  stop("Script halted: Could not connect to the database.")
} else {
  # 1. Query Master Cruise Details for target year
  cruise_id <- dbGetQuery(
    connec,
    paste0("
      SELECT
        a.CRUISE6 AS CRUISE_ID,
        a.YEAR,
        b.SEASON,
        b.STARTDATE,
        b.ENDDATE,
        d.PURPOSE,
        c.VESSEL_NAME,
```

```
      e.GEAR_DEFINITION
    FROM SVDBS.MASTER_CRUISE_DETAILS a
    INNER JOIN SVDBS.MASTER_CRUISE_TABLE b
      ON a.SVVESSEL = b.SVVESSEL AND a.VESSEL_NUM = b.VESSEL_NUM AND a.YEAR = b.YEAR AND a
    INNER JOIN SVDBS.SV_VESSEL c ON a.SVVESSEL = c.VESSEL_ABBREV
    INNER JOIN SVDBS.SVCRUISE_PURPOSE d ON a.PURPOSE_CODE = d.PURPOSE_CODE
    INNER JOIN SVDBS.SVGEAR e ON b.SVGEAR = e.SVGEAR
    WHERE (UPPER(e.GEAR_DEFINITION) LIKE '%HABCAM%' OR UPPER(e.GEAR_DEFINITION) LIKE '%AUV
      AND a.YEAR = ", year, "
    ORDER BY CRUISE_ID"
  )
)

# 2. Query Data Identifier Reference Table
data_identifier <- dbGetQuery(connec, "SELECT * FROM HABCAM.HABCAM_DATA_IDENTIFIER")

# 3. Lazy Connection to Annotation & Metadata Tables
ann_memory <- tbl(connec, in_schema("HABCAM", "HABCAM_ANNOTATIONS"))

if (gear == "Habcam-V4") {
  img_memory <- tbl(connec, in_schema("HABCAM", "HABCAM_IMAGE_METADATA"))
} else if (gear == "AUV") {
  img_memory <- tbl(connec, in_schema("HABCAM", "HABCAM_IMAGE_METADATA_LRAUV"))
} else if (gear == "Habcam-V3") {
  img_memory <- tbl(connec, in_schema("HABCAM", "HABCAM_IMAGE_METADATA_V3"))
} else {
  stop("Invalid gear selected.")
}

# 4. Filter Image Metadata (Lazy Query)
img_lazy <- img_memory |>
  filter(CRUISE_ID %like% paste0(!!year, "%")) |>
  filter(ALTIMETER_ALTITUDE_METER <= !!alt_cutoff) |>
  filter(FIELD_OF_VIEW_SQ_METER > 0) |>
  filter(SHIP_LATITUDE > 0) |>
  filter(SHIP_LONGITUDE >= -180) |>
  filter(CTD_VEHICLE_DEPTH_METER > 5) |>
  filter(abs(VEHICLE_PITCH_ANGLE) < 15) |>
  filter(abs(VEHICLE_ROLL_ANGLE) < 15)

# 5. Join Annotations to Filtered Image Telemetry & Download
start_time <- Sys.time()
ann <- ann_memory |>
  filter(CRUISE_ID %like% paste0(!!year, "%")) |>
  filter(DATA_IDENTIFIER %in% !!data_identifier_list) |>
```

```
    filter(ANNOTATION_DEPRECATED == 0) |>
    select(
      CRUISE_ID, DATA_IDENTIFIER, ASSIGNMENT_ID, IMAGE_NAME,
      ANNOTATION_TIMESTAMP, ANNOTATOR_USER_ID, GEOMETRY_TEXT, CLASS_ID
    ) |>
    mutate(GEOMETRY_TEXT = sql("CAST(GEOMETRY_TEXT AS VARCHAR2(4000))")) |>
    inner_join(img_lazy, by = c("IMAGE_NAME", "CRUISE_ID")) |>
    select(-HABCAM_PK) |>
    arrange(desc(IMAGE_NAME)) |>
    collect()
  end_time <- Sys.time()
  print(paste0("Finished annotation query. Downloaded ", nrow(ann), " rows."))

  # 6. Download Full Filtered Image Metadata
  start_time <- Sys.time()
  img <- img_lazy |>
    select(-HABCAM_PK) |>
    arrange(desc(IMAGE_NAME)) |>
    collect()
  end_time <- Sys.time()
  print(paste0("Finished image metadata query. Downloaded ", nrow(img), " rows."))

  # 7. Close Database Connection
  DBI::dbDisconnect(connec)
  print("Finished all queries; Oracle database connection closed successfully.")
}
```

24.4 4. Data Cleaning & Unusable Image Filtering

Images flagged as `"unusable"` during annotation (due to turbidity, dark frames, or deck frames) and all annotations associated with those images must be removed prior to analysis.

```
# Join downloaded annotations with master species list
ann <- ann |>
  left_join(species_list, by = "CLASS_ID")

# Calculate summary of total vs. unusable image and annotation counts
dashboard_ann_summary <- tibble(
  Category = c("Images", "Annotations"),
```

```
  Total    = c(n_distinct(ann$IMAGE_NAME), nrow(ann)),
  Unusable = c(
    n_distinct(ann$IMAGE_NAME[ann$CLASS_NAME == "unusable"]),
    sum(ann$IMAGE_NAME %in% ann$IMAGE_NAME[ann$CLASS_NAME == "unusable"])
  ),
  Usable   = Total - Unusable
)

knitr::kable(dashboard_ann_summary)

# Extract list of unusable image names and filter them out
images_to_remove <- ann |>
  filter(CLASS_NAME == "unusable") |>
  distinct(IMAGE_NAME)

ann_usable <- ann |>
  anti_join(images_to_remove, by = "IMAGE_NAME")

# Verify that usable counts match expected totals
dashboard_ann_summary |>
  mutate(
    ann_usable = c(n_distinct(ann_usable$IMAGE_NAME), nrow(ann_usable)),
    Match      = (Usable == ann_usable)
  ) |>
  knitr::kable()
```

24.5 5. Target Species Isolation Pipelines

The pipeline branches into two specialized processing tracks depending on whether the query targets **Live Sea Scallops** or **Other Taxonomic Groups**.

24.5.1 5.1 Track A: Live Sea Scallop Pipeline

Isolates live sea scallops, removes clappers/dead shell counts, and builds a zero-count image manifest (`ann_scallop_zero`) containing one record per usable image where no live scallops were observed.

```
# 1. Filter all scallop annotations
ann_scallop <- ann_usable |>
  filter(CLASS_CATEGORY_PHYLUM == "Scallop")
```

```r
# 2. Exclude dead/clapper classes
ann_scallop_live <- ann_scallop |>
  filter(!str_detect(CLASS_NAME, regex("clapper|dead", ignore_case = TRUE)))

# 3. Create zero-count image manifest (one row per image with no live scallops)
ann_scallop_zero <- ann_usable |>
  anti_join(ann_scallop_live, by = "IMAGE_NAME") |>
  distinct(IMAGE_NAME, .keep_all = TRUE) |>
  mutate(
    GEOMETRY_TEXT         = NA,
    CLASS_ID              = 1102,
    CLASS_NAME            = "Sediment Unassessed",
    CLASS_CATEGORY_PHYLUM = "Substrate"
  )

# 4. Verify that live_images + zero_images = total_usable_images
dashboard_ann_summary |>
  mutate(
    ann_scallop_live           = c(n_distinct(ann_scallop_live$IMAGE_NAME), nrow(ann_scallop
    ann_scallop_zero           = c(n_distinct(ann_scallop_zero$IMAGE_NAME), NA),
    ann_scallop_live_plus_zero = ann_scallop_live + ann_scallop_zero,
    Match                      = (Usable == ann_scallop_live_plus_zero)
  ) |>
  knitr::kable()

# 5. Compile output list
output_list <- list(
  cruise     = cruise_id,
  identifier = data_identifier,
  usable     = ann_usable,
  live       = ann_scallop_live,
  zero       = ann_scallop_zero,
  img        = img
)
```

24.5.2 5.2 Track B: Other Species Pipeline

Isolates target fish or invertebrate species (via `class_id_list` or `class_phylum_list`) and constructs a corresponding zero-count image manifest.

```
# 1. Filter target species annotations by Class ID or Phylum
if (using_what_to_query == "class_id_list") {
  ann_species <- ann_usable |>
    filter(CLASS_ID %in% class_id_list)
} else {
  ann_species <- ann_usable |>
    filter(CLASS_CATEGORY_PHYLUM %in% class_phylum_list)
}

# 2. Create zero-count image manifest
ann_species_zero <- ann_usable |>
  anti_join(ann_species, by = "IMAGE_NAME") |>
  distinct(IMAGE_NAME, .keep_all = TRUE) |>
  mutate(
    GEOMETRY_TEXT         = NA,
    CLASS_ID              = 1102,
    CLASS_NAME            = "Sediment Unassessed",
    CLASS_CATEGORY_PHYLUM = "Substrate"
  )

# 3. Verify image count alignment
dashboard_ann_summary |>
  mutate(
    ann_species           = c(n_distinct(ann_species$IMAGE_NAME), nrow(ann_species)),
    ann_species_zero       = c(n_distinct(ann_species_zero$IMAGE_NAME), NA),
    ann_species_plus_zero = ann_species + ann_species_zero,
    Match                 = (Usable == ann_species_plus_zero)
  ) |>
  knitr::kable()

# 4. Compile output list
output_list <- list(
  cruise     = cruise_id,
  identifier = data_identifier,
  usable     = ann_usable,
  species    = ann_species,
  zero       = ann_species_zero,
  img        = img
)
```

24.6 6. Output File Structure & Export

24.6.1 6.1 Pipeline Output Definitions

The pipeline generates structured R list objects (`output_list`) or exported `.csv` files.

Output Element	Description & Contents
`cruise`	Summary table of active cruise IDs, season, operational dates, vessel, and gear definitions.
`identifier`	Reference table of `HABCAM_DATA_IDENTIFIER` descriptions used in the query.
`img`	Telemetry and metadata for all collected images meeting core environmental criteria.
`ann`	Raw annotations matching image filtering criteria, joined with class names and phyla.
`usable` / `ann_usable`	Primary dataset of all annotations associated strictly with usable images.
`live` / `ann_scallop_live`	Live sea scallop annotations on usable images (excluding clappers and dead shells).
`species` / `ann_species`	Target species annotations on usable images for requested Class IDs or Phyla.
`zero` (`ann_scallop_zero` / `ann_species_zero`)	Manifest of usable images containing zero target species (one record per zero-count image).

24.6.2 6.2 Export Output Code

Saves finalized pipeline outputs as serialized `.rds` workspace files or flat `.csv` spreadsheets:

```
if (output_type == "rds") {
  saveRDS(
```

```
    output_list,
    file = file.path(output_dir, paste0(output_name, ".", output_type))
  )
}

if (output_type == "csv") {
  if ("Live Scallop" %in% class_phylum_list) {
    write_csv(
      ann_scallop_live,
      file = file.path(output_dir, paste0(output_name, "_ann_live.", output_type))
    )
    write_csv(
      ann_scallop_zero,
      file = file.path(output_dir, paste0(output_name, "_ann_zero.", output_type))
    )
  } else {
    write_csv(
      ann_species,
      file = file.path(output_dir, paste0(output_name, "_ann_species.", output_type))
    )
    write_csv(
      ann_species_zero,
      file = file.path(output_dir, paste0(output_name, "_ann_zero.", output_type))
    )
  }
}
```

Chapter 25

Optical Data Management

25.1 Purpose

This document outlines standard procedures for managing, storing, and analyzing optical data and related data products to ensure accuracy, consistency, and compliance with regulatory and organizational standards.

25.2 Scope

This Standard Operating Procedure (SOP) applies to all personnel involved in collecting, processing, and maintaining optical data across the **NOAA NEFSC HabCam** and **Tethys AUV** teams.

25.3 Key Definitions

Term	Definition
Raw Optical Data	Unprocessed imagery collected via optical instruments (e.g., machine vision or stereo cameras) and stored in RAW formats such as Grayscale TIFFs or Bayer-encoded files.

Term	Definition
Processed Optical Data	Optical images that have undergone stereo calibration, stereo rectification, debayering, color correction, lightmap calibration, or lightmap smoothing.
Annotated Optical Data	Optical images manually annotated using HabCam annotation software, with recorded data stored in HabCam relational databases.
Raw Data	Unprocessed oceanographic data directly collected from instruments on HabCam or Tethys stereo imaging payloads.
Image Metadata	Raw sensor streams processed into standardized formats for ingestion into HabCam relational databases and processing pipelines.
Relational Databases	PostgreSQL or Oracle database systems containing survey image metadata and annotated optical datasets.

25.4 Roles & Responsibilities

- **Technicians:** Execute live data collection and perform initial data validation.
- **Data Managers:** Oversee proper storage architecture, access permissions, and maintenance of optical data repositories and relational databases.
- **Quality Assurance Team:** Periodically review data accuracy and compliance across all operational phases—including calibration, lightmapping, processing, database ingestion, and manual annotations.

Chapter 26

Data Analysis

[Insert analytical R scripts, population modeling pipelines, swept-area calculations, and stock report workflows here]

Part V

Staging & Surveys

Chapter 27

Annual Operational Calendar

Overview of the 12-month lifecycle starting 6 months out from a Mid-May cruise departure

NOVEMBER ~6 Months Out Calibrations & Procurement / Confirm ship time; send CTDs/puck to WA for RMA calibration; order long-lead supplies; ship damaged gear for repair.	DECEMBER ~5 Months Out Inventory & Ordering / Track repair shipments, audit spare parts inventory, and finalize pre-cruise ordering lists.	JANUARY ~15 Weeks Out IT & Security Audit / Check in with ITD to ensure software security patching is applied to all field machines prior to surveys.	FEBRUARY ~12 Weeks Out Software & Workstations / Setup 2026 software directories/GUIs, install R packages, verify IP addresses, and load laptops.
MARCH ~10 to 8 Weeks Out DB & Bench Assembly / Clevis burnout; create Postgres DB; assemble vehicle; check fuses/O-rings; pull 16 inHg vacuum on main bottle.	APRIL ~6 to 4 Weeks Out Tank Tests & Staffing / WHOI tank calibration test; replace pinger batteries; send selection emails; build watch bills & assign berthing.	MAY Survey Month Staging & Departure / Mount weak link; hand off .GPX file to bridge; hold pre-cruise meeting; execute active field survey operations.	JUNE Post-Survey Data Offload & Breakdown / Offload primary raw image/sensor data, wash down vehicle, log maintenance, and store vehicle in Net Loft.
JULY Data Processing Data Verification / Verify complete offload of prior survey image databases and clear/reformat server backup drives.	AUGUST Maintenance Hardware Refurbishment / Inspect frame integrity, service mechanical connectors, and log spare parts consumption.	SEPTEMBER Planning Phase Ship Time Allocation / Confirm upcoming year survey dates with NOAA marine operations and review past cruise performance logs.	OCTOBER Pre-Season Prep RMA Preparation / Prepare Return to Manufacturer Authorization (RMA) paperwork for upcoming November calibration shipments.

Chapter 28

Pre-Cruise

28.1 Pre-Cruise Operational Checklist

28.1.1 6 Months Out

- ☐ Confirm ship time and set cruise date
- ☐ Send Seabird 37, Seabird 49 CTDs and Wetlabs ecotriplet puck to Washington for Calibrations [Return to Manufacturer Authorization (RMA) form]
- ☐ Begin ordering equipment and supplies needed before upcoming cruise [itemized ordering guide here]
- ☐ Send out any equipment that was damaged from last survey

28.1.2 15 Weeks Out

- ☐ If ITD hasn't reached out to have computer software security added to machines prior to surveys (i.e. patched), check-in

28.1.3 12 Weeks Out

- ☐ Set up all servers with 2026 software directories and GUIs
- ☐ Collect flashdrive from POPDY and Install R packages on annotation machines
- ☐ Verify that IP address can be changed on annotation machines
- ☐ Ensure sonar software and coastal explorer are installed on 2 field laptops

28.1.4 10 Weeks Out

- ☐ Hand-off Clevis to Shellfish group to have the epoxy burned out at Reidar's in New Bedford
- ☐ Ensure Seabird 37, Seabird 49 and Wetlabs ecotriplet puck have returned from Washington
- ☐ Ensure any damaged equipment sent for repairs has been returned
- ☐ Create a new HABCAM database on Postgres servers
- ☐ Confirm previous years data has been offloaded
- ☐ Clear server drives
- ☐ Reformat back-up drives

28.1.5 8 Weeks Out

- ☐ Prepare HABCAM for WHOI Sundance Tank Calibration Test
- ☐ Pack supply kit for WHOI sundance tank calibration test
- ☐ If applicable, install chanos onto tow sled
- ☐ Science center staffing call

28.1.6 6 Weeks Out

- ☐ Take vehicles to WHOI sea water tank test to calibrate equipment
- ☐ Repair or Resolve any issues that came up during the WHOI tank test
- ☐ Store Vehicle at netloft until staging
- ☐ Replace batteries in Pinger (Benthos UAT 376 acoustic transponder)

28.1.7 5 Weeks Out

- ☐ Send "You are scheduled Email" notifying staff they have been selected for the survey. Include:
  - ☐ Survey expectations
  - ☐ Medical clearance forms
  - ☐ Dietary restrictions
  - ☐ Packing list (i.e. foul weather gear)
  - ☐ Preparing for sea packet
- ☐ Notify watch chiefs that they will be sailing in leadership positions
- ☐ Create balanced watches with cruise personnel list
- ☐ Assign berthing

28.1.8 1 Week Out

- ☐ Email the scientific party a few days prior to departure letting them know about departure times, meeting points, parking, and weather contingencies.

28.1.9 Staging

- ☐ Mounting the weak link
- ☐ Splice fiber optics
- ☐ Do the pull test
- ☐ Load vehicle
- ☐ Load equipment
- ☐ Setup stations
- ☐ Provide the captain/bridge with the .GPX file of station locations so they can import it into their navigation software and check tracklines and make recommendations for any shifts in waypoints.
- ☐ Chanos sensor dissolved inorganic carbon sensor (PH, DIC, total alkalinity)

28.1.10 Day of Departure

- ☐ If leaving from WH store vehicle keys in the locker by the Survey refrigerator. Parking forms must be filled out and placed on the car dashboard.
- ☐ When heading to Newport, RI, keep the ship informed about the science party's arrival time.
- ☐ Ensure all members of scientific party are aboard
- ☐ Muster to bridge 30 min prior to departure time
- ☐ Inspect underwater cables for damage and/or voltage leaks.
- ☐ Host an on-board, pre-cruise meeting with scientific staff
- ☐ Track federal employee time

28.2 Detailed Staging & Survey Procedures

28.2.1 6 Months Out (~November)

- ☐ Confirm ship time and set cruise date
- ☐ Send Seabird 37, Seabird 49 CTDs, and Wetlabs ECO-Triplet puck to Washington for calibrations [Return to Manufacturer Authorization (RMA) form]
- ☐ Begin ordering equipment and supplies needed before upcoming cruise [itemized ordering guide here]
- ☐ Send out any equipment that was damaged from last survey

28.2.1.1 Confirm Ship Time & Cruise Date

[Insert details on contacting NOAA Marine Operations, confirming vessel availability, and setting firm survey departure/return windows]

28.2.1.2 CTD & ECO-Triplet Calibration Shipments (RMA)

[Insert RMA submission instructions, packaging requirements, Sea-Bird/WET Labs shipping addresses, and tracking protocols]

28.2.1.3 Equipment & Supply Procurement

[Insert itemized ordering guide, long-lead component lists, vendor contacts, and requisition procedures]

Quick Task:

- ☐ Send out any equipment that was damaged from last survey.

28.2.2 15 Weeks Out

- Check in with ITD regarding computer security patching if they haven't reached out

28.2.2.1 Patched

[Insert details explaining OS patch requirements, security software updates, firewall rules, and offline field machine clearance steps]

28.2.3 12 Weeks Out

- ☐ Set up all servers with software directories and GUIs
- ☐ Collect flashdrive from POPDY and Install R packages on annotation machines
- ☐ Verify that IP address can be changed on annotation machines
- ☐ Ensure sonar software and Coastal Explorer are installed on 2 field laptops

28.2.3.1 Server Directory & GUI Setup

[Insert server folder structure templates, database pathing, GUI executable placements, and directory permission settings]

28.2.3.2 POPDY Flash Drive & R Package Installation

[Insert steps for retrieving the POPDY flash drive, offline R package repository paths, dependency installations, and workstation scripts]

28.2.3.3 Annotation Machine IP Configuration

[Insert network adapter settings, static IP assignment procedures, subnets, and verification commands for annotation machines]

28.2.3.4 Field Laptop Sonar & Navigation Software

[Insert software installation steps for Coastal Explorer, BlueView sonar drivers, license key verifications, and laptop field profiles]

28.2.4 8 Weeks Out

- ☐ Prepare HABCAM for WHOI Sundance Tank Calibration Test
- ☐ Pack supply kit for WHOI sundance tank calibration test
- ☐ If applicable, install chanos onto tow sled
- ☐ Science center staffing call

28.2.4.1 Prepare HabCam for WHOI Sundance Tank Calibration Test

- ☐ Attach/Install CTDs, ecotriplet puck, and Blueview forward looking sonar
- ☐ Open main electronics bottle, check fuses, inspect O-rings, replace desiccant and pull a vacuum
- ☐ After 24 hours, confirm PSI has remained stable: Pressure holds at 16 inHg. If leak or change in pressure is detected, likely to reinspect O-rings.
- ☐ Pull the camera plate and inspect O-rings in both housings. Refocus camera lenses if necessary.
- ☐ Attach altimeter and attitude sensors onto camera plate. Re-attach camera plate to HABCAM.
- ☐ Inspect underwater cables for damage and/or voltage leaks
- ☐ Regrease Connector Faces & Plug Into Sensors
- ☐ Power On Main Electronics Bottle & Verify Engineering GUI Data
- ☐ Connect to Stereocameras & Verify Camera/Strobe Trigger Rates

28.2.4.2 Pack Supply Kit for WHOI Sundance Tank Calibration Test

[Insert packing list, required hand tools, spare O-rings, calibration targets, and test gear inventory here]

28.2.4.3 Install Chanos Sensor onto Tow Sled (If Applicable)

[Insert mechanical mounting, bracket alignment, cable harness routing, and power/data integration steps for the Chanos sensor]

28.2.4.4 Science Center Staffing Call

[Insert staffing call protocol, participant application deadlines, skill set requirements, and roster collection guidelines]

28.2.5 6 Weeks Out:

- ☐ Take vehicles to WHOI sea water tank test to calibrate equipment
- ☐ Repair or Resolve any issues that came up during the WHOI tank test
- ☐ Store Vehicle at netloft until staging
- ☐ Pinger: Benthos UAT 376 acoustic transponder - replace batteries

28.2.6 Calibration

28.2.6.1 Reserving the Tank

[Insert reservation procedures, point of contact at WHOI, scheduling lead times, and facility access requirements here]

28.2.6.2 Transporting the Vehicle

[Insert vehicle transport logistics, truck/trailer loading steps, strapping protocols, and staging at the Sundance Tank here]

28.2.6.3 Setting Up Calibration

[Insert tank placement, water fills, sensor submersion setup, power hookups, and initial system checks here]

28.2.6.4 Collecting Calibration Data

1. Checkerboard Target Data Collection:

- **Common Area Positioning:** Ensure the entire checkerboard is placed in the overlap zone (common area) between the left and right cameras' fields of view so it appears fully on both cameras.

Checkerboard Positioned Within the Overlapping Common Viewing Area

- **Grid Coverage:** Divide the common viewing area into 9 equal regions and take photos in all 9 regions to ensure full spatial coverage.

9-Region Division Grid Across Common Field of View

- **Glare & Tilting:** Avoid direct strobe glare on the board (glare images will be rejected by estimation software). Tilting the board slightly is preferred over lying flat to minimize glare.
- **Altitude & Angle Sampling:** Capture images across 2–3 different vehicle altitudes and multiple angles (both flat and tilted).

Figure 1: Acceptable Checkerboard Sample Images Across Angles and Altitudes from the 2015 Calibration (Best)

2. Scallop Target Data Collection:

- **Placement:** Lay the scallop target board flat on the bottom of the dunk tank for accurate altimeter reference. Lower and raise the vehicle using the overhead crane.

Physical Scallop Calibration Target Board with Known Shell Heights

- **Field-of-View Distribution:** Divide the full field of view into 9 regions. Capture images with the target in 5 regions of the left field of view and 2 regions at the far right of the right field of view (or 5 right / 2 far left).

/ 5 Regions Left FOV / 2 Regions Far Right	/ 5 Regions Right FOV / 2 Regions Far Left

- **Varied Sampling:** Take photos at 2–3 distinct altitudes and varying tilt angles.

3. Altitude Logging: * Ensure the Engineering GUI is actively running and logging live altimeter depth/altitude data during all test passes.

28.2.6.5 Evaluation Criteria & Processing Iterations

Criteria for a Successful Calibration:

1. **Visual Integrity:** Processed images must be undistorted with minimal black edge bands (black border pixels must represent $< 5\%$ of total image pixels).

/ Bad Calibration: Heavy Distortion & Black Edge Bands (>10%)	/ Good Calibration: Minimal Edge Distortion (

2. **Stereo Coverage Rate:** Successful stereo altitude estimates must be generated for $> 95\%$ of total captured images.
3. **Measurement Accuracy:** Estimated scallop shell lengths (calculated via stereo altitude) must fall within $\pm 3\%$ of true physical measurements.

Step-by-Step Parameter Estimation Workflow: 1. Estimate initial calibration parameters from selected checkerboard images. 2. Process all scallop target images using the candidate calibration parameters. 3. Inspect processed output images visually for optical distortion. 4. Calculate the percentage of black edge pixels ($< 5\%$ threshold). 5. Determine the percentage of total images receiving valid stereo altitude estimates. 6. Annotate the scallop target board images in numbered sequence (1 through 9).

Required Numbered Sequence (1–9) for Scallop Target Board Annotations

7. Calculate average percentage bias in estimated scallop shell height compared to ground truth.
8. Iterate image selection and parameter fitting until all three success criteria are satisfied.

28.2.7 The 2018 Experiment

28.2.7.1 Rationale

The paired tows of NEFSC Habcam, NEFSC dredge, and VIMS dredge were conducted at the same time at 19 overlapping stations in Elephant Trunk (ET) in 2018 (Figure 1). These pair tows were designed to study the dredge filling issue at high density area and the difference between 10 minutes and 15 minutes tows.

Figure 1. The paired tow locations for NEFSC Habcam, NEFSC dredge, and VIMS dredge survey, along with the locations for the entire VIMS dredge and WHOI Habcam survey in ET (the right gray polygon is ET-Open and the left gray polygon is ET-Flex) in 2018.

However, a problem for the NEFSC Habcam data emerged when analyzing the paired tow data - the mean scallop shell height collected by NEFSC Habcam estimated using the altimeter altitudes (122mm in ET-Open and 119mm in ET-Flex) was 13-16% higher than the mean scallop shell height collected by the NEFSC dredge (108mm in ET-Open and 103mm in ET-Flex; Figure 2). The shell height distributions from the NEFSC dredge and Habcam survey should be nearly identical since they were surveying overlapping locations and there is only one dominant cohort in ET.

The WHOI Habcam data have a similar problem compared to the VIMS dredge data, the mean shell height estimated using stereo altitudes was 6-8% higher (Figure 2). Even though the two surveys were not completely overlapped (Figure 1), a 6-8% difference in mean shell height is not expected in ET. If the altitudes from the Habcam data are accurate, the comparison of dredge and Habcam data in areas with only a single cohort are expected to be similar to Figure 3.

Figure 2. Shell height frequencies, the number of shells measured, and mean shell height from multiple surveys in ET in 2018.

Figure 3. Shell height frequencies, the number of shells measured, and mean shell height from VIMS dredge and CFF Habcam survey (altimeter altitude) in the Nantucket Light Ship South Deep area in 2018.

Because the NEFSC Habcam altimeter was not collecting accurate altitudes, the usefulness of stereo altitudes was explored. However, of 119,645 NEFSC Habcam images annotated in 2018, only 3,627 images have stereo altitudes. The estimated stereo altitudes are either close to the altimeter altitudes or centered at 0.93 meters, which means that the stereo altitudes were estimated poorly and also biased (Figure 4).

Figure 4. Comparison of altimeter and stereo altitudes for NEFSC Habcam data (annotated images) in 2018. Red line is a one to one line.

The comparison between Habcam and dredge survey shell heights were further examined from 2011 to 2018 for all areas surveyed to see if this is a single year problem or it persists over years. The list of the type of altitudes (and field of view calculations) used to estimate scallop shell heights for the Habcam data for each year is below (Burton Shank personal comm.):

- 2019 stereo altitudes only.
- 2018 sonar altitudes only.
- 2017 sonar altitudes only; 386 stereo altitudes in database.
- 2016 stereo altitudes for 36,844 images of 97,266 images; sonar for the rest.

- 2015 sonar altitudes only; no direct FOV calculation, had to use altitude-based second-order polygon model; should fix this in the future.
- 2014 sonar altitudes only. Direct FOV calculation for all except 723 images with FOV filled in with polygon model.
- 2011 to 2013 working on this.

The problem of altitudes for the NEFSC Habcam started in 2017. There are no accurate altitudes recorded or estimated for 2017 and 2018 NEFSC Habcam data. The problem is still not resolved to this date. This forced us to use stereo altitudes for the 2019 data. However, even though the bias of shell heights calculated using stereo altitudes are less than 5% in 2019, it is still not as good as the altimeter altitudes if the altimeter is working properly (Figure 3) because the stereo altitudes are estimates and estimates have errors.

28.2.7.2 Goals

1. Find a method to examine the accuracy of the stereo altitudes estimated from the calibration - at least one set of altitude needs to be accurate, if not both.
2. Identify a method to correct the 2017 and 2018 NEFSC Habcam data.

28.2.7.3 Experiment Design

1. Three targets were made to explore the accuracy of the stereo altitudes, one with scallop shells and two with plastic disks to represent smaller scallops shell heights.

2. Images of the 3 targets were taken during the 2018 post-cruise calibration at 2 different altitudes and 5 different locations of the left field of view.

3. For each target, 2 images at each location and altitude were processed and annotated, resulting in a total of 64 images analyzed.
4. The 64 selected target images were processed using the calibration estimated using the checkerboard images taken during the experiment (2018 post-cruise calibration, 20181102), as well as calibrations from the past:
    - 2018 post-cruise calibration (20181102)
    - 2018 pre-cruise calibration (20180507)
    - 2018 during-cruise calibration (20180527)
    - 2017 pre-cruise calibration (20170501)
    - 2015 post-cruise calibration (20150801)
    - 2015 pre-cruise calibration (20150501)
5. Each of the 64 target images was processed 6 times (using the above 6 calibrations). All of these processed images were annotated by two annotators.

6. The scallop shell heights (or disk sizes) estimated using the stereo altitudes were compared to their true sizes and percent biases were calculated using the following equation:

$$\%\text{Bias} = \frac{\text{Estimated Size} - \text{True Size}}{\text{True Size}} \times 100\%$$

7. The percent biases were compared by calibrations, annotators, scallop or disk sizes, 3 targets, 2 altitudes, and 5 locations in the field of view.
8. Once the best calibration is identified, biased measurements from other calibrations were reestimated using the stereo altitudes from the best calibration to check if biased measurements can simply be corrected by reprocessing.
9. To test the sensitivity of checkerboard images, 200 calibrations were estimated from randomly selected checkerboard images iteratively 200 times.

28.2.7.4 Results

1. Below is the summary table of the number of images that have stereo altitude estimated:

Calibration	# of images with stereo estimates	Percent with stereo estimates
2018 post-cruise (20181102)	64	100%
2018 pre-cruise (20180507)	14	22%
2018 during-cruise (20180527)	41	64%
2017 pre-cruise (20170501)	25	39%
2015 post-cruise (20150801)	61	95%
2015 pre-cruise (20150501)	53	83%

2. There are significant differences in percent biases by calibration. Mean bias ranged from +14% (2018 pre-cruise) down to -2% (2015 post-cruise; Figure 5).

Figure 5. Density plot of percent biases from all measurements (gray bars) and measurements by calibrations. Dash lines are the mean percent bias by calibrations.

3. The best calibration is the 2015 post-cruise calibration (20150801) - it estimated 95% of stereo altitudes with a mean bias of -2%. Reprocessing reduced mean percent biases to within ±3% (Figure 6).

Figure 6. Density plot of percent biases reestimated using stereo altitudes from the 2015 post-cruise calibration (20150801).

4. Target locations in the field of view caused slight differences in bias (e.g., top-right corner was slightly more biased; Figure 7).

Figure 7. Density plot of percent biases by target locations in FOV and altitudes (1.92m and 2.41m).

5. The 200 calibrations estimated from randomly selected checkerboard images show significant variance in performance, ranging between ±20% bias (Figure 8).

Figure 8. Density plot of percent biases across 200 random checkerboard calibration iterations.

28.2.7.5 Conclusions

1. Calibration performance varies dramatically; thorough validation is required before official deployment.
2. The 2015 post-cruise calibration (20150801) is the best performing calibration tested.
3. Calibration accuracy is sensitive to the specific checkerboard images chosen rather than minor camera hardware changes over time.
4. Biased measurements from bad calibrations can be corrected by reprocessing images with a validated calibration model without needing full re-annotation.

28.2.8 5 Weeks Out

- ☐ Send "You are scheduled Email" notifying staff they have been selected for the survey. Include:
  - ☐ Survey expectations
  - ☐ Medical clearance forms

- ☐ Dietary restrictions
- ☐ Packing list (i.e. foul weather gear)
- ☐ Preparing for sea packet

☐ Notify watch chiefs that they will be sailing in leadership positions
☐ Create balanced watches with cruise personnel list
☐ Assign berthing

28.2.8.1 Send "You Are Scheduled" Email & Participant Packet

Send formal notification emails to all selected scientific personnel. Attach the complete participant packet containing required forms and guidelines.

28.2.8.1.1 1. Survey Expectations

[Insert survey expectations overview: 12-hour watch shifts, zero tolerance policies, data logging protocols, deck safety, and operational conduct]

> **Example Survey Expectations Summary:** * Scientific staff stand 12-hour daily watch rotations (6-on / 6-off or 12-on / 12-off). * Active participation in gear deployment, image annotation quality control, and deck operations is required. * Strict adherence to NOAA safety guidelines, including mandatory hard hats and safety vests during deck work.

28.2.8.1.2 2. Medical Clearance Forms

[Insert instructions for completing NOAA Health Services Form NHS 56-29, submission deadlines, and regional medical officer contact info]

Medical Clearance Form Checklist: * NOAA Form 56-29 (Report of Physical Examination) * Tuberculosis (TB) Screening / PPD Test Verification * Current prescription medication disclosures

28.2.8.1.3 3. Dietary Restrictions Form

Participant Name	Dietary Requirement (Vegetarian, Vegan, Gluten-Free, etc.)	Severe Food Allergies (Nuts, Shellfish, Dairy, etc.)
[Participant Name]	[List Dietary Restrictions]	[List Medical Allergies / Anaphylaxis Risk]

28.2.8.1.4 4. Packing List (Foul Weather Gear)

Essential Packing List: * **Personal Protective Equipment (PPE):** Steel-toe boots (ANSI rated), commercial foul weather gear (bibs and jacket). * **Work Attire:** Layered thermal clothing, non-slip deck shoes, work gloves, wool socks. * **Personal Items:** Government-issued ID/passport, personal prescription medications (1.5x cruise duration supply), toiletries, bed linen/towel.

28.2.8.1.5 5. Full "Preparing for Sea" Packet

[Insert full text/download link for the official NOAA / HabCam "Preparing for Sea" orientation packet]

28.2.8.2 Create Balanced Watch Bills

Watch Team	Shift Hours	Watch Chief	Lead Annotator	Watch Standers
Watch A (Day)	0000–0600 / 1200–1800	[Name]	[Name]	[Name], [Name], [Name]
Watch B (Night)	0600–1200 / 1800–2400	[Name]	[Name]	[Name], [Name], [Name]

28.2.8.3 Assign Berthing

Stateroom #	Deck / Location	Berth Type	Assigned Watch	Occupants
Room 101	Main Deck / Port	Double Bunk	Watch A (Day)	1. [Name] / 2. [Name]
Room 102	Main Deck / Port	Double Bunk	Watch B (Night)	1. [Name] / 2. [Name]
Room 103	01 Deck / Starboard	Double Bunk	Watch A (Day)	1. [Name] / 2. [Name]
Room 104	01 Deck / Starboard	Double Bunk	Watch B (Night)	1. [Name] / 2. [Name]

Quick Task:

- ☐ Notify watch chiefs that they will be sailing in leadership positions.

28.2.9 1 Week Out

- ☐ Email the scientific party a few days prior to departure regarding departure times, meeting points, parking, and bad weather contingencies.

28.2.9.1 Pre-Departure Logistics Email

28.2.9.1.1 1. Departure Times & Meeting Points

Example Meeting & Arrival Details: * **Vessel Location:** Pier 2, NOAA Marine Operations Dock / WHOI Facility, Woods Hole, MA. * **Mandatory Boarding Time:** 0700 EDT on Departure Day (Vessel departs promptly at 0900 EDT). * **Pre-Sailing Assembly Point:** Main pier loading bay adjacent to the survey gear staging area.

28.2.9.1.2 2. Parking Procedures

Example Parking Instructions: * All long-term vehicles must be parked in designated Scientist Lot C. * Complete the NOAA/WHOI temporary parking slip and place it face-up on the driver's side dashboard. * Leave spare vehicle keys in the designated key locker near the Survey Refrigerator prior to boarding.

28.2.9.1.3 3. Weather Protocols & Operational Limits

Example Operational Weather Thresholds: * **Max Sailing Wind (Transit Limit):** Sustained winds $>$ 35 knots or seas $>$ 10 ft trigger a departure delay assessment with the Captain. * **Max Operational Wind (HabCam Deployment Limit):** Sustained winds $>$ 25 knots or seas $>$ 6 ft require halting vehicle towing operations and recovering the sled. * **Weather Contingency Communication:** In the event of forecasted adverse weather, updates will be issued via email/SMS every 6 hours by the Chief Scientist.

28.2.10 Staging Protocols

- ☐ Mounting the weak link
- ☐ Splice fiber optics
- ☐ Do the pull test
- ☐ Load vehicle
- ☐ Load equipment

- ☐ Setup stations
- ☐ Provide the captain/bridge with the .GPX file of station locations so they can import it into their navigation software and check tracklines and make recommendations for any shifts in waypoints.
- ☐ Chanos sensor dissolved inorganic carbon sensor (PH, DIC, total alkalinity)

28.2.10.1 Mounting the Weak Link

Refer to 02 Weak Link Assembly for detailed instructions.

28.2.10.2 Splice Fiber Optics

Refer to 02 Fiber Termination for detailed instructions.

28.2.10.3 Do the Pull Test

Refer to 02 Fiber Pull Test Protocol for detailed instructions.

28.2.10.4 Load Vehicle

[Insert details for vehicle deck loading, crane hoisting protocols, tie-down locations, and secure deck placement here]

28.2.10.5 Load Equipment

[Insert details for loading auxiliary crates, spare parts, toolboxes, calibration targets, and laboratory hardware here]

28.2.10.6 Setup Stations

HabCam operations aboard the R/V *Henry B. Bigelow* are distributed across four main laboratory spaces: the Acoustic Lab (Primary Control Center), Dry Lab, Chemistry Lab, and Cutting Room.

28.2.10.6.1 1. Acoustic Lab Command Center Overview

The Acoustic Lab serves as the primary HabCam flight and monitoring hub. It houses 4 distinct stations operated by NOAA Fisheries personnel and Ship Survey Technicians:

- **Watch Chief Station:** Located in the rear row with a 2-monitor setup directly behind the flight team.
- **Co-Pilot Station:** Located in the front row (left) with a dedicated 4-monitor array.
- **Pilot Station:** Located in the front row (right) with a 4-monitor array, vehicle joystick, and winch controller.
- **Survey Technician Station:** Located adjacent to the flight team to monitor shipboard multibeam, EK-80, and vessel navigation.

Acoustic Lab Wall of Monitors & Operator Workstations Overview

28.2.10.6.2 2. Watch Chief Station

Operated on 12-hour rotating shifts (2 Watch Chiefs per cruise) to maintain continuous leadership, maintenance oversight, and quality control.

- **Hardware Setup:** 2-monitor workstation positioned behind the Pilot/Co-Pilot.
- **Primary Responsibilities:** Watch roster assignment, training new scientists on flight operations, reviewing image quality, and beginning preliminary image annotations.

Watch Chief Workstation (Positioned Behind Pilot/Co-Pilot)

28.2.10.6.3 3. Co-Pilot Station

Positions the Co-Pilot to the left of the Pilot to continuously monitor imaging telemetry, software health, and incoming seabed features.

Co-Pilot Workstation Array

Monitor #	Application / Display	Operational Function & Image Placement
Monitor 1	EK-80 Scientific Echo Sounder	Mirrored feed from Survey Techs displaying water column echograms, seafloor depth, and early warnings for incoming sand waves. / /

Monitor #	Application / Display	Operational Function & Image Placement
Monitor 2	Engineering GUI & Event Logger	Live raw sensor feeds (AHRS, CTD, Oxygen, Fluorometer), GPS coordinates, fathometer depth, active file logging status, and event tagging. / /
Monitor 3	Coastal Explorer Navigation	Mirrored navigation software displaying planned tracklines, waypoints, vessel speed, and potential subsea hazards/wrecks. / /
Monitor 4	Image Viewer (dsvimview)	Live stereo camera image waterfall; allows toggling port/starboard feeds, adjusting display rates, and checking camera trigger health. / /

28.2.10.6.4 4. Pilot Station

Positions the Pilot to the right of the Co-Pilot with direct access to the vehicle control joystick and hydraulic winch controls.

Pilot Workstation Overview & Controls

Monitor #	Application / Display	Operational Function & Image Placement
Monitor 5	Live Winch & Deck Video (VxOpsCenter)	CCTV camera feeds monitoring the oceanographic winch drum, wire spooling, and aft deck operations. / /
Monitor 6	Bottom Tracking (dsvimview / Dixon Screen)	Real-time vehicle altitude, computed bottom profile, fish depth, and altimeter filtering for flight adjustments. / /

Monitor #	Application / Display	Operational Function & Image Placement
Monitor 7	BlueView Forward Sonar	Acoustic sector sweep displaying subsea obstacles, boulders, and terrain relief ahead of the vehicle. / /
Monitor 8	Pentagon Winch Control Display	MacGregor / Rapp Marine interface displaying cable tension (tons), line speed (m/min), and paid-out cable length (m). / /

28.2.10.6.5 5. Annotation & Auxiliary Laboratory Stations

Image processing, secondary scientific analysis, and physical sample handling are distributed across three auxiliary spaces:

- **Dry Lab:** Contains Annotation Station 1, Annotation Station 2, the Backup Annotation Station, and the primary timeserver rack.
- **Chemistry Lab:** Contains Annotation Station 3, Annotation Station 4, chemistry fume hood, and science freezers.
- **Cutting Room:** Dedicated HabCam workspace (Cutting Table 1) for physical biological sampling, sorting belt operations, and sample freezer access.

28.2.10.7 Bridge .GPX File & Station Transects

[Insert details on assigning survey transects, saving/exporting the .GPX file, delivering tracklines to the captain/bridge, and protocols for modifying waypoints based on captain's feedback here]

28.2.10.8 Chanos Sensor (pH, DIC, Total Alkalinity)

[Insert details for Chanos sensor mounting, reagent preparation, fluidic connections, and calibration protocols here]

28.2.11 Day of Departure Protocols

- ☐ If leaving from WH store vehicle keys in the locker by the Survey refrigerator. Parking forms must be filled out and placed on the car dashboard.
- ☐ When heading to Newport, RI, keep the ship informed about the science party's arrival time.
- ☐ Ensure all members of scientific party are aboard
- ☐ Muster to bridge 30 min prior to departure time
- ☐ Inspect underwater cables for damage and/or voltage leaks.
- ☐ Host an on-board, pre-cruise meeting with scientific staff
- ☐ Track federal employee time

28.2.11.1 Host On-Board Pre-Cruise Meeting

Host the mandatory pre-cruise orientation meeting with all scientific staff prior to departure to establish shift leadership, review safety expectations, and outline initial survey logistics.

- Perform Introductions; Thank everyone for participating
- **Identify Watch Chief for first shift; they must be listened to concerning FSCS procedures**
- Announce area of operation, general cruise direction, and ETA to first station, and sequence of events at station
- Emphasize importance of 'Safety First' (i.e. hard hats and safety vests)
- Go over protocols for injuries
- Relieve other watch 10 mins early
- Remind Scientific Staff to notify Watch Chief and Chef of any food allergies
- Ensure watch times on day of departure/arrival are divided equally for pay purposes

28.2.11.1.1 Sequence of Events at Station

[Insert detailed sequence of events at station here, including vehicle deployment steps, logging initiation, recovery protocols, and watch handover procedures]

28.2.11.1.2 Safety Protocols

[Insert detailed safety protocols here, including mandatory personal protective equipment (PPE) requirements such as hard hats, safety vests, steel-toe boots, and deck safety rules]

28.2.11.1.3 Injury Protocols

[Insert detailed injury protocols here, including immediate medical response steps, notifying the Chief Scientist and Vessel Medical Officer, and completing official incident documentation]

28.2.11.1.4 Track Federal Employee Time

[Insert detailed instructions on how to track federal employee time here, including daily timekeeping accounting, agency codes, sea pay/overtime rules, and pay period submission protocols]

Chapter 29

Field Team Roles Table

Cruise Manning Structure & Shift Operations: The HabCam scientific party typically consists of 9 to 10 personnel operating on 24-hour continuous sea trials divided into two 12-hour watch rotations (e.g., 11:30 AM to 11:30 PM and 11:30 PM to 11:30 AM). Each shift is managed by a Watch Chief and includes a rotating team of Pilots, Co-Pilots, and Annotators who cycle stations every 30 to 60 minutes to prevent screen fatigue. The Chief Scientist maintains active oversight across both watches (6 AM to 6 PM baseline plus as-needed response).

Role	Manning / Shift	Primary High-Level Responsibilities
Chief Scientist	1 Person (Floating / Day)	Primary liaison between scientific party, vessel crew, and shore support; leads survey trackline design, staffing, daily safety briefings, and on-the-spot decision making.
HabCam Lead	1 Person (Technical Lead)	Technical authority for hardware, fiberoptic terminations, server configuration, GPS time syncing, navigation integration, light maps, and data redundancy.

Role	Manning / Shift	Primary High-Level Responsibilities
Watch Chief	2 People (1 per 12-hr shift)	Direct shift supervisor; manages station role rotations, controls core flight software startup/shutdown, trains new scientists, and conducts shift QA/QC.
Pilot	Rotating (30–60 min blocks)	Actively flies vehicle via hydraulic winch joystick; maintains target altitude above seafloor using live camera feeds, flight GUI, and forward-looking sonar.
Co-Pilot	Rotating (30–60 min blocks)	Second pair of eyes for Pilot; monitors dual camera feeds, telemetry, and sonar for sand waves/boulders; optimizes camera/sonar settings; mitigates software glitches.
Annotator	Rotating (30–60 min blocks)	Frontline image processor; logs primary targets (scallops/fish) and secondary substrate/habitat data from incoming 10-minute image batches.
QA/QC Lead	1 Person / PopDyn	Hosts pre-cruise training, maintains identification guides, instructs scientists on protocols, and audits/certifies ~30 baseline images per annotator.

Role	Manning / Shift	Primary High-Level Responsibilities
Population Dynamics	1–2 People	Quantitative scientist; collaborates on trackline design, executes camera calibrations, verifies altimeter confidence, and synthesizes population data.

29.1 Field Team Roles Overview

29.1.1 Chief Scientist

Acts as the primary liaison between the scientific party, the vessel's crew, and shore-based support. Responsible for overall cruise execution, safety, trackline planning, and maintaining an open-door policy.

- **Submit Ship Time Requests:** Draft and submit formal administrative requests to secure vessel availability aligned with scientific objectives.
- **Plan the Survey and Tracklines:** Lead overarching scientific design, mapping out geographical scope and adjusting routes dynamically for weather or bathymetry.
- **Personnel Selection and Staffing:** Recruit and assemble a balanced scientific team with the necessary technical and watchstanding expertise.
- **Cruise Plan Updates:** Finalize operational schedules and delegate updates to co-leads for uninterrupted operational continuity.
- **Inter-Crew Communication & Captain Liaison:** Facilitate daily communication with the Captain and ship's command regarding navigation, weather, and safety.
- **Brief Watch Chiefs & Shift Transitions:** Proactively update shift leaders on scientific priorities, altered tracklines, and operational changes prior to watch changes.
- **Active Oversight & Daily Safety Briefings:** Maintain active presence for at least half of every watch shift and lead mandatory daily safety meetings at sea.

29.1.2 HabCam Lead

The primary technical authority responsible for vehicle hardware deployment, data integrity, and subsea/shipboard IT infrastructure.

- **Execute Fiberoptic Termination:** Perform precise fiberoptic splicing and clevis termination through the hydraulic winch line to guarantee low-loss telemetry.
- **Configure IT Infrastructure & Servers:** Set up main HabCam servers, control stations, annotation computers, and local network switches.
- **Install GPS Time Server & Navigation Integration:** Configure master GPS time synchronization and link shipboard GPS/fathometer feeds to vehicle logging systems.
- **Server Maintenance & Data Monitoring:** Verify storage drive mounting, Apache web server status, and real-time raw image/sensor data writes.
- **Build Light Maps & Oversee Image Pipeline:** Generate illumination correction maps and verify processed images serve smoothly to annotation workstations.
- **Metadata QA/QC & Continuous Data Backups:** Verify PostgreSQL database integrity and execute automated redundant backups onto spare drives.
- **Vehicle Pre-Dive Checks & Troubleshooting:** Lead pre-deployment deck inspections and rapidly resolve electrical, mechanical, or software faults.

29.1.3 Watch Chief

Direct shift supervisor managing personnel, flight software operations, and biological data collection during assigned 12-hour rotations.

- **Manage 12-Hour Operational Shifts:** Command continuous watch shifts and execute thorough handovers with the counterpart Watch Chief.
- **Assign Station Rotations:** Create and enforce station schedules (Pilot, Co-Pilot, Annotator) rotating personnel every 30–60 minutes to manage fatigue.
- **Control Flight Software Systems:** Safely launch, monitor, and shut down pilot and co-pilot control software and logging interfaces.
- **Train & Onboard Scientific Staff:** Conduct standardized training for new annotators and flight standers using reference slide decks.
- **Integrate into Active Rotation:** Lead by example by taking regular turns at pilot, co-pilot, and annotation stations.
- **Shift Data QA/QC:** Review logged species identifications and habitat classifications to correct entry errors before shift sign-off.

29.1.4 Pilot

Responsible for the active, real-time underwater navigation and altitude control of the HabCam vehicle above the seabed.

- **Execute Timed Rotations (30–60 Minutes):** Operate in short, high-concentration blocks to maintain sharp reflexes and situational awareness.
- **Maintain Safe Seafloor Altitude:** Continuously adjust winch controls to keep the vehicle at the target photographic altitude (typically ~2m) above bottom.
- **Operate Winch Joystick:** Command vertical vehicle movement using the physical winch joystick to pay out or haul in tow cable.
- **Monitor Telemetry & Forward Sonar:** Synthesize live camera video, flight GUI metrics (pitch, roll, tension), and forward-looking sonar to detect and avoid subsea hazards (boulders, wrecks, drop-offs).

29.1.5 Co-Pilot

Acts as the technical support partner and second pair of eyes for the Pilot to ensure flight safety and sensor data quality.

- **Execute Timed Rotations (30–60 Minutes):** Rotate alongside the Pilot to maintain continuous vigilance across telemetry displays.
- **Monitor Dual Cameras & Sensor Telemetry:** Verify camera lighting, exposure, and continuous data logging across all environmental sensors (CTD, Oxygen, Fluorometer).
- **Sonar Hazard & Topography Analysis:** Interpret sonar displays for upcoming hazards and terrain changes (e.g., sand waves) requiring rhythmic winch adjustments.
- **Pilot Verbal Communication:** Maintain clear, concise verbal alerts to notify the Pilot immediately of upcoming obstacles or depth changes.
- **Real-Time Instrument Optimization & Software Mitigation:** Adjust camera frame rates, sonar ranges, and winch settings; immediately troubleshoot station software freezes or network glitches.

29.1.6 Annotator

Frontline scientific data processor responsible for identifying, counting, and recording biological and habitat targets from incoming imagery.

- **Execute Timed Rotations (30–60 Minutes):** Process imagery in focused blocks alongside co-annotators to maintain high visual accuracy.

- **Process Image Assignments:** Rapidly review 10-minute image batches, logging primary targets (commercial species/fish) and secondary environmental parameters (substrate, encrusting organisms).
- **Complete Mandatory Protocol Training:** Complete pre-cruise or under-sail identification training to align species classification standards with project guidelines.

29.1.7 QA/QC Lead

Primary auditor and educator guarding scientific data integrity across the annotation team.

- **Host Pre-Cruise Training Overview:** Lead mandatory orientation outlining data quality standards, species identification guides, and target priorities.
- **Instruct Standardized Protocols:** Teach exact identification criteria, substrate classifications, and efficient software operation.
- **Audit & Certify Annotators:** Review, grade, and verify a baseline sample set of ~30 images for every annotator to calibrate accuracy before independent logging.

29.1.8 Population Dynamics Specialist

Lead quantitative scientist linking field data acquisition with statistical modeling, stock assessment, and spatial analysis.

- **Strategic Trackline Planning & Dynamic Routing:** Collaborate with Chief Scientist to design survey routes based on depth stratification and historical distribution, updating lines dynamically for weather or time constraints.
- **Integrate into Operational Watch Rotation:** Stand regular shifts as Pilot, Co-Pilot, and Annotator to maintain direct understanding of data collection conditions.
- **Camera Calibrations & Altimeter Verification:** Oversee tank/field camera calibrations, calculate altimeter confidence intervals, and determine exact swept-area scale metrics.
- **Data Synthesis & Reporting:** Aggregate biological annotations, identify spatial coverage gaps, summarize population trends, and present preliminary statistical findings to leadership.

29.2 Detailed Field Team Roles

29.2.1 CHIEF SCIENTIST

As the Chief Scientist, you act as the primary liaison between the scientific party, the vessel's crew, and shore-based support. You must be prepared to make decisive, on-the-spot decisions at any time of day or night to ensure the success and safety of the mission. Maintaining an open-door policy is essential to foster a communicative and supportive environment.

In addition to these duties, you are responsible for:

- **Submit Ship Time Requests:** Take charge of the administrative process required to secure vessel availability. This involves drafting and submitting formal requests to the appropriate scheduling bodies or funding, ensuring that the requested dates align with the mission's scientific objectives and seasonal requirements.
- **Plan the Survey and Decide on Tracklines:** Lead the overarching scientific design of the survey. You will map out the geographical scope of the survey, establish precise tracklines (routes) for data collection, and determine the most efficient deployment of instruments. You must also be ready to adapt these plans dynamically in response to weather, sea conditions, or preliminary findings.
- **Personnel Selection and Staffing:** Assemble a highly capable scientific team. You are responsible for identifying, recruiting, and selecting qualified researchers, technicians, and watchstanders, ensuring the team possesses a balanced mix of skills, experience, and expertise necessary to achieve the cruise's specific goals.
- **Complete and Delegate Cruise Plan Updates:** Maintain the official cruise plan as a living document. You will finalize all operational schedules, scientific protocols, and logistical details, keeping the document updated as parameters change. You must pass these updates onto your co-lead to ensure uninterrupted operational continuity and shared leadership.
- **Facilitate Communications Between Crews:** Serve as the central communication hub. You must bridge the gap between the scientific party and the maritime professionals operating the vessel, ensuring that both groups understand each other's needs, limitations, and daily objectives to maintain a harmonious and productive working environment.
- **Interact with the Vessel Captain and Crew:** Cultivate a strong, collaborative relationship with the ship's command. This requires regular, professional meetings with the Captain and senior crew to discuss navigation, logistical feasibility, impending weather systems, and the safe execution of complex scientific operations.
- **Update Watch Chiefs:** Empower your shift leaders. You are responsible for consistently briefing the Watch Chiefs on the broader scientific objectives, current equipment status, and any deviations from the primary

plan, ensuring they have the information required to effectively manage their respective teams.

- **Communicate Changes for the Upcoming Watch:** Guarantee smooth shift transitions. Before a new watch takes over, you must proactively communicate any immediate shifts in strategy, altered tracklines, weather-related delays, or new scientific priorities so the oncoming team can hit the ground running without confusion.
- **Maintain Active Oversight During Watches:** Provide hands-on leadership and support. You are expected to be awake, physically present, and actively engaged for at least half of every single watch shift (day and night). This ensures you are available to troubleshoot equipment, guide data collection, and make immediate judgment calls.
- **Conduct Daily Safety Briefings at Sea:** Champion a culture of safety. You will lead mandatory daily meetings with the scientific team to review general emergency protocols, address specific operational hazards anticipated for that day's operations, and ensure that all personnel remain vigilant and secure while at sea.

29.2.2 HABCAM LEAD

As the HABCAM Lead, you are the primary technical authority responsible for the deployment, operation, and data integrity of the imaging system. You manage both the physical hardware of the vehicle and the IT infrastructure required to process its data.

Your responsibilities include:

- **Perform Fiberoptic Termination:** Execute precise fiberoptic splicing and termination through the hydraulic winch line to ensure high-speed data transmission between the underwater HABCAM vehicle and the shipboard servers. This requires minimizing signal loss to guarantee reliable, real-time communication and video feeds.
- **Configure HABCAM IT Infrastructure:** Lead the setup of all specialized hardware and networking. This includes deploying and configuring the main HABCAM servers, pilot and co-pilot control stations, and annotation machines, ensuring seamless network integration and communication across the entire system.
- **Install and Calibrate the GPS Time Server:** Configure the dedicated GPS used for the network's time server. This guarantees that all scientific data, images, and metadata collected by the HABCAM are precisely time-stamped and synchronized to a single, highly accurate master clock.
- **Integrate Shipboard Navigation Systems:** Establish stable, reliable connections between the HABCAM system and the vessel's primary navigation instruments, specifically the ship's GPS and fathometer (depth sounder). This integration is crucial for accurate geographical and depth referencing of all collected imagery.

- **Maintain Server Operations:** Verify that all essential server infrastructure is fully operational at all times. This requires routinely checking that all necessary storage drives are correctly mounted and ensuring the Apache web server is running smoothly to host and distribute data to the science team without interruption.
- **Monitor Raw Data Acquisition:** Monitor the data pipeline to confirm that all raw imagery and sensor telemetry captured by the HABCAM are being successfully and continuously written to the primary storage servers, preventing any loss of data.
- **Build and Optimize Light Maps:** Generate and implement accurate light maps for the underwater camera systems. This calibration process corrects for uneven illumination caused by the strobes, ensuring consistent exposure and quality, evenly lit visuals for accurate scientific analysis and annotation.
- **Oversee Image Processing Workflow:** Supervise the real time image processing. You must verify that incoming raw images are being correctly processed, formatted, and reliably served to the annotation machines so that the scientific team can seamlessly identify and log marine life and habitats.
- **Perform Quality Assurance on Metadata:** Conduct quality control checks within the PostgreSQL database. This involves regularly verifying the accuracy, completeness, and integrity of all metadata (e.g., timestamps, location coordinates, altitude, depth) associated with the HABCAM imagery to ensure the final dataset is scientifically valid.
- **Execute Continuous Data Backups:** Implement and oversee a data redundancy protocol. You are responsible for continuously backing up all raw data, processed images, and database records onto spare, secure drives to safeguard against hardware failure, data corruption, or loss during the survey.
- **Conduct Pre-Deployment Vehicle Checks and Troubleshooting:** Lead the pre-dive inspection of the physical HABCAM vehicle. You must test all mechanical and electronic systems on deck, quickly identify potential faults, and troubleshoot any electrical or software issues before deployment to ensure a safe, successful, and uninterrupted dive.

29.2.3 WATCH CHIEF

As a Watch Chief, you are the direct shift supervisor responsible for managing the personnel, operations, and data collection during your assigned rotation. You ensure that the scientific objectives set by the Chief Scientist are executed safely and efficiently while maintaining the standards required for accurate data logging.

Your responsibilities include:

- **Manage 12-Hour Operational Shifts:** As one of exactly two Watch

Chiefs assigned to the cruise, you are responsible for commanding a continuous 12-hour shift. You must maintain operational efficiency during your watch and ensure a seamless, detailed handover of information and system status to your counterpart during shift changes.

- **Assign and Manage Rotating Station Roles:** Orchestrate the team's workflow to maximize focus and prevent fatigue. You will create and enforce a schedule, rotating your assigned scientific personnel through the active stations (Pilot, co-pilot, and image annotator). You are also responsible for verifying that all team members are present, rested, and prepared for their assigned blocks.
- **Control Core Operational Software:** Serve as the gatekeeper for the vehicle's flight systems. You hold the direct responsibility for properly and securely starting up, monitoring, and shutting down the specialized pilot and co-pilot software systems used to fly and monitor the HABCAM, ensuring all programs are stable and communicating correctly with the hardware.
- **Train and Onboard New Scientists:** Act as the primary shift educator. You will conduct comprehensive training sessions for new or inexperienced scientists using standardized materials, such as the annotation PowerPoint. You must ensure they thoroughly understand the established protocols for identifying marine life, geological features, and logging accurate metadata.
- **Actively Participate in the Operational Cycle:** Lead by example. Rather than solely acting in a supervisory capacity, you are expected to fully integrate into the active rotation schedule. You will regularly take your turn at the pilot, co-pilot, and annotator stations, bearing a portion of the continuous operational workload alongside your team.
- **Perform Quality Assurance and Quality Control (QA/QC):** Safeguard the integrity of the scientific data. You will actively review the biological and habitat annotations logged by your team members during the shift. This involves cross checking complex species identifications, identifying and correcting entry errors, and ensuring that the data collected under your watch meets the survey's scientific standards.

29.2.4 PILOT

You are directly responsible for the safe, precise, and active underwater navigation of the HABCAM vehicle. Because this role requires intense concentration and quick reflexes to protect scientific equipment, you operate in short, focused intervals.

Your responsibilities include:

- **Execute Rotational Shifts (30 to 60 Minutes):** Maintain concentration and prevent operational fatigue. Because flying the vehicle requires

uninterrupted attention, you will rotate into the pilot's seat for strictly timed blocks of 30 minutes to one hour. This rotation ensures your reflexes and situational awareness remain sharp during your entire time at the helm.

- **Maintain Safe Altitude Over the Seafloor:** Safeguard the physical integrity of the HABCAM vehicle. Your primary, overriding objective is to "fly" the vehicle at the optimal target altitude for photography (typically just a few meters above the bottom). You must constantly adjust to sudden changes in bathymetry to prevent catastrophic impacts with the seabed or entanglement in benthic structures.
- **Operate the Ship's Winch System via Joystick:** Manually control the vertical movement of the towed vehicle. Utilizing a specialized joystick interface, you will directly command the heavy machinery of the ship's winch. You must actively pay out or haul in the tow cable, making precise, real time micro-adjustments to the HABCAM's depth as the vessel moves forward through the water column.
- **Monitor Live Camera Feeds and Flight Telemetry (GUI):** Maintain constant visual and data awareness. You must continuously process multiple streams of information simultaneously, keeping your eyes locked on both the real time, downward-facing video feed from the HABCAM and the graphical user interface (GUI) of the winch flight software. This allows you to react to both visual terrain cues and telemetry data, such as cable tension, vehicle pitch, and exact depth.
- **Scan Forward-Looking Sonar for Obstacle Avoidance:** Anticipate and evade underwater hazards before they become a threat. Because visibility is limited underwater, you will continuously read and interpret the forward-looking sonar display to detect upcoming obstacles. Such as steep drop-offs, large boulders, shipwrecks, or abandoned fishing gear; well before they enter the camera's field of view, allowing you to proactively haul in the winch and avoid a collision.

29.2.5 CO-PILOT

As the Co-Pilot, you act as the essential second pair of eyes and primary technical support for the active Pilot. While the Pilot focuses entirely on the physical maneuvering of the vehicle, you are responsible for comprehensive situational awareness, system optimization, and hazard detection to ensure both the safety of the HABCAM and the quality of the scientific data.

Your responsibilities include:

- **Execute Rotational Shifts (30 to 60 Minutes):** Maintain vigilance as part of a synchronized team. Just like the Pilot, you will rotate into the Co-Pilot station for short, focused intervals of 30 minutes to one hour. This frequent rotation prevents screen fatigue and ensures you remain sharply attuned to subtle changes in sensor data and visual feeds.

- **Monitor Dual Camera Feeds and Sensor Telemetry:** Oversee the vehicle's diagnostic health. You are tasked with continuously watching the live video streams from multiple cameras to ensure clear visibility and proper illumination. Simultaneously, you must verify that all onboard environmental sensors (such as temperature, depth, and altimeters) are actively logging and transmitting accurate, uninterrupted data back to the ship.
- **Analyze Sonar for Obstacles and Topographical Shifts:** Serve as the early warning system for navigational hazards. You will actively interpret the sonar displays to map out the terrain ahead. This includes not only spotting immediate, dangerous obstacles (like boulders or wrecks) but also identifying complex seafloor topography, such as rolling sand waves, which require the Pilot to make continuous, rhythmic depth adjustments.
- **Communicate Critical Concerns to the Pilot:** Facilitate seamless, high-stress teamwork. You must maintain clear, constant, and concise verbal communication with the Pilot. If you detect an upcoming obstacle on the sonar or notice a sudden drop in sensor performance, you are responsible for immediately alerting the Pilot so they have the maximum possible time to react and adjust the winch.
- **Optimize Hardware and Software Settings in Real-Time:** Manage the configuration of the HABCAM's instruments. As environmental conditions change, you will actively fine tune the system's parameters. This includes adjusting sonar ranges for better resolution, modifying camera frame rates to match the vehicle's speed and altitude, and tweaking settings within the winch flight software to ensure smooth data capture.
- **Troubleshoot and Mitigate Station Software Issues:** Act as the immediate first line of defense for control station stability. If the pilot/co-pilot software freezes, an application crashes, or a localized networking glitch occurs, you must diagnose and resolve the issue on the fly. Your mitigation prevents operational downtime and ensures the Pilot does not lose control or visibility of the vehicle.

29.2.6 ANNOTATOR

As an Annotator, you are the frontline data processor of the scientific survey. Your primary responsibility is to translate the raw visual imagery captured by the underwater HABCAM into quantifiable, actionable scientific records. Because this requires sharp visual acuity and rapid decision making, you operate within a fast paced, collaborative workflow.

Your responsibilities include:

- **Execute Rotational Shifts (30 to 60 Minutes):** Maintain visual sharpness and combat screen fatigue. Accurately identifying camouflaged marine life and complex benthic habitats requires continuous concentration. To ensure the highest data quality, you will rotate through the

annotation station in 30- to 60-minute intervals. You will frequently work concurrently alongside several other annotators to divide the workload and maintain a steady processing speed.

- **Process Primary and Secondary Image Assignments:** Analyze a rapid, continuous stream of visual data. You are tasked with reviewing batches of newly captured seafloor images, which are automatically pushed to your workstation every 10 minutes. You must swiftly and accurately log specific targets according to your assigned priorities; this typically involves identifying and counting primary species of interest (such as specific fish or scallops) while also recording secondary environmental details, like substrate types and other invertebrate life.
- **Complete Mandatory Identification and Protocol Training:** Guarantee standardized data collection and scientific validity. Before taking on active shifts, you must complete a specialized training program, conducted either prior to departure or while the vessel is under sail to the survey site. This essential onboarding ensures your identification skills are calibrated with the rest of the scientific party, and that you are fully proficient in utilizing the survey's specific annotation software and classification guidelines.

29.2.7 QA/QC LEAD

As the QA/QC lead, you are the primary guardian of data integrity for the survey. Your main objective is to ensure that all visual data processed by the scientific team is highly accurate, consistent, and scientifically sound. You act as both an educator and auditor for the annotation team.

Your responsibilities include:

- **Host the Comprehensive Pre-Cruise Training Overview:** Organize and lead the foundational orientation for the scientific party. Before the vessel departs or active data collection begins, you will host a mandatory, large-scale training session. This overview sets the baseline expectations for data quality, distributes visual reference guides, and aligns the entire team on the specific biological and ecological targets of the upcoming survey.
- **Train Annotators on Standardized Protocols:** Serve as the primary, ongoing educator for the data processing team. You are responsible for thoroughly instructing all annotators on the exact, standardized protocols required for the mission. This includes teaching precise species identification criteria, benthic habitat classification, and the correct, efficient operation of the customized annotation software to ensure absolute consistency across all shifts.
- **Audit and Confirm Baseline Annotation Accuracy:** Conduct targeted quality control checks on individual performance. For every annotator on the team, you will review, grade, and verify a sample set of

approximately 30 assigned images. This initial audit allows you to catch individual misidentifications early, provide constructive, one-on-one feedback, and officially certify that each annotator is fully calibrated and ready to process large volumes of data independently.

29.2.8 POPULATION DYNAMICS

As the Population Dynamics specialist, you are the lead quantitative scientist for the survey. Your primary role is to ensure that the data collected is statistically robust and biologically meaningful for stock assessments and ecological modeling. You bridge the gap between active field operations and high level data analysis.

Your responsibilities include:

- **Assist in Strategic Trackline Planning and Dynamic Routing:** Collaborate directly with the Chief Scientist to design mathematically sound survey routes. You provide input based on statistical requirements, historical population distributions, and depth stratifications. During the cruise, you actively participate in real-time decision-making, recalculating and updating tracklines on the fly when the team faces severe weather, time constraints, or unexpected biological discoveries.
- **Actively Participate in the Operational Cycle:** Maintain a hands-on connection to the raw data collection process. Rather than solely working on backend analysis, you are expected to fully integrate into the rotating shift schedule. You will take regular turns functioning as a HABCAM Pilot, Co-Pilot, and Annotator alongside the rest of the scientific party, ensuring you intimately understand the operational conditions and visual complexities of the data you will later analyze.
- **Execute System Calibrations and Altimeter Verification:** Guarantee the spatial accuracy of the survey's visual data. You will oversee camera calibration procedures and analyze the resulting calibration annotation data. You must calculate the statistical confidence of the altimeter's positioning to ensure that the exact scale of the images is known, which allows for precise measurements of organism density and total seafloor area swept. You will leverage these metrics to continually assist cruise planners in developing efficient tracklines for each leg of the journey.
- **Analyze Data, Identify Gaps, and Synthesize Results:** Serve as the central hub for data synthesis. You are responsible for aggregating and mathematically analyzing the massive volume of biological and habitat annotations generated by the shift teams. This requires meticulously auditing the dataset to identify any spatial or temporal gaps in coverage, summarizing population trends, and formally presenting these preliminary statistical results to leadership for ongoing survey adjustments and future stock assessments.

Chapter 30

HabCam Pre-Deployment Procedures

Complete the following procedural checklist prior to deploying the HabCam vehicle into the water:

Vehicle Checklist: Complete the Pre-Deployment Vehicle Checklist. [Fill out details for pre-deployment vehicle checklist procedures here]

Power Inverter Activation: Power up the vehicle via the power inverter located at the Watch Chief Station.

Flip the Right switch first, then flip the Left switch.

Image Viewer Configuration: Open two image viewers, minimize one, and drag it over to the Pilot Station display:

Adjust frame rate to 6.00 Hz.

Set gain to [Fill out details for gain setting].

Stereo Acquisition Launch: Click stereo acquisition to start real-time image processing.

Verify that a small black terminal window opens on the Co-Pilot's left monitor displaying the live data feed.

Troubleshooting: If the image viewer freezes, verify runVimCam status or click Stereo Acquisition to relaunch.

[Fill out details for additional pre-deployment checks, sensor verifications, and deck communications here]

Chapter 31

After-Cruise Checklist

31.1 Day of Destaging

31.1.1 Data Backup

- ☐ Verify with the HabCam lead that HabCam survey data is backed up
- ☐ Ensure the ship data is copied onto a USB drive and given to Chief Scientist
  - ☐ Copy to `FSCS_RAWDATA`, following completion of the cruise.
- ☐ If CTDs were performed, the crew will provide the CTD Logs for Oceanography.
- ☐ Make sure all special requests samples are taken off the ship at the completion of the cruise, coordinate pick-up of samples from researchers (this may be done pre-cruise; check with Advanced Tech lead or previous leg's CS).

[Insert details on data offload verification, server network paths, external drive formatting protocols, and photos of backup storage setups here]

31.1.2 HabCam Vehicle Teardown

- ☐ Cut tow cable one foot before termination (prior to black boot)
- ☐ Drain junction box
- ☐ Rinse down HabCam with freshwater
- ☐ Detach tail piece while HabCam is still on the vessel.
- ☐ Detach clevis [Will need to be taken to Reidar's Manufacturing in New Bedford to break down the socket lock]

- ☐ Disconnect and take the power converter.

[Insert details on freshwater washdown procedures, junction box oil draining, clevis transport logistics to Reidar's, and photos of vehicle disassembly here]

31.1.3 Workstations Breakdown

- ☐ Break down monitors and put in tote bags
  - ☐ Put cables in designated boxes
- ☐ Break down the server.
- ☐ Place computers into designated pelican cases.
- ☐ Disconnect GPS puck from flying bridge.
- ☐ Place ethernet cables into Ethernet tote.
- ☐ Remove and consolidate plywood.

[Insert details on server rack dismounting, cable packing inventories, flying bridge access procedures, and photos of packed Pelican cases/totes here]

31.1.4 Foul Weather Gear

- ☐ Place NMFS labeled gear back into mesh bags
- ☐ Place PFD's into a long black tote.

[Insert details on gear laundering, safety inspection, drying guidelines, and photos of storage totes here]

31.1.5 Transportation & Logistics

- ☐ Ensure everyone (including volunteers and Teachers at-Sea) have proper travel arrangements following completion of cruise.
- ☐ If needed, work with onshore support staff/branch chief to ensure a vehicle is available for a provision run in between cruise legs.
- ☐ Divide vans for what is going to Woods Hole (keep WHOI and NMFS stuff separate) and what is going to Netloft

[Insert details on shoreside transport routes, Woods Hole vs. Netloft unloading schedules, equipment segregation rules, and photos of van staging here]

Part VI

Maintenance & Inventory

Chapter 32

Maintenance Schedule

[Insert annual hardware inspection schedules, sensor re-certification timelines, and service logs here]

Chapter 33

Maintenance

33.1 Replace Doors

1. Drill out three holes down the side of the door, with a 5/16 drill bit (you can use an old door as a template).[cite: 1]

   Side view of HABCAM door panel mounted inside frame.

2. Attach the door to HABCAM by threading a 3/8 bolt through a washer and fender washer (fender washer against door, washer on top of fender washer) with the door on the inside of the door frame and the shank facing away from the vehicle. Secure the shank end of the bolt with washer, lock washer and nut.[cite: 1]

   Door panel alignment showing cable harness routing.

3. Position the door and door latches, so that the top of the door is on the first hole (from the outside edge) on the door latch and the bottom of the door is on the second hole (from the outside edge).[cite: 1]

   Bottom latch assembly and bolt placement view.

4. On door, Measure with a tape measure and/or mark hole placement with a sharpie.[cite: 1]

5. Remove door, and use a 3-16 thread tapper to drill a shallow hole for the top and bottom door latch bolt to sit in.[cite: 1]

6. Reattach door to door frame (door on the inside of the door frame and the shank facing away from the vehicle) by threading bolt through washer and fender washer. Prior to adding the stainless-steel washer, lock washer and nut, apply aqua shield over all of the threads on the side of the door (no need to do this on the top bolts).[cite: 1]

7. Secure door on top and bottom latch using a 3/8 bolt with washer and lock washer (washer against head of bolt). Use a 3/8 ratcheting wrench to pinch the door “closed” into a fixed position.[cite: 1]

33.2 Sand HabCam

[Placeholder: Insert instructions for sanding HabCam frame/doors here]

33.3 Splice Cables

[Placeholder: Insert instructions for cable splicing here]

33.4 Fabricate Camera Plate

[Placeholder: Insert instructions for fabricating camera plate here]

Chapter 34

Inventory

Location	Box	Item	Quantity

Chapter 35

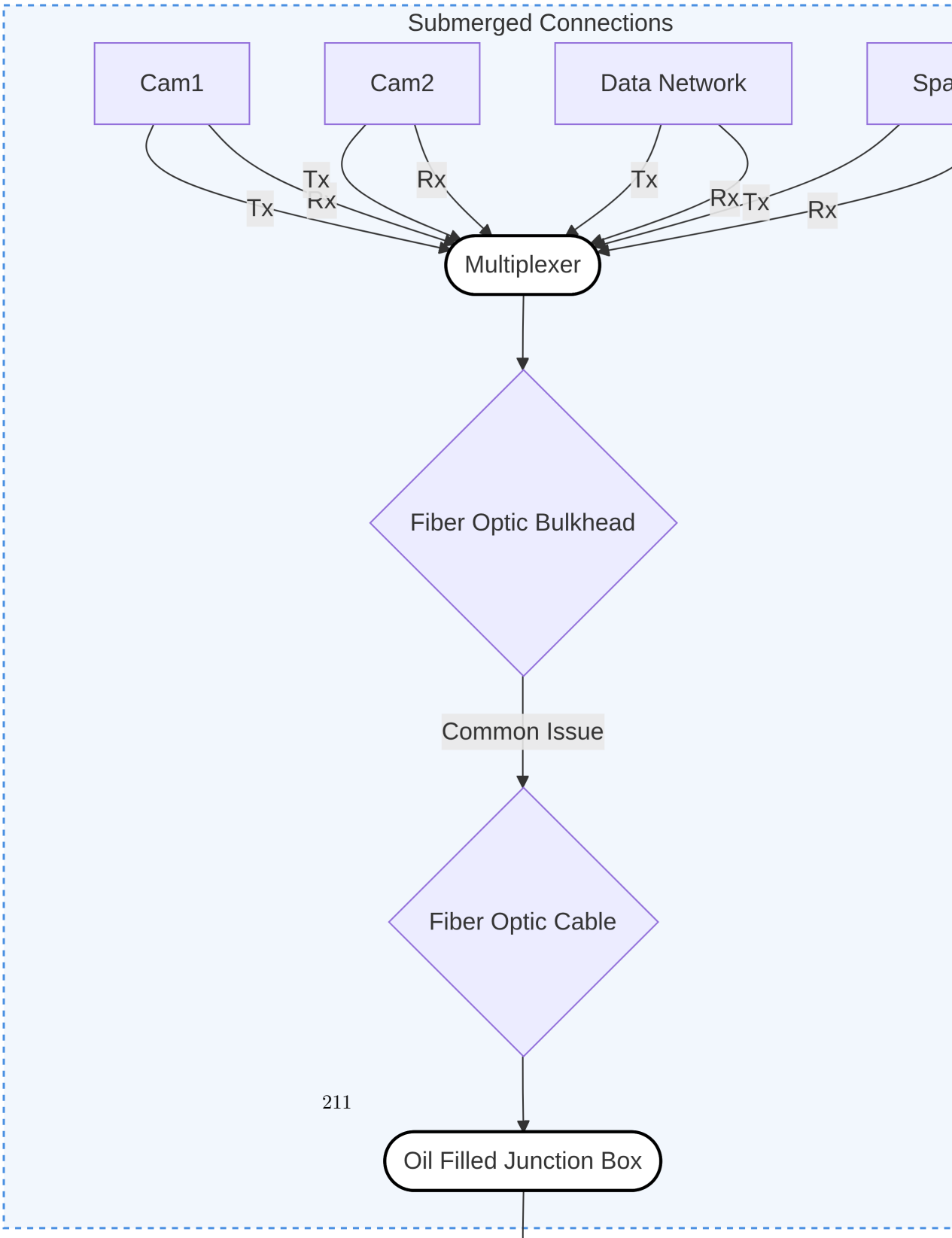
Troubleshooting

35.1 Fiber Optics

Inside the HabCam main electronics bottle there are four media converters: one for each camera, one for the data network (all other sensors, sonars, moxas, gardasofts, expansion bottle, and the chanos sensor), and a spare port not actively used. When the vehicle is powered on, full link lights on the front of each media converter indicate functional fiber optic connectivity per channel.

Multiple junctions (coupling points) exist throughout the system where improper seating or contamination will cause data loss.

35.1.1 Connection Flowchart



Inspection Checklist

Carefully inspect designated high-failure points:

- **OptoLink bulkhead & mating cable** on the main electronics bottle
- **Ship's slip ring** assembly
- **Dry lab junction box to demultiplexer** interconnects

Cleaning Protocols

- **ST Connectors:** Clean using isopropyl alcohol and CLETOP fiber optic wipes. Use fiber optic insert cleaning pens for couplers.
- **Subconn OptoLink Series:** Ensure mating surfaces are dry and clean using **only** isopropyl alcohol and Kimtech wipes. Apply **Molykote Series 55** o-ring grease to o-rings.

35.2 OptoLink Connectors

Follow these instructions carefully to ensure correct use of your OptoLink connector and penetrator harnesses.[cite: 1]

35.2.1 Assembly

35.2.1.1 OptoLink A-CCP to OptoLink A-FCR, BCR or CCR Receptacle

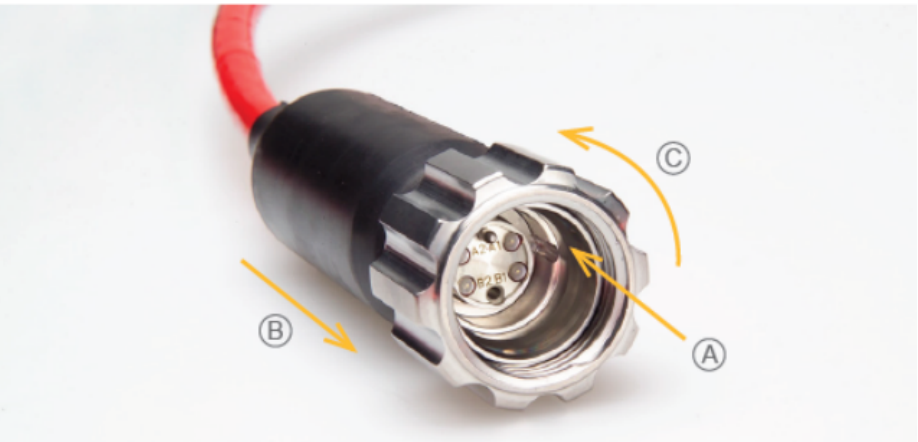


Figure 35.1: OptoLink Assembly

- A: Keyway visual alignment; B: Push connection path; C: Locking sleeve rotation direction.*

1. Inspect the connector face and O-rings. Grease the O-rings with Molykote 111.
2. Inspect the connector surface. If the connector surface is dirty, follow the OptoLink maintenance instructions below.
3. Visually align the key way on the CCP connector with the key pin tip on the receptacle before mating (illustration A).
4. Do not twist the connector to align it during mating. Push to mate the connector (illustration B).
5. Engage connector whilst turning the locking sleeve (illustration C).

> The connector must be hand tightened.

35.2.2 Maintenance

35.2.2.1 Receptacle Maintenance (A-CCP, A-FCR, BCR or CCR)

- **Dirty Lenses:** Clean with a dry cotton swab and isopropyl alcohol.

- **Greasy Lenses:** Clean with an isopropyl alcohol soaked cotton swab followed by a dry cotton swab. Repeat the cleaning process if necessary.
- **O-Ring Care:** Check O-rings before mating/mounting; replace if showing signs of damage. Always grease O-rings with Molykote 111.

> Always install dust caps when connectors are unmated.

35.2.2.2 OptoLink Penetrator Harnesses

- O-rings must be checked before mounting and will require replacing if they show signs of damage.
- Always grease O-rings with Molykote 111.

35.2.3 Torque & Mounting Specifications

35.2.3.1 Thread Locking

For locking metallic connectors, the use of Loctite 243 is recommended.

35.2.3.2 Bulkhead Connectors Tightening Force

Type	Material	Rec. Torque (Nm)
5/8”-18 UNF	Brass, aluminium	29.0
5/8”-18 UNF	Stainless steel, titanium	41.0
7/8”-14 UNF	Brass, aluminium	60.0
7/8”-14 UNF	Stainless steel, titanium	80.0

35.2.3.3 Pressure Balanced Systems

FCR and BCR are rated to 600 bar back pressure. MacArtney recommends DC-200/350 or PMX-200/350 in oil compensated systems.

35.3 SubConn Connectors

Follow these handling instructions carefully to ensure correct operation and longevity of your SubConn underwater environment connectors.

35.3.1 General Handling Guidelines

- **Greasing Standard:** Connectors must be greased with **MOLYKOTE 44 Medium** before every mating.
- **O-Ring Care:** Always grease O-rings on BH, BCR, and FCR connectors with **Molykote 111**.
- **Disconnect Technique:** Pull straight out when disconnecting; do not pull at an angle.
- **Cable Safety:** Do not pull directly on the cable and avoid sharp bends at the cable entry point.
- **Bulkhead Loads:** Ensure there are no angular loads on bulkhead connectors and apply recommended torque when tightening bulkhead nuts.
- **Environmental Exposure:** Do not expose connectors to extended periods of heat or direct sunlight. If a connector becomes very dry, soak it in fresh water before use.
- **Electrical & Sleeves:** All current ratings are based on the connector being submerged in water. The locking sleeve must be hand-tightened only.

35.3.2 Greasing and Mating Procedures

Above Water (Dry-Mate)

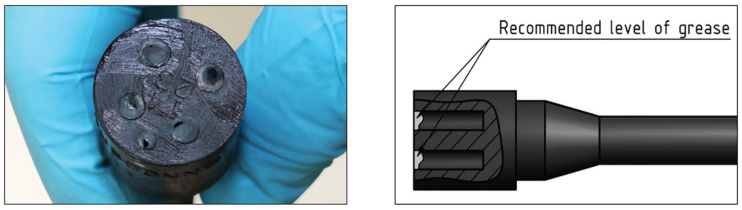


Figure 35.2: Dry-Mate Greasing Level

1. Apply a layer of MOLYKOTE 44 Medium corresponding to a minimum of **1/10 of the socket depth** to the female connector.
2. Ensure the inner edge of all sockets is completely covered, leaving a thin layer of grease visible on the connector face.
3. Fully mate the male and female connectors to distribute grease across all contacts and sockets.
4. De-mate to verify grease coverage on every male pin, then re-mate.

Under Water (Wet-Mate)

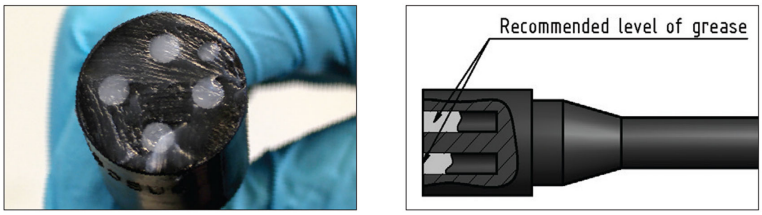


Figure 35.3: Wet-Mate Greasing Level

1. Apply a layer of MOLYKOTE 44 Medium corresponding to approximately **1/3 of the socket depth** to the female connector.
2. Completely seal all sockets, leaving a transparent layer of grease visible on the connector face.
3. Fully mate the connectors underwater and wipe away any excess grease extruded from the joint.

35.3.3 Maintenance & Thread Locking

35.3.3.1 Cleaning Protocol

- **General Cleaning:** Remove accumulated contaminants using spray-based contact cleaner (Isopropyl alcohol) or liquid soap with hot water.
- **Re-Greasing:** Fresh MOLYKOTE 44 Medium must be re-applied prior to mating whenever connectors are cleaned.

35.3.3.2 Thread Locking Specifications

Ensure all threads and threaded holes are clean and completely free of foreign particles prior to application.

- **Metallic Connectors:** Apply **LOCTITE 243**.
- **Non-Metallic (PEEK) Connectors:** Apply **LOCTITE 5910**.

> ⚠ **Critical PEEK Warning**
>
> Never use Acetone on PEEK non-metallic connectors, as it causes severe stress-induced cracking in the material.

35.3.4 Bending Radius & Coax Specifications

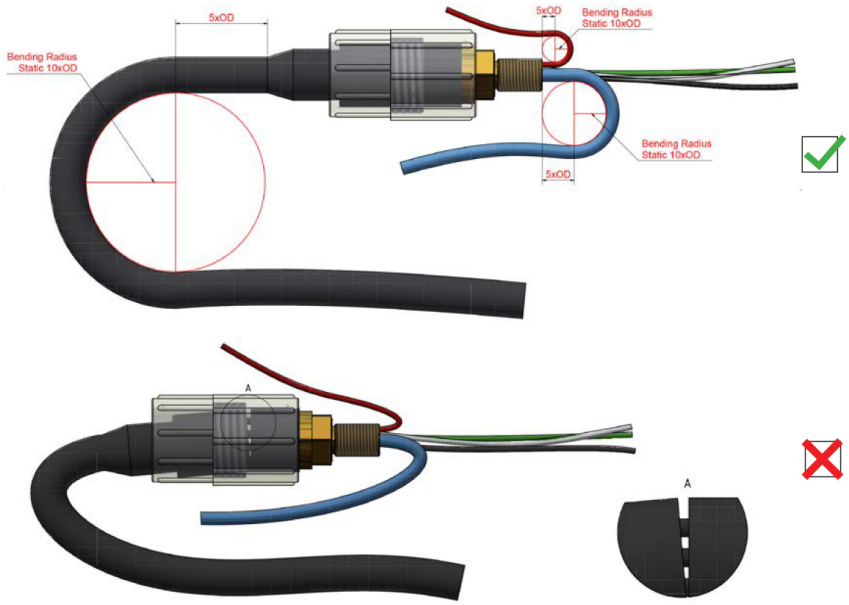


Figure 35.4: Inline Cable Bending Radius Guidelines

Minimum static cable bend radius must be 10x cable diameter (top); avoid sharp bends at cable entry (bottom).

35.3.4.1 Recommended Cable Bending Radius

- **Static Minimum Bend: 10× Cable Outer Diameter** ($10 \times$ **OD**).
- Avoid sharp localized bends directly behind the connector strain relief boot.

35.3.4.2 Coax Connector Requirements

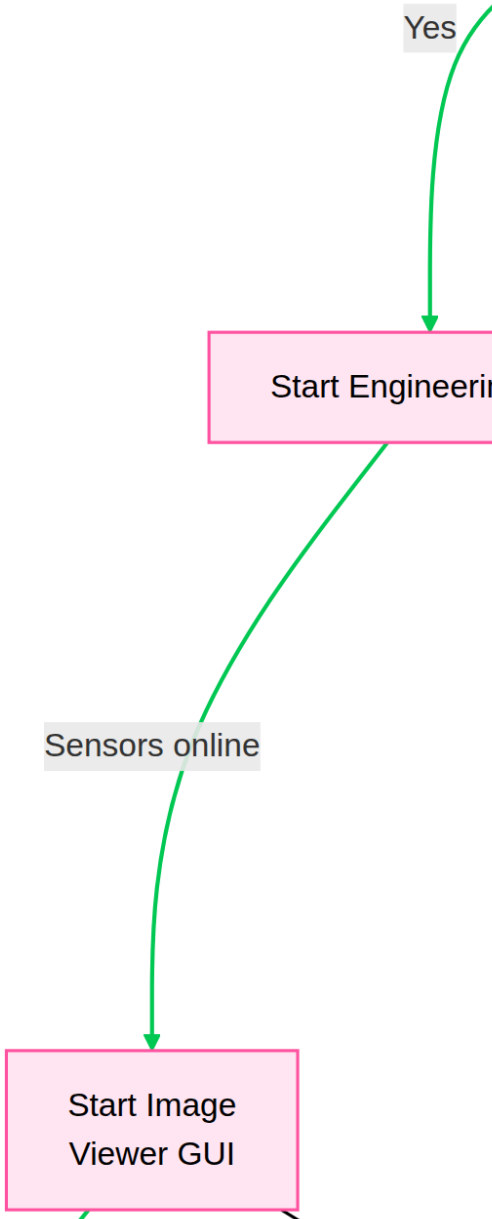
- **Mating Limit: Dry mating only**. Not suitable for open-face pressure or water-block applications.
- **Hardware:** Locking sleeves are strictly required, and dummy connectors must be installed whenever coax connectors are unmated.
- **Greasing Coax:** Avoid applying grease to the actual central coaxial contact. Grease only the outer rubber socket components. De-mate after

greasing to verify no grease contaminated the coax center before final re-mating.

35.3.5 Pressure Balanced Systems

- **Compensated Systems:** MacArtney recommends **DC-200/350** or **PMX-200/350** fluid in oil-compensated pressure-balanced systems.

35.4 Vehicle Start-Up Flow Chart



35.5 Troubleshooting During Survey

During active surveys, software programs may encounter imaging freezes, application dropouts, or setting failures. Use these steps to resolve issues without interrupting survey operations.

35.5.1 Examples & Solutions

Example 1: Co-Pilot Image Viewer Frozen (One or Both Cameras)

If the image viewer freezes on one or both cameras at the Co-Pilot position:

1. Check **RunVimCam**.
2. Double-click and restart **Stereo Acquisition**.
3. Confirm that **Rate** is set back to **6.00 Hz** and restore gain to preferred survey settings.

Example 2: Pilot Image Viewer Frozen

If only the Pilot station image viewer freezes:

1. Hover over any indicator lights that turned red to see if they return to green.
2. Click **"show and un-show"** on `port` or `stbd` (located under the two running image waterfalls) to restart data flow.

Example 3: Pilot Winchfly Frozen

If Winchfly freezes at the Pilot station:

1. Hover over the running application to confirm it has not entered sleep mode.
2. Click **"un-show and show winchfly"** (located under the two running image waterfalls) to open a new Winchfly window.
3. Re-adjust settings back to operational standards.

Example 4: Total Application Crash (Viewers & Winchfly Down)

If both image viewers AND Winchfly crash simultaneously, stereo acquisition has likely stopped:
If both image viewers AND Winchfly crash simultaneously, stereo acquisition has likely stopped:

1. Double-click the **Stereo Acquisition** app icon to restart application capture and re-adjust settings.
2. **Fallback Procedure:** If restarting fails, close the image viewer completely, open a fresh viewer instance, reset rate back to **6.00 Hz**, and verify all settings.